

DATA SCIENCE FOUNDATIONS

Navigating digital insight

Stephen Mariadas and Ian Huke



bcs

The
Chartered
Institute
for IT

DATA SCIENCE FOUNDATIONS

Navigating digital insight

Stephen Mariadas and Ian Huke



A brilliant guide to data science. Covers the full lifecycle with clarity and practical insight – perfect for newcomers but a fantastic reference for all.

Chris Weston, Technology Consultant, NashTech

Data Science Foundations is the perfect beginner's guide, offering a clear, engaging overview of key concepts with real-world examples. A must-read companion for anyone eager to dive into data science.

Jake Wilson PhD, Data Scientist, GHGSat Inc.

This is a very topical subject, affecting everyone's lives, and a book on its foundations could have been very dry, but the authors have struck the right blend of sharing the concepts, principles and approaches and bringing it to life with real-world examples and scenarios. My company is delivering several projects for clients using IoT (Internet of Things) which extensively use data science to deliver a service, and more broadly our teams get involved in many projects where there is a data science aspect (obvious or discrete). This book will definitely be the one I hand to our team members who are about to embark on a new project. Highly recommended.

***John Opsblum, Co-Founder and Chief Consulting Officer,
Illuminet Solutions***

This was the book that I needed! An excellent, thoughtful and concise grounding in data science, packed with relatable examples, practical tips and a comprehensive framework for achieving successful data science projects. The explanations of the link between data science, machine learning and AI, as well as the overview of the different data science models, were invaluable in enhancing my understanding. This book will become a trusted companion, one that I am sure I will return to again and again.

Paul Williamson, Senior Business Analyst (Private Equity)

A must-read for data professionals, this book blends technical depth with practical guidance. It refines model selection strategies and enhances communication, turning raw data into meaningful, actionable insights.

Raimondo Fanale MBCS, R&D Manager, Intuisco Ltd

For anyone who views data science as a dark art, but wants to improve their understanding of it, this is a 'must read' book. It guides the reader through data science concepts and practices, making it real through its extensive use of case studies and real

world examples. Refreshingly jargon free, it offers practical advice and humanises the topic. Strongly recommended.

Nigel Turner, *Principal Information Management Consultant EMEA, Global Data Strategy*

Data Science Foundations is a welcome addition to the resources available to those beginning their journey or those already practicing advanced data analysis. It manages to be both comprehensive and readable; all areas of data science are covered, but no prior knowledge is assumed. A splendid piece of work by the authors.

Martin Bell, *CEO and Chief Data Scientist, BEXIT Ltd*

A fantastic introduction to data science, covering problem definition, statistics, model selection, communication and visualisation. Clear and concise, it's an excellent starting point for anyone new to the field!

Richard Fritz-Hewer, *Data Scientist, CMC Partnership*

DATA SCIENCE FOUNDATIONS

BCS, THE CHARTERED INSTITUTE FOR IT

BCS, The Chartered Institute for IT, is committed to making IT good for society. We use the power of our network to bring about positive, tangible change. We champion the global IT profession and the interests of individuals, engaged in that profession, for the benefit of all.

Exchanging IT expertise and knowledge

The Institute fosters links between experts from industry, academia and business to promote new thinking, education and knowledge sharing.

Supporting practitioners

Through continuing professional development and a series of respected IT qualifications, the Institute seeks to promote professional practice tuned to the demands of business. It provides practical support and information services to its members and volunteer communities around the world.

Setting standards and frameworks

The Institute collaborates with government, industry and relevant bodies to establish good working practices, codes of

conduct, skills frameworks and common standards. It also offers a range of consultancy services to employers to help them adopt best practice.

Become a member

Over 70,000 people including students, teachers, professionals and practitioners enjoy the benefits of BCS membership. These include access to an international community, invitations to a roster of local and national events, career development tools and a quarterly thought-leadership magazine. Visit www.bcs.org to find out more.

Further information

BCS, The Chartered Institute for IT,

3 Newbridge Square,

Swindon, SN1 1BY, United Kingdom.

T +44 (0) 1793 417 417

(Monday to Friday, 09:00 to 17:00 UK time)

[**www.bcs.org/contact**](http://www.bcs.org/contact)

shop.bcs.org/

publishing@bcs.uk

bcs.org/qualifications-and-certifications/certifications-for-professionals/



DATA SCIENCE FOUNDATIONS

Navigating digital insight

Stephen Mariadas and Ian Huke



© BCS Learning and Development Ltd 2025

The rights of Stephen Mariadas and Ian Huke to be identified as authors of this work has been asserted by them in accordance with sections 77 and 78 of the Copyright, Designs and Patents Act 1988.

All rights reserved. Apart from any fair dealing for the purposes of research or private study, or criticism or review, as permitted by the Copyright Designs and Patents Act 1988, no part of this publication may be reproduced, stored or transmitted in any form or by any means, except with the prior permission in writing of the publisher, or in the case of reprographic reproduction, in accordance with the terms of the licences issued by the Copyright Licensing Agency. Enquiries for permission to reproduce material outside those terms should be directed to the publisher.

All trade marks, registered names etc. acknowledged in this publication are the property of their respective owners. BCS and the BCS logo are the registered trade marks of the British Computer Society charity number 292786 (BCS).

Published by BCS Learning and Development Ltd, a wholly owned subsidiary of BCS, The Chartered Institute for IT, 3

Newbridge Square, Swindon, SN1 1BY, UK.

www.bcs.org

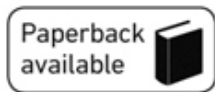
EU GPSR Authorised Representative: LOGOS EUROPE, 9 Rue
Nicolas Poussin, 17000 La Rochelle, France.

Contact@logoseurope.eu

Paperback ISBN: 978-1-78017-6994

PDF ISBN: 978-1-78017-7007

ePUB ISBN: 978-1-78017-7014



British Cataloguing in Publication Data.

A CIP catalogue record for this book is available at the British
Library.

Disclaimer:

The views expressed in this book are of the authors and do not
necessarily reflect the views of the Institute or BCS Learning
and Development Ltd except where explicitly stated as such.
Although every care has been taken by the authors and BCS

Learning and Development Ltd in the preparation of the publication, no warranty is given by the authors or BCS Learning and Development Ltd as publisher as to the accuracy or completeness of the information contained within it and neither the authors nor BCS Learning and Development Ltd shall be responsible or liable for any loss or damage whatsoever arising by virtue of such information or any instructions or advice contained within this publication or by any of the aforementioned.

All URLs were correct at the time of publication.

Publisher's acknowledgements

Reviewers: Sophie Carr, Adam Ben Hill, Jagdev Bhogal

Publisher: Ian Borthwick

Commissioning Editor: Heather Wood

Production Manager: Florence Leroy

Project Manager: Sunrise Setting Ltd

Copy-Editor: Mary Hobbins

Proofreader: Barbara Eastman

Indexer: David Gaskell

Cover Design: Alex Wright

Cover Image: iStock/natalia lagutkina

Sales Director: Charles Rumball

Typeset by Lapid Digital Services, Chennai, India

CONTENTS

[List of figures and tables](#)

[Authors](#)

[Foreword](#)

[Acknowledgements](#)

[Abbreviations](#)

[INTRODUCTION](#)

[Practical tips](#)

[STAKEHOLDERS](#)

[Identifying stakeholders](#)

[Investigating stakeholders](#)

[Managing stakeholders](#)

[Practical tips](#)

[PROJECT DELIVERY](#)

[Data analysis lifecycle](#)

Managing the lifecycle

Practical tips

ETHICS AND LAWFULNESS

When to consider?

Who might be impacted?

Ethics

Legislation and regulation

Internal policies

Other governance

Commercial constraints

Intellectual property

Risk mitigation strategies

Practical tips

DISCOVERY

The team

Understanding the domain

Resources

Problem definition

Developing the hypothesis

Effective enquiry.

Practical tips

PROPERTIES OF DATA

The importance of the right properties

Bias or skewed data?

Quantitative research

Qualitative research

Things to note

SOURCING

Big Data

Data collection

Principles of data: data types

Finding data

Storing data

Moving data

Data governance

Data quality

Metadata

Practical tips

PREPARATION

Form

Properties of data sets

Quality risks

Quality checks

Mitigation strategies

Practical tips

BASIC CONCEPTS

Averages

Measures of spread

Probability

Navigating the Cartesian plane and measuring distances

Conclusion

MODEL SELECTION

Which model to use?

Descriptive analytics: understanding the past

Predictive analytics: peeking into the future

Prescriptive analytics: deciding what to do

How they work together

Understanding the data for your model

Understanding the hypothesis

How to choose between different models

Predicting value over time: time series analysis

Classification

Clustering

Association

VISUALISATIONS

Considerations

User experience in data visualisation

Bar chart

Histogram

Line graph

Area chart

Scatter plot

Box and whisker plot

Heat map

Stem-and-leaf plot

Combining charts

Practical tips

MODEL EVALUATION

What is probability?

What is a p -value?

Hypothesis testing: a step-by-step guide

Is this chance? Understanding significance

How much does the model explain? Comparing train and test

How to interpret the model

COMMUNICATION

Principles of user experience

Improving user experience

Know your audience

Domain context

Information to knowledge

Storytelling

Risks in storytelling

Visualising a story.

Recommendations

Operationalise

Practical tips

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Machine learning

Data science and machine learning

Neural networks

Artificial intelligence

Foundational models

Practical tips

CASE STUDIES

Innovation Factory

Smart Container Co

Cognitive Business

Good With

SMARTabg

CONCLUSION

The power of data science

Ethical considerations in data science

The future of data science

A final thought

Glossary

References

Index

FIGURES AND TABLES

Stakeholders in the research looking at the link between smoking and cancer

The stakeholder wheel

Dynamic pricing PESTLE

A generic power interest grid

Power interest grid for phone project

Communication plan for phone project

Data analysis lifecycle

Phone fault data analysis lifecycle showing iterations

Generic data science project roles

NOIR data types

Positively skewed data

Negatively skewed data

The 3Vs of Big Data

Types of data (availability and creation)

ETL and ELT

CRUD, the four ages of data

The three lenses of data quality

The Cartesian coordinate plane

Visualising distance calculation

Example of standard deviation measures from the mean in a data set

Chart of hours gaming trend

Plot chart of battery capacity

Linear regression of battery capacity

Decision tree for budget and brand

k-means clustering example

Tooltips: offer additional information when hovering over data points

Filters: allow users to customise views and focus on specific data subsets

Example monthly sales bar chart

Example histogram

Example line graph

Example area chart

Example scatter plot

Example box and whisker plot

Example heat map

Example stem-and-leaf plot

Example scatter plot and line graph combined

The DIKW model

The user experience honeycomb

Data storytelling pyramid

The four pillars of data storytelling

Draft storyboard

The relationship between data science, machine learning and artificial intelligence

A neural network

Hierarchy of AI definitions

Examples of data classification

Data with fault type in one column

Data with fault types recorded in separate columns

Data on contacts in two tables

Data on contacts combined into one table

Fault types

Narrative comparison for contact centre performance investigation

AUTHORS

Stephen Mariadas is CEO of an institute of technology. He has a depth of understanding in technical skills and a wide range of experience in both industry and education. Having held the position of chief information officer for a European airline he moved into education 12 years ago. He has been instrumental in building digital and data apprenticeships across the South West region. Steve was awarded a Technical Teaching Fellowship by the Royal Commission for the Exhibition of 1851 on delivering data skills to industry. He also works closely with a number of multi-nationals on their data skills strategy.

Ian Huke qualified as a lawyer, specialising as a litigator before starting to develop legal software. Moving into financial services, he worked for both start-up companies and leading insurance and banking firms. Here he created and led several business analysis functions engaged in all aspects of change. Having become an independent consultant, he found his work was increasingly focused on data science, leading to a large dose of imposter syndrome. This book is inspired by his search for the knowledge to enable him to flourish in this domain.

FOREWORD

In my years of navigating the mathematically beautiful waters of artificial intelligence and more recently surfing the waves of societal and business interest in generative AI in particular, I've seen firsthand how the currents of skills that make up 'Core' data science practice then feed in to create the most powerful AI models and the most interesting applications.

In both the past and present, it's clear that a strong foundation in data science skills and knowledge isn't just beneficial – it's essential – it is the foundations on which so many emerging technologies are based, whether that's AI or other emerging technologies like quantum computing. *Data Science Foundations* provides that bedrock and it's the kind of text I wish we had had to hand in the early days of training data scientists in industry.

What strikes me most about this book is its emphasis on democratising data science and considering projects holistically. It goes beyond the algorithms and coding, addressing the critical importance of understanding stakeholders, ethical considerations and effective communication. In today's world, where AI systems can have profound societal impacts, this

broader perspective is non-negotiable. We can no longer afford to train data skills in a vacuum.

In today's rapidly evolving world, data science stands as a beacon of innovation and insight. This book is designed to introduce you to the fascinating realm of data science, guiding you through the essential techniques and models that define the field. More importantly, it aims to show you how to harness the power of data science to achieve success within your organisation. This book is unique in the way that it covers all the practical concepts of getting a data science project up and running, without the need for deep technical knowledge.

The book's structure mirrors the data analysis lifecycle, a framework that while seemingly straightforward, is often oversimplified in educational settings. It correctly highlights the iterative nature of this process, reminding us that real-world data science is rarely a linear progression. This emphasis on iteration and adaptability is crucial for anyone working with data, where experimentation and refinement are the name of the game.

While data science is inherently technical, it is ultimately driven by people. The success of any data science project hinges on the skills and creativity of the individuals involved. From

data engineers to machine learning experts, each role is vital in the journey from raw data to actionable insights.

As data science continues to grow in influence, addressing its ethical implications is critical. This book highlights the importance of fairness, privacy, transparency and accountability in data science practices. Looking ahead, advancements in AI, big data and cloud computing promise to further revolutionise the field, making data science more accessible and impactful.

I hope this book inspires you to explore the vast opportunities within data science and empowers you to make a real difference in the world.

Dr Vicky Crockett

AI Portfolio Director, QA Ltd

ACKNOWLEDGEMENTS

A thank you to all the apprentices we have had the pleasure of meeting, for all we have learnt from you. Also, to everyone who contributed to the case studies, we appreciate your time and for allowing us to share your journeys.

ABBREVIATIONS

artificial intelligence

autoregressive integrated moving average

customer relationship management

create, record, update, delete

density-based spatial clustering of applications with noise

development operations

data, information, knowledge, wisdom

data protection impact assessment

exploratory data analysis

extract, load, transform

enterprise resource planning

extract, transform, load

General Data Protection Regulation

Information Commissioner's Office

Internet Protocol

interquartile range

information technology

key performance indicator

local interpretable model-agnostic explanations

mean absolute error

mean absolute percentage error

machine learning

mean squared error

nominal, ordinal, interval, ratio

net promoter score

Office of National Statistics

political, economic, socio-cultural, technological, legal,
environmental

root mean squared error

SHapley Additive exPlanations

user experience

INTRODUCTION

Brad Pitt might not be the first person you think of when talking about data science. However, he both produced and starred in the hit movie *Moneyball* (Miller, 2011). This told the tale of a minor baseball team and how it used data science to compete with the major teams with significantly larger budgets. It's a great underdog film, but what is really interesting is how the data science was adopted and changed baseball, in fact changed all sports. If movies aren't your thing, the film is based on a book, which tells the true story (Lewis, 2004).

You can see the impact of data science not only in sport but also in the world around you. Want to stream some TV or music? You are bound to have recommendations made to you, but have you ever thought that the TV programmes themselves are only made because the 'data' says they will be popular? All courtesy of data science. Of course they are the visible manifestation of its use. It has applications across society. Look closely at any sector and you'll find data science at work.

Data science has, and continues to have, a huge impact on the society we live in. It is hard to separate debate around artificial intelligence (AI) from the movie *Terminator* (Cameron, 1984), where the robots take over and wage war on humankind. Of

more immediate concern might be the effect of AI on the nature of work. We are not here to talk about AI in great detail, but we are keen to explain the foundations to the data science that allows AI to have the capability that it does. Although largely outside the scope of this book, AI might be seen by some as the current pinnacle of data science. We cover their relationship in [Chapter 14](#).

We have written this book as an introduction to data science – to help you unlock its potential and be part of this force that is fundamental to shaping the society in which we live. Before we explore this, let's first consider what we mean by data science.

There is debate as to where the term data science originates, and no consensus on its exact meaning. However, the definitions tend to focus on two aspects: the disciplines involved in undertaking the analysis; and the type of analysis undertaken.

Advancements in digital technology have been a key driver behind its development, enabling the more effective use of mathematical and statistical methods in understanding and interacting with the world around us. However, for this to have value, it requires domain knowledge to identify the problems and opportunities to which it is applied. As you will discover,

data science is a multidisciplinary activity that will require you to develop skills and knowledge in all the above.

Data science requires data analysis to be undertaken. Data analysis is often broken down into three components:

- **Descriptive analytics:** where data is analysed to explain what has happened.
- **Predictive analytics:** where data is analysed to predict what might happen.
- **Prescriptive analytics:** where data is used to determine a course of action.

Sticking with sport, let's take football (also known as soccer) – by many measures the biggest sport in the world. Descriptive analytics would be the number of goals scored by a player. You might be curious and want to know how many goals a player might be expected to score in a season. In doing this, you would be entering the realm of predictive analytics. Taking this a stage further, you could look to use the data to pick a team based on, among other things, the number of goals scored – this would be prescriptive analytics.

These three categories provide a useful model when we talk about data analytics. You can use them to help communicate

what you can achieve with data. However, it is important that we don't forget that the world is complex and, although data science can be seen as the predictive and prescriptive analytics, it cuts across them all and some data analytics doesn't easily fit within one of these descriptions.

For this book, we have defined data science as using analytical methods, in an endeavour to gain insight from diverse data. This definition might be useful for writing a book, but we admit isn't very exciting and engaging. If asked:

'What is data science?'

you might like to answer

'The art of finding patterns in data.'

Why art? Data science involves creative thinking, about the challenges to be addressed, the sources, preparation and visualisation of the data and, of course, how it is communicated to stakeholders.

What are patterns? Earlier we mentioned streaming services. If you looked at the number of people streaming each month over a period, you might see that the numbers increase by a certain amount each month. This regular increase is a pattern and

might be used to predict the number of people streaming in the future. In this case, the pattern is an upwards trend. Another pattern could be in the types of films you like to watch or music you like to listen to. By looking at patterns in the characteristics of these, held in the data, films and music with similar characteristics can be found and recommended to you. So, we can say a pattern is an arrangement or sequence of data normally found through examining a model.

Data science can be compared to an iceberg, which is mostly hidden in the ocean. Most people only see a very small part of what is going on. For every hypothesis reported on and visualisation shared, many, potentially complex, steps have been taken. We will explore these steps in detail.

We have already noted the societal impact of data science. This raises questions as to when it is appropriate to use data science and how the insight generated is used. As early as the 1950s the links between smoking and cancer were known. However, this wasn't something that the tobacco industry welcomed, leading to the use of public relations, lawyers and untruths to refute this. We will look at the regulatory and ethical issues related to data science.

Data science is often applied with a very specific audience in mind. How do you communicate findings in a way that is both accurate and meaningful to them? Your intended audience might not be the only people with an interest in or to be affected by the work. Therefore, we will also consider stakeholders. Take a moment to consider the smoking example: who do you consider to be stakeholders? We will come back to this in [Chapter 2](#).

The software tools for data science are wide ranging. There are numerous specialist proprietary systems that can be used. Tools are also prevalent in spreadsheets and readily available as open-source software. Given the number of tools available and their many differences, we will not be looking at any specific software. We will, however, examine the different methods that you can apply, using the tools of your choice, to answer questions and gain insight. The value of your work will depend upon the selection of the right methods to address the problem or opportunity presented, and the quality and properties of the data they are applied to.

Data science in practice can take many forms, from major pharmaceutical research lasting years to a simple forecast of sales using Excel. For the purposes of this book, we will use the

catch-all term ‘project’ to describe data science initiatives and activities in all their forms.

As we previously discussed, data science is an amalgamation of different disciplines, which in turn have their own branches and specialisms. This makes it a large and amorphous topic, which is rapidly forging new ground. In this book we have taken the essential core knowledge and set it out to form the foundations of data science. However, we don’t want this to be a dusty academic tome. So, to aid you in applying this knowledge to the real world, we have included case studies and practical tips.

PRACTICAL TIPS

- You will always be learning; data science is a big topic.
- To be a data scientist, you need to embrace a range of skills and expertise.
- If in doubt about what someone means, ask them to clarify:
- People have different understandings of data science.
- Terminology can have different meanings in different contexts.

STAKEHOLDERS

When we do something, other people normally have a stake in it. The stake they have can vary. It might be that they contribute to it, benefit from it, are affected by it or have some other interest in it. These people are stakeholders. This is true of data science projects, which can have a wide range of stakeholders, as we will see in this chapter.

It is fair to say that not all stakeholders are equal; not only are the reasons they have a stake different, but they also have different levels of interest in the project and different levels of ability to influence it. Let us imagine we work for a company that wants to understand whether the phones issued to employees are likely to develop a fault in the next six months. The person responsible for phones is going to be very interested in the outcome. They are likely to benefit from the project by being able to make better informed decisions in their role. They are also likely to have a lot of influence over the project as they have responsibility for the area that is being investigated. Conversely, if you have a phone provided by the company, you could be affected by the outcome of the investigation – your phone's reliability might be improved. However, it is unlikely that you, as one phone user, will have influence over the project and it is likely that other things will be higher on your priority

list than this project. These different levels of interest and influence – or power, as influence is often termed – determine how we engage and communicate with the various stakeholders.

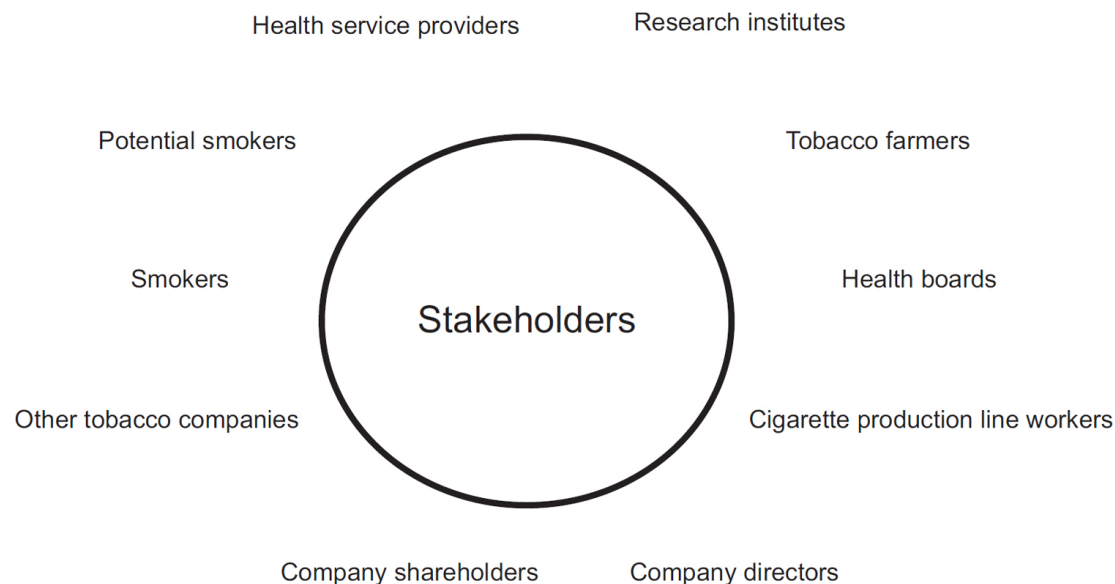
Data science projects can have a far-reaching impact, as we have seen in [Chapter 1](#), when looking at smoking and cancer. This means that the stakeholders can be quite removed from the immediate project. We saw above that employees were likely to have little interest or power over the investigation of phone reliability, but what about former employees, people who have left the company, but whose phone usage data is used in the analysis? They are further removed but still a stakeholder.

As we will see in [Chapter 4](#), when looking at ethics and lawfulness, understanding the wider group of stakeholders is essential for good data science. This is particularly important when people's data is being used, and they should be seen a key stakeholder in any project regardless of their power over and interest in it.

IDENTIFYING STAKEHOLDERS

In [Chapter 1](#) we suggested you think about the stakeholders in the analysis of the link between smoking and cancer. Who did you come up with? One simple view of the stakeholders is shown in [Figure 2.1](#). We have made it generic and from the standpoint of one tobacco company.

Figure 2.1 Stakeholders in the research looking at the link between smoking and cancer



The question you might be asking is why each of the groups shown is a stakeholder in the analysis. The simple answer is that they all have a stake in undertaking the analysis and/or its outcome. The stakes in our example are:

- **Research institutes:** These are the institutes carrying out the analysis for the tobacco company. They are doing the work.

- **Tobacco farmers:** If the analysis shows (as it did) that smoking is harmful it could impact their businesses.
- **Health boards:** The outcome of the analysis is likely to influence public policy and health care provision.
- **Cigarette production line workers:** Like the tobacco farmers, their livelihoods could be impacted by the analysis.
- **Company directors:** The analysis will influence decisions they make, and hence the future of the company.
- **Company shareholders:** The return they get on their investment in the company could be impacted by the research.
- **Other tobacco companies:** The findings of the analysis could impact their business in the same way as it impacts the tobacco company commissioning the analysis.
- **Smokers:** If the analysis shows that smoking is linked to cancer, they might change their smoking habits. They also need to provide their data to make the data science happen.
- **Potential smokers:** Like smokers, the findings might influence their view of smoking and ultimately whether they take it up or not. Their data is also needed for the research.
- **Health service providers:** These are the bodies that provide health care, which might be public or private. Showing a link with cancer could impact how they treat patients with smoking-related cancers and the health care advice they give.

The stakeholders go much wider than this. If we think about the health implications of smoking, then the funders of health care, whether insurance companies or taxpayers, will be interested. This in turn would create interest from people paying for services. In fact, the public's interest in the first instance is likely to be driven by wider health concerns rather than the financial implications. By this point, both politicians and governments will be showing an interest.

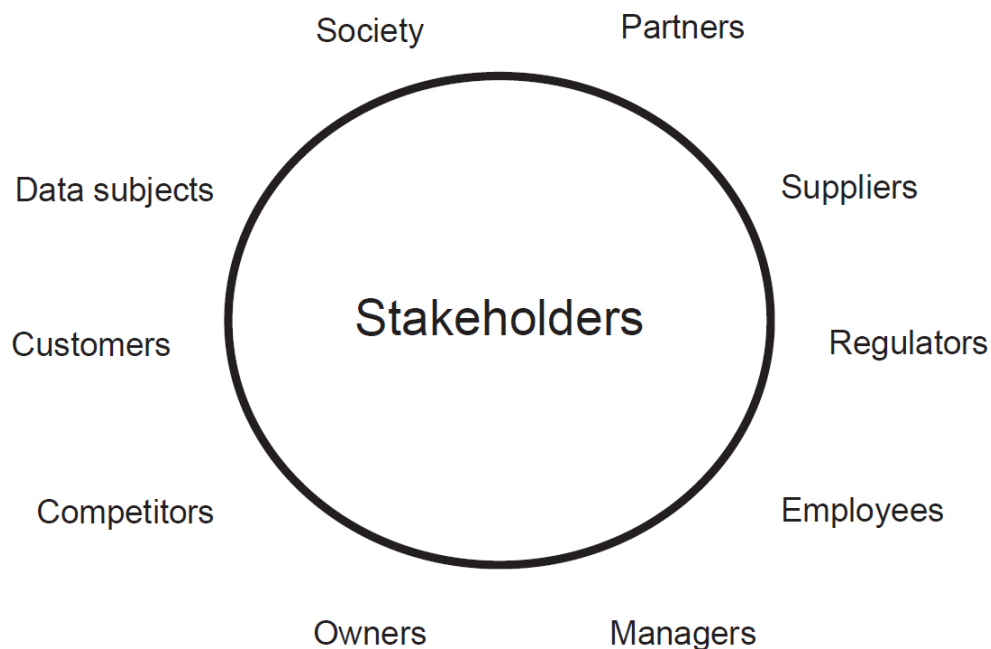
It isn't always obvious who is going to be interested in a data science project. It is unlikely those who developed the dynamic pricing for concert tickets would have expected this to draw comment from the United Kingdom's prime minister. Those comments came following the public reaction to prices charged for the Oasis 2025 tour (McArthur and Youngs, 2024).

From the above, we can see that a wide range of people have a stake in the analysis. Understanding who they are and why they have a stake in the analysis is important. This can affect how a problem is framed and how the outcome is communicated, including who to. More importantly, who the stakeholders are needs to be understood to identify any ethical issues – if they are not clear, ethical considerations might not be apparent.

The stakeholder wheel

The representation of the smoking and cancer stakeholders above uses the ‘stakeholder wheel’ ([Figure 2.2](#)). This is a useful framework for helping to identify the stakeholders in any piece of analysis. It sets out the different categories of stakeholder, and these can be considered to see if they apply to your project.

Figure 2.2 The stakeholder wheel



The categories are:

- **Partners:** These are organisations that provide specialist services to an organisation. These include organisations to whom data services are outsourced.
- **Suppliers:** These are organisations that provide the goods and services an organisation uses. These could be impacted

by analysis that is undertaken, for example changes in demand for their goods and services or in contractual terms because of a data science project.

- **Regulators:** Most organisations are subject to regulation that might be general or specific to the sector. Most organisations are subject to regulations relating to their use of data. These regulations are shaped, implemented and enforced by regulatory authorities.
- **Employees:** These are the people who work within an organisation and, like suppliers, could be impacted by the analysis undertaken.
- **Managers:** This group are the people commissioning and acting on the data analysis.
- **Owners:** There are several different ways an organisation can be owned. For a commercial organisation, it could be direct ownership by individuals or through shares. This group tends to provide oversight and can set strategy. Not-for-profit organisations tend to be 'owned' by trustees. Data science can affect their investment decisions and how they shape the organisation's future.
- **Competitors:** Most organisations have competitors, whether it be for customers or funding. The industry they are in could be impacted by insight gained by another organisation (as

with tobacco), or how that organisation acts in response to the insight gained (such as launching a new product).

- **Customers:** This group might change their behaviours because of data science. This could be through consuming more product (e.g. streaming another movie based on a recommendation) or stop using a product (as many people have done with tobacco).
- **Data subjects:** These are the people whose data is used by a project. For the smoking and cancer analysis, it would be the smokers and non-smokers who were subject to the research. It is important to identify this group of stakeholders as you need to understand them to ensure their data is used ethically and lawfully.
- **Society:** This is a bit of a catch-all. Is the project likely to have a wider impact on society? If so, who might be interested? In considering this, you might want to focus in on specific segments of society. For example, if looking at the accident and emergency waiting times for a particular hospital, there might be little national interest but lots of local interest.

The stakeholder wheel is a framework that can be adapted to the projects you undertake.

What changes might we make if we were working on data science projects for a recruitment company? A common

stakeholder is likely to be candidates for the roles being recruited. To state the obvious, just because they don't easily fit into any categories on the stakeholder wheel, it doesn't mean they shouldn't be considered.

INVESTIGATING STAKEHOLDERS

It is all very well us talking about the importance of stakeholders and the need to engage with them, but how do we practically work out who they are? In some instances, it might be obvious, we might have undertaken a very similar project before and the person commissioning the work or the project manager should be able to give some pointers. However, the stakeholder landscape does change, so it is beneficial to keep them under review both at the start of a project, however familiar, and as a project progresses.

Identifying stakeholders requires some research. This research takes two forms: talking to people and checking out documents. The two go hand in hand – talking to people can reveal sources to check out, and looking at sources can reveal people to talk to.

A good starting point for identifying internal stakeholders is an organisation chart and the organisation's intranet. This can show departments and people who might be interested. This

then provides a basis for talking to colleagues to gain any insight they have. Your initial research should help to engage them in the conversation, as you will be able to question and explore what they have to say and they won't feel that you are just leaning on them for an easy life.

At this point, you should have a list of potential stakeholders. It is then a matter of reaching out to them. As you reach out to people, they can often advise you of other people who will have an interest in the project, resulting in an expanding network. This can be prevalent in a larger organisation with complex structures and reporting lines. It is also worth remembering that people who don't consider themselves stakeholders early on in a project might become interested later. Stakeholders can be an ever-moving feast.

Conversations and research can not only identify the internal stakeholders, but also external ones. People in an organisation normally understand who their external stakeholders are. If there are questions in relation to this, a PESTLE analysis can help. This looks at the external factors relating to a project by considering the following elements:

- **Political:** Including policies implemented and those advocated by political parties and interest groups.

- **Economic:** Covering both the current position and trends and expectations, for example in levels of wealth and availability of finance.
- **Socio-cultural:** Reflecting changes in the population and their interests.
- **Technological:** The availability of and expected advances in technology.
- **Legal:** Laws and regulation, including expected changes.
- **Environmental:** Current themes and interest in the natural world.

Figure 2.3 briefly looks at dynamic concert ticket pricing in this context. When looking at these elements, you can see how dynamic ticket pricing could attract a lot of interest from a range of stakeholders. Hindsight is a wonderful thing.

Figure 2.3 Dynamic pricing PESTLE

Political	A new government with a focus on giving people a better deal.
Economic	People suffering from a cost of living squeeze.
Socio-cultural	Loyal fan bases and interest in stadium acts.
Technological	Ticket sales online with advancement in connectivity speeds and processing power.
Legal	Data protection considerations in use of people's data. Competition authorities interested in market dominance of ticket agencies.
Environmental	Growing desire to reduce environmental impacts.

MANAGING STAKEHOLDERS

As we have discussed, a data science project can have a wide group of disparate stakeholders. They will have different stakes in the project, which means that we will need to deal with them

differently. This poses the problem of how to work out how to manage each stakeholder effectively to achieve good project outcomes. Fortunately, we have some frameworks that can help us in this. The first is the power interest grid, which in turn can help us to develop a communication plan.

Power and interest

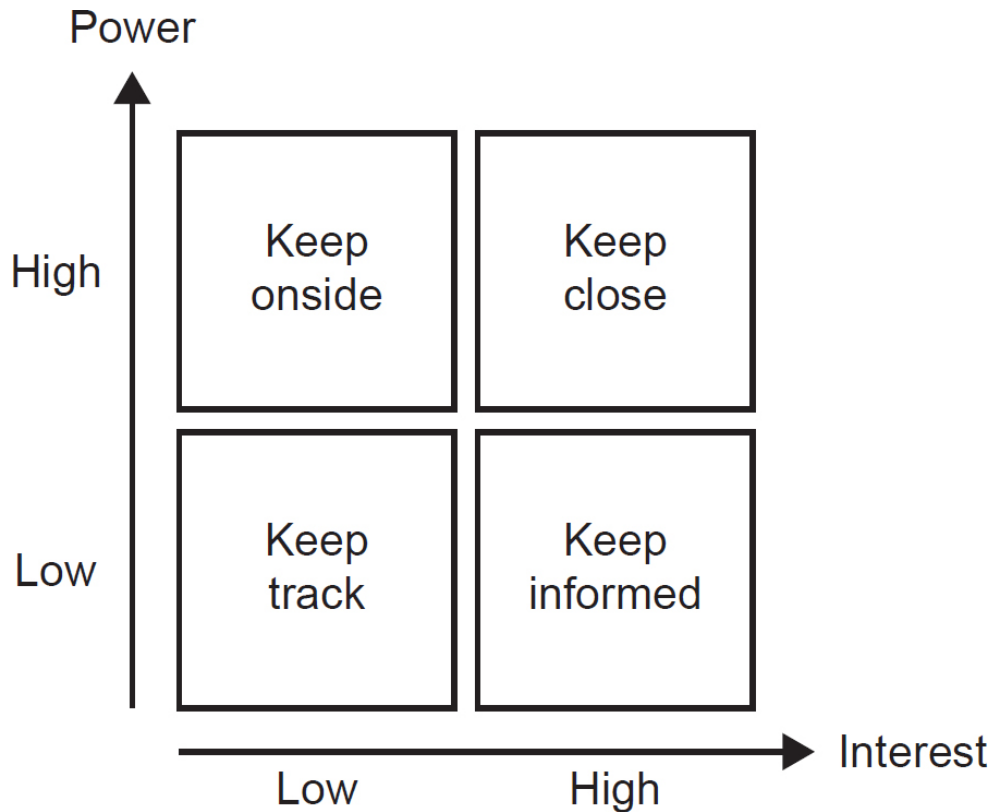
We have already emphasised the different degrees of power over the project and interest in it that stakeholders have. In our phone project, the phone manager, who we will refer to as the sponsor of the project, will have lots of power and a high level of interest. Compare this with employees who will have little interest and little power (at least initially). In terms of managing these stakeholders, it would be sensible to put more effort into the sponsor than the employees. However, life isn't simple, and you have stakeholders, such as regulators, who have substantial powers but little interest in specific projects. The regulator could impose large fines if employee data was misused in the phone analysis, but it is unlikely to be concerned with the project unless things go wrong. As for employees, if the project results in changes to the phone service provided (e.g. the adoption of a more reliable but a less popular phone), they might suddenly have a lot of interest although still very little power.

To help us navigate this, we can consider different strategies depending on the level of power and interest a stakeholder has. Potential strategies include:

- **Keep close:** This is where you keep your stakeholder closely informed of what is happening so that their views can be understood and considered. This strategy tends to be applied with high power, high influence stakeholders.
- **Keep onside:** In this instance, enough is done to ensure that the stakeholder is satisfied that everything is in order and they don't need to get involved. This is often used for stakeholders who have a lot of power but little interest.
- **Keep informed:** This strategy is employed where stakeholders have lots of interest but little power. In this instance, it is about providing sufficient information and engagement to satisfy their interest.
- **Keep track:** If a stakeholder hasn't any real power or interest, it is sensible to keep track of them to see if this changes. If it does, one of the other strategies can be employed.

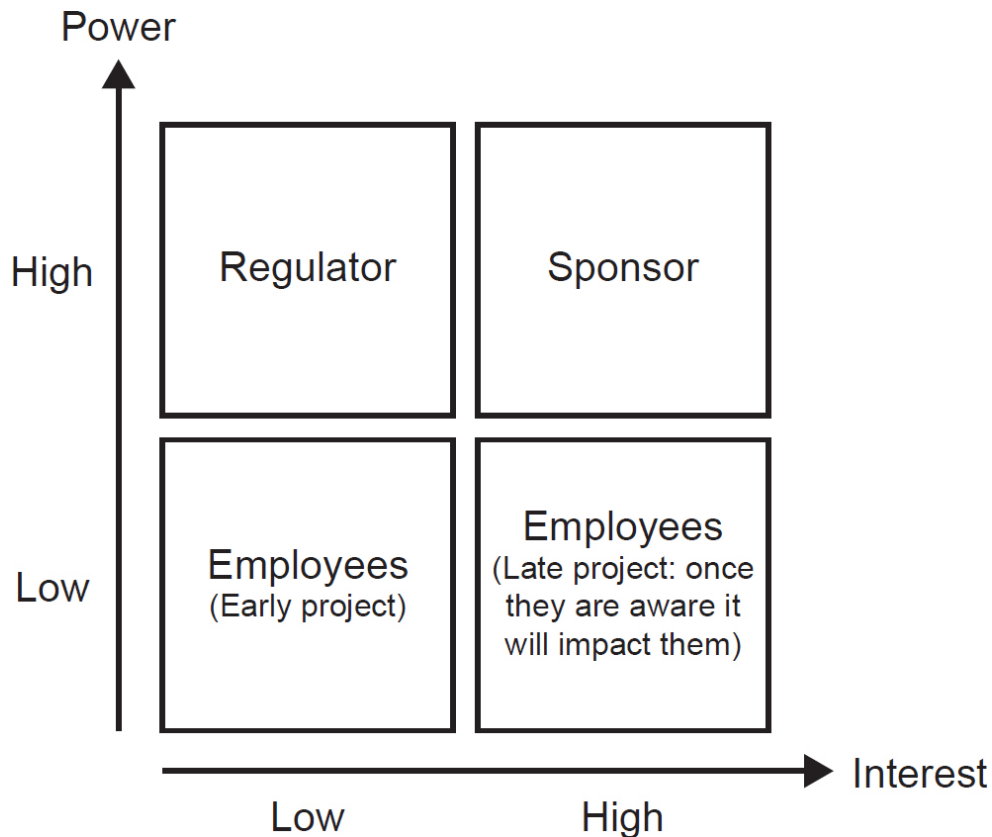
This can be visualised in a power interest grid (Newcombe, 2003), illustrated in [Figure 2.4](#). You will find various versions of this grid, some which are more complex, containing a wider range of management strategies.

Figure 2.4 A generic power interest grid



Our phone project stakeholders are shown in [Figure 2.5](#). It is a simple example, and you could have multiple stakeholders in each of the boxes for clarification. Why have we placed the stakeholders in the particular boxes?

Figure 2.5 Power interest grid for phone project



- **Sponsor:** It is their project, and they are accountable for the scope, resources and delivery. Therefore, they have high power and high interest.
- **Regulator:** The regulator has limited, if any interest, in the project, but could intervene if they wish – giving them high power, but low interest.
- **Employees (late project):** At this point, the employees understand the impact on them (in this case a move to a less popular phone). They have a high interest because of this, but little power.

- **Employees (early project):** The impact on employees isn't known, so they have little interest and little power.

Communication plan

The strategies identified using the power interest grid can be used to create a communication plan. This simply sets out how the stakeholders will be communicated with. This could be through one-to-one sessions, meetings or written updates. These could be set up specifically for the project or be existing communication channels.

An example of the communication plan for the phone analysis is shown in [Figure 2.6](#). The plan shows how each stakeholder group will be communicated with. Having a communication plan mitigates the risk that different stakeholders aren't appropriately engaged in the project. As with stakeholders, the communication plan should be kept under review throughout the project – it is a living document. We consider communication of the analysis in [Chapter 13](#).

It is worth noting in this example that the communication plan focuses on the compliance team, who represent the regulator in this project. They understand the regulations and, if needed, would represent the project to the regulator.

Figure 2.6 Communication plan for phone project

Sponsor	A weekly meeting will take place with the sponsor where they will be given an update on progress, and any key issues will be raised with them.
Regulator	The compliance team will be updated at the monthly compliance board meeting.
Employees (Early project)	HR will be asked to inform the project of any questions raised in relation to phones.
Employees (Late project: once they are aware it will impact them)	Updates will be provided weekly on the intranet and team leaders will communicate the proposed changes in team meetings.

One final point on stakeholders: they will have different levels of understanding of the problem and data science. It is important to understand this and adapt your language and explanations to reflect this.

PRACTICAL TIPS

- Consider who your stakeholders are at the start of the project.
- Stakeholders can be inside and outside your organisation.
- Keep your stakeholders under review throughout the project.
- Consider the power and interest of stakeholders to inform how you engage with them.
- Create a communication plan (even if it is just a regular catch-up with your manager).
- Think through any risks related to wider stakeholder groups.

PROJECT DELIVERY

Projects that employ data science come in many different forms. Some, such as pharmaceutical research, might take many years, others can be delivered in an afternoon. However, they all have some common activities.

If you are in a small business and looking to predict profitability for the next year, you need to: work out what you are doing; find and prepare your data; determine the methods to be used to analyse it; build your model; evaluate the outcome; and then share your findings. The same activities are required if you are working for a pharmaceutical company on medical trials. However, in the first instance you might do everything in a day, in the latter case it is likely to take a team of people years.

Therefore, you need to consider how the activities are going to be planned and managed. This includes understanding the problem space to be explored along with the resources and time needed to do it. If, having done this, the project looks feasible, you then need to bring those resources together, creating a team to deliver and manage the activities to fruition.

DATA ANALYSIS LIFECYCLE

People can get very excited about the outcome of data science, but frequently don't appreciate the effort that goes into that result. The data analysis lifecycle is a framework that looks at the steps needed to successfully deliver a data project. It can help in planning work and managing stakeholder expectations.

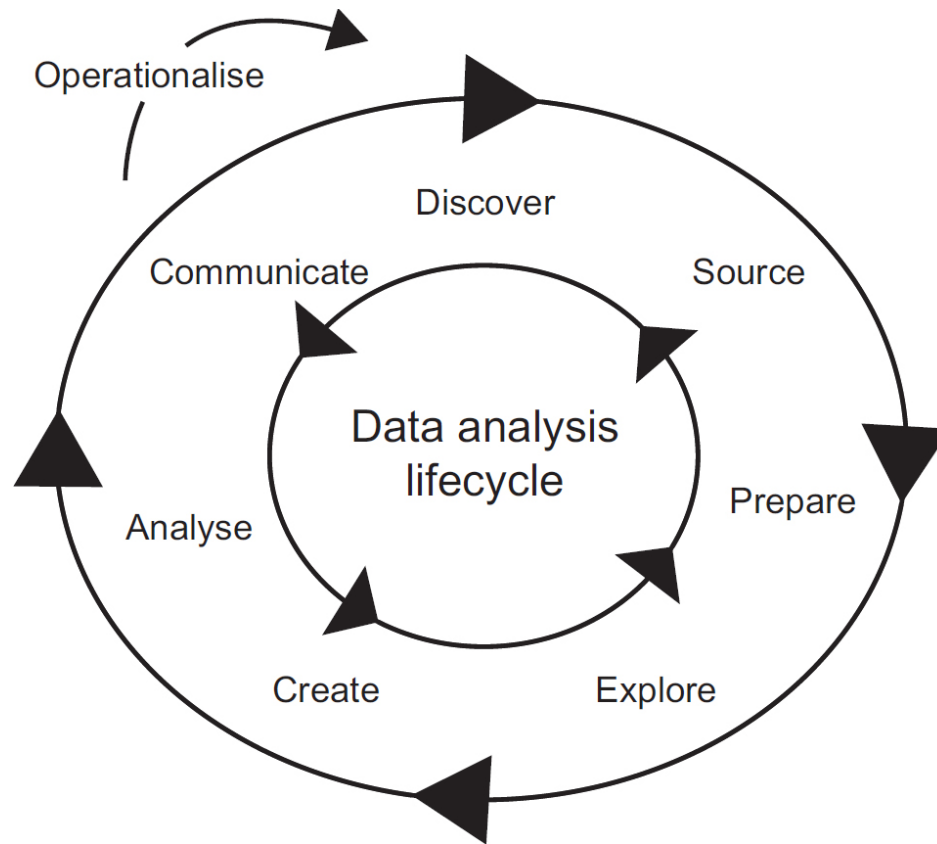
There are many versions of the lifecycle, and, like all frameworks, it can be adapted for your needs. The one we have used in [Figure 3.1](#) describes the steps in terms that easily relate to data science.

The outer ring shows the overall flow through the steps from discovery to communication, while the inner ring shows that it is iterative. A step may be returned to and revisited at any point and any number of times. The circular nature also reflects that in answering one question others will arise, hence the flow from communication back to discovery.

Each step contributes to the success of the data science initiative:

- **Discover** considers and defines the problem that the analysis will explore and the resources that will be needed to undertake it.

Figure 3.1 Data analysis lifecycle



- **Source** looks for and at the data that is available to answer the question.
- **Prepare** ensures the data is of a suitable quality and format to be used in the analysis.
- **Explore** investigates appropriate methods to answer the questions and the ability to use the available data to build effective models.
- **Create** builds the actual models to be used in the analysis.
- **Analyse** evaluates the models created.
- **Communicate** shares the insight gained from the analysis of the models.

- **Operationalise** embeds the analysis into the regular operation of an organisation. Many data science projects are a one-off investigation and therefore this step doesn't always take place.

Let us return to our phone fault example. What might the steps in the data analysis lifecycle look like?

- **Discover:** We might receive a request by email or via a portal. A discussion might take place with the sponsor to clarify what they need from the analysis. This then might be confirmed by email.
- **Source:** Having considered the data we need to answer this question, we might look at the data available to identify suitable data sets to be used in the analysis.
- **Prepare:** Once we have the data sets, we might not need all the data within them, so we might remove the data that isn't needed. We might also find that some records are inaccurate, leading us to correct or remove them. Parts of the data could also be in an inappropriate format, for example the age of the phone is held as a string not an integer (these are data types in the database). Often, preparation is the most time intensive activity in a data science project.
- **Explore:** Having thought through the tools and methods available, we select the most appropriate ones, taking

account of our data. This might involve some prototyping of different models.

- **Create:** We then build the model using the selected methods and data.
- **Analyse:** The model outputs are considered and interpreted to allow us to draw conclusions on which phones are likely to develop a fault in the next six months.
- **Communicate:** The results of the analysis can then be communicated by us to those asking the question.
- **Operationalise:** The exploration of phone reliability is a one-off project, so it does not need to be embedded into the operation.

This scenario makes it look very easy to move through the flow, but in practice there is often significant iteration. We can consider some of the problems we might encounter in looking at phone faults ([Figure 3.2](#)):

1. When trying to source the data, we can only find it for some phones. We therefore go back to those asking the question to confirm they want us to go ahead with a more limited scope.
2. On preparing the data, we find that, although phone faults are contained in our data set, the details of the phone are contained in another data set that needs to be sourced.

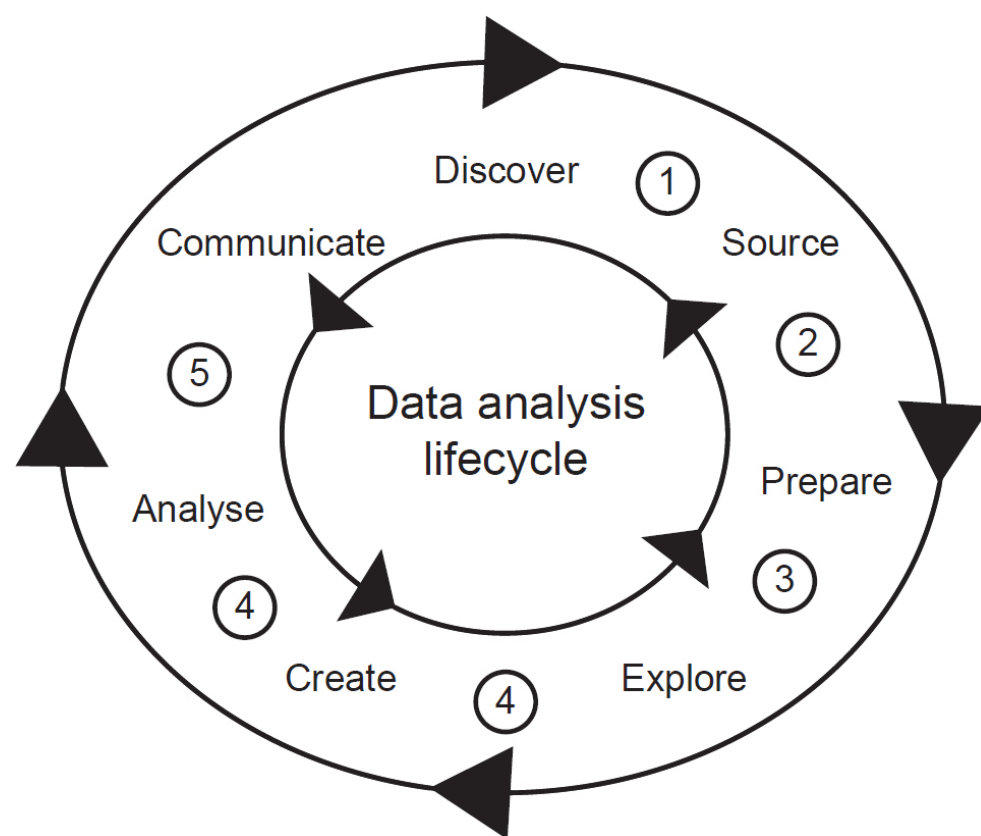
3. When exploring the methods, we find that the method needs the different makes of phone to be recorded as a separate data field, rather than all the makes being held in one data field, resulting in the need to restructure the data.
4. On analysing the results of the model, we find that it is weak and requires further exploration, perhaps leading us to add or remove data fields to strengthen the model. In this case, going back two steps in the framework.
5. On communicating the results, a further question is asked – whether a previous repair makes any difference – which, although in the model, isn't something we considered, meaning that we need to analyse the model outputs further to respond to the question.

Gates might be used to move from one step of the data analysis lifecycle to the next. This is where the project is reviewed to determine whether a step is sufficiently advanced to move on to the next (recognising the iterative nature of data science projects). This might be an informal discussion or, for larger more complex undertakings, a formal review to determine whether the gate should be passed through.

MANAGING THE LIFECYCLE

Those commissioning projects often want a level of certainty from their projects: what will be delivered, when and at what cost. These three factors interact; when one changes, the others change. Data science projects are no different, with the added complication that, although an answer to the question posed might be delivered, it could have a level of uncertainty or might not give the answer the commissioner wants.

Figure 3.2 Phone fault data analysis lifecycle showing iterations



To understand the three key aspects of a project – scope (the problem), the time and the cost – a feasibility exercise is often

undertaken before a project starts or at the start of the project. This looks at what will be needed to undertake the project and whether it is feasible. This is closely aligned with the discovery step in the data analysis lifecycle, and is covered in more detail in [Chapter 5](#).

Once a decision has been made to go ahead, the commissioner will want to track progress. Often this is delegated to a project manager. They will adopt an approach to managing the project's progress. This might be a linear approach, or an iterative and incremental one. In many instances, the approach will be a combination of the two.

It is also worth remembering that the data science might be part of a larger project or programme of work. In this case, the data science part of the project will fall within the governance of the overall programme of work. If we were feeling entrepreneurial, we might want to launch an app that builds a perfect CV for any role, based on a person's own unique career history. This will clearly involve data science, creating the capability to understand what a good CV looks like for any role. However, there is other development activity required, for example developing the user interface, not to mention the sales and marketing required to launch the app. This illustrates that data science doesn't take place in a vacuum.

Linear delivery

Linear project delivery is often referred to as Waterfall – the idea being, as you complete one activity, you cascade to the next. This requires a clear understanding of all three factors, scope, time and cost, to allow a clear delivery plan to be created at the start of the project. Changes can be made in all these factors through change control. When a change is desired, it goes through the change control process to ensure the impact of the change is clearly understood. The change can then be accepted, with the plan being amended to reflect it, or the change can be declined.

Iterative and incremental delivery

Iterative and incremental development involve a repeating process that incrementally addresses the problem. This often involves the use of sprints and backlogs. A backlog is a catalogue of work that needs to be done, and a sprint is a period to do the work. A project will have multiple sprints, with each sprint often being one or two weeks long. At the start of each sprint, it will have a number of the work items from the backlog assigned to it. If the sprint doesn't deliver a backlog item, it is placed back in the backlog. This gives a high level of flexibility, as the content of the backlog can change as the

project progresses and items in the backlog can be prioritised for each sprint. Agile adopts these methods.

Hybrid delivery

It is possible to take a hybrid approach. This might typically consist of a high-level linear plan with the detail being managed through sprints. In this situation, the data analysis lifecycle can provide the framework for the linear plan, showing the focus at any one time. The iterations required to address the problem are managed at the granular level through the backlog and sprints.

PRACTICAL TIPS

- Plan your activity and time – the data analysis lifecycle can help with this.
- Allow time to go back to previous steps – the data analysis lifecycle is iterative.
- Planning can help to manage stakeholder expectations.
- Consider how the project fits with other initiatives.

ETHICS AND LAWFULNESS

Data science opens a vast wealth of possibilities and opportunities. However, just because we can do something, doesn't mean we should. What would your response be if asked to develop a model to predict the health outcomes for people like you? This might have lots of positive benefits in improving health within the population, including you. That in turn might lead to economic gains, from which you benefit. It could also be good for the company you work for if, for example, it was an insurance company who determined premiums based on this information. Good for society, good for your company and potentially good for you.

But would it be good for everyone? Will individuals be affected? Potentially even you. It could cause you embarrassment if your medical details were shared. There is a risk of losing out financially if, for example, your insurance premiums rose as a result. Worse still, you might be denied access to services. This could limit both your quality of life and life expectancy. How do you balance the different interests?

This balance can be considered at several different levels. We might start by asking: is it legal to do this? Our ability to do so could be constrained by regulations or contractual obligations.

Then, do we have a professional duty? Perhaps one imposed by our membership of an organisation or group. Finally, is it morally right? A question of ethics.

WHEN TO CONSIDER?

Issues can arise with the ethical and lawful use of data at all stages of a project. Their consideration should be ongoing, with thought given as to whether issues might already have arisen or might arise later in the project.

As you will see, there are no clear rules on what is ethical, and lawfulness can be subject to interpretation. However, there are questions you can ask at each stage of the data analysis lifecycle to help identify potential ethical or legal issues.

Discovery

Has a data impact assessment taken place, or an ethical review?

A data protection impact assessment, often referred to as a DIPA, will consider data protection risks and set out how they will be mitigated. For research projects, ethical approval may be required, which often entails an ethical review and then approval by an ethics board. Both of these activities provide a reference point for the project's appropriate use of data.

Source

Is the quality of the data understood? Poor-quality data can lead to ethical issues, for example if it has bias in it.

Is the lineage of the data known? If you don't know the lineage of the data, how do you know that you have permission to use it?

Prepare, explore, create and analyse

Are you using the data as intended? Referencing a DIPA, ethical approval and the lineage can assist in this.

Are you doing anything to it that will impact the data quality?

Are you using data that you don't really need? In this case, should you be using it?

Is your analysis objective?

If using personal information, do you need to?

Communicate

Is the analysis being used for the intended purpose?

Are the outcomes of the analysis objectively presented and interpreted?

Can you explain the model to your stakeholders?

Operationalise

Has the data or its use changed on being operationalised?

How will the model be kept under review?

Let us consider a project to help recruitment, looking to identify good candidates who would be good at a job. Potential data sources could well be biased. If we used that data, could it prejudice or harm future applicants for the job? If so, it would seem unethical to use the data. We could also ask ourselves whether people giving data to apply for a job would be happy to have their data used to help predict whether future applicants would be successful in that role. They might have signed a consent clause, but is this really consent to the use of the data in this way? This might mean the data's use in this way isn't lawful. In fact, both questions might make us consider the use of the data both unethical and unlawful.

Looking forward, what are the implications of undertaking the analysis? We might predict health outcomes to improve the

health of the general population. However, as we have already alluded to, it might also be used to target specific individuals and treat them differently. Again, they could be prejudiced by this, raising the question of whether this is ethical and lawful.

WHO MIGHT BE IMPACTED?

The stakeholder wheel we discussed in [Chapter 2](#) is a useful tool for identifying stakeholders in a project, including those who provide data and benefit from the analysis. Considering lawfulness and ethics in relation to each stakeholder can help to identify any risks that need to be addressed.

ETHICS

You may recall the stakeholders we looked at in the analysis of the link between smoking and cancer. One of those stakeholders was the tobacco companies. How did they react to the research findings? Ultimately, they lied about them – a very clear example of unethical behaviour related to data science.

However, the ethics of a situation are not always so clear-cut. Let's consider another example. In 2020, exams for A-levels were cancelled due to the Covid pandemic (BBC, 2020). The question, then, was how to award grades to those who were due to sit them. Data science to the rescue: look at the historic

performance of schools and then apply a similar profile of grades to the current cohorts in any school – a rigorous scientific approach compared to the alternative of relying on teachers' assessment.

This might have been fair for a lot of students, but what about those whose grades were downgraded from their teacher's prediction? Add to this that certain schools were too small to be included in the model, so their pupils' grades weren't downgraded. An ethical question was whether to use the model, given the perceived unfairness to some individuals. The result was a backtrack and use of teacher assessment, an approach that could also be critiqued.

What are ethics?

Ethics are frequently discussed, but often not defined. What are they? This is something that was debated in ancient civilisations and is still debated now. For our purposes, we will define them as the principles by which we and society judge our and others' behaviours. It is important to recognise that ethics change over time; some argue that ethical rules should never be broken, others argue that they are flexible and depend on context.

Although there is no legal sanction for behaving unethically if that behaviour does not break the law, it is important as it is fundamental to trust and in turn reputation. If you are not concerned about reputation, it's worth remembering the £37 billion loss of value in Facebook when questions were raised as to how data was collected and used by Cambridge Analytica in election campaigns (BBC, 2018).

Ethical questions

The subjectiveness of ethics means we can't set out a clear set of ethical rules to follow. To help navigate this tricky subject, we suggest asking the following two questions:

Will this harm anyone?

Are people happy for their data to be used in this way?

Will it harm?

Harm can occur in many forms. It could be direct harm, or it could be indirect. Harm doesn't have to be physical; it could be harmful to other aspects of our lives, including educational prospects, financial well-being and social relationships. When thinking about harm, we need to consider the risk of harm and

that others would regard as harmful what we might consider to be benign.

What ethical considerations might there be if we wanted to explore minority groups' experiences of services to support them? This should be beneficial in understanding how services could be improved. However, it could also have adverse consequences for those involved in the research. It might lead to the targeting of those contributing to and undertaking the research by those opposed to those minority groups. This doesn't mean the research shouldn't take place, but steps should be taken to mitigate the risk of harm. Many organisations will have processes in place to review the ethics of research projects and, where they are not in place, individuals and teams should undertake appropriate reviews. Here, we have an example of an ethical consideration in relation to the collection of data.

Are people happy?

Elon Musk raised concern at his private jet's movements being shared on social media, as it would make it easier for someone to attack his family (Halpert and Moloney, 2022). Here, he was concerned about harm, but there is also a question of someone's right to control the use of data that relates to them.

Do you want your data used to identify goods and services you might be interested in? Are you happy for it to be used in medical research? How about using it to promote political adverts to influence how you vote? Here, we are considering the ethical question, not the legal one. The fact that a box has been ticked giving consent doesn't mean someone would be happy for their data to be used that way. The use to which the data is to be put might not have been envisaged by them at the time they ticked the box. They might also have had little choice but to consent to access goods, services or employment. As we will see, ethics and lawfulness overlap.

Having read this section on ethics, you might think it has provided no answers and just raised questions. You would be right, and that is the point of it – to think before launching into a data science project and during the project. Think about the data used, think about the people involved and think about what might be done with the insight generated.

LEGISLATION AND REGULATION

Ethics is subjective; helpfully, legislation and regulation are a little more objective. These are limits on the use of data that society has set as a standard, that have legal consequences if not complied with. Different countries take different

approaches to setting these standards. Some will have overarching data protection laws, such as the UK, others regulate by specific sector, for example the USA. We will focus on the UK's overarching legislation: the Data Protection Act 2018.

Data Protection Act

The Data Protection Act 2018 gave effect to the European General Data Protection Regulation (GDPR). Rather than set out rules as to how data can be used, it has a set of principles that should be followed. The advantage of principles is that they can be applied to different situations. Rules tend to address specific scenarios, making them complex and inflexible. This does, however, mean the principles require interpretation.

GDPR principles

This is a brief introduction to the GDPR principles, and we would encourage you to explore this subject further to better understand the detail and nuances. You will find that these align with many of the ethical considerations we have discussed. You should remember that these principles apply to living persons and not to other entities, such as companies or those who are dead.

Lawful, fair and transparent

- **Lawful:** Data should only be processed where lawful under the act and where it doesn't breach other laws such as the Equality Act. Processing is only lawful under the act where there is genuine consent or a contractual need, there is a legal obligation or public interest, or where there is a legitimate interest in processing it and this is not overridden by the individual's interest. Legitimate interest provides a wide basis for the use of data.
- **Fair:** However, the use must also be fair, so the data should be used in line with the person's expectations and should not have an undue adverse effect on them.
- **Transparent:** The individual also needs to be informed of the purpose for which their data will be processed and the consequences of that processing.

Limited purpose

Data should be collected for specific purposes and not processed in a manner incompatible with those purposes. However, there are circumstances where the purpose can be expanded, including where the new purpose is compatible with the old purpose.

Data minimisation

Data should be limited to that necessary for the purpose for which it is being processed. If a data item is not needed to achieve the purpose, it should not be used.

Accuracy

Data should not be inaccurate or misleading.

Storage limitation

Data should only be kept for as long as is necessary for the purpose for which the data is being processed. However, this doesn't apply where the person can no longer be identified within the data.

Integrity and confidentiality

Data should be kept safely and securely. This means protecting it against accidents and malicious acts.

Accountability

Organisations need to be able to show compliance with the above principles. This accountability is to regulators, to the subjects of the data and to the courts.

Special category data

Special category data is a subset of personal data that is regarded as more sensitive. It includes data relating to:

- race and ethnicity;
- political opinions, religious or philosophical beliefs and trade union membership;
- genetics;
- biometrics;
- health;
- sex life and sexual orientation.

For special category data, specific conditions are set out for lawful processing that could be summarised as specific consent being given or a societal interest in the data being processed.

The ICO

The Information Commissioner's Office (ICO) can investigate and take enforcement action for breaches of GDPR. This isn't just against big companies; a school was reprimanded in relation to its implementation of facial recognition technology for cashless catering (ICO, n.d.).

INTERNAL POLICIES

It is good practice for organisations to have policies in place relating to the use of data. These policies allow the application of data protection principles, and in some cases ethical considerations, to be set out within the organisation. For example, the GDPR principles don't explicitly set out how long data can be retained; a policy can clearly set this out, taking account of the circumstances.

OTHER GOVERNANCE

You may find that there are other restrictions on the use of data that apply to the specific sector in which you work. This might be put in place by a regulatory or professional body. An example of this is the Caldicott Principles that apply to health and social care. BCS, The Chartered Institute for IT also has a professional code of conduct for members. This has four key principles:

- You make IT for everyone.
- Show what you know, learn what you don't.
- Respect the organisation or individual you work for.
- Keep IT real, keep IT professional, pass IT on.

Within the code, the first duty relates to public interest and includes having 'due regard for ... privacy, security and

wellbeing of others’.

COMMERCIAL CONSTRAINTS

So far, much of our consideration on limiting the use of data has focused on individuals and their personal data. However, restrictions on the use of data can be contained in contracts. For example, if a contract is agreed to outsource the provision of a contact centre, the rights to use the data generated in the operation of the contact centre are likely to be set out. This might provide that the data can be used to improve the performance of the contact centre, but not shared with other customers of the contact centre. Likewise, if third-party data is acquired for use in the contact centre, the licence to use that data might restrict the purpose and time for which it can be used, for example the right to use the details to email customers over a three-month period.

INTELLECTUAL PROPERTY

It is important to remember that intellectual property exist in some artefacts that can be represented as data, for example copyright in literary works, music and images. When working with such data, it is important to understand these rights and whether the data can be used in the way you intend.

RISK MITIGATION STRATEGIES

There are several strategies that can be adopted to reduce the risk of unethical or unlawful use of data.

Follow policy

Where your organisation has a policy in place, follow it. This means you need to know where the policies are, what is in them and who to ask if something isn't clear. Just keep in mind that the policy might not properly reflect the legal and regulatory standards to which you will be held.

Data classification

Where your organisation has a data classification policy, understand the classification that applies to the data you plan to use and apply the appropriate classification to the data sets you create. Data classification in this context means applying a label to the data, which then sets out how the data items should be used and managed. Example classifications are given in [Table 4.1](#).

These are examples, and your organisation might have different classification levels and controls related to them. It is also worth remembering that data classification is sometimes

used to refer to whether data is structured or unstructured, and whether it is qualitative or quantitative. There are also classification methods used to analyse data. To avoid confusion, check the context of its use.

Table 4.1 Examples of data classification

Data	Classification	Rationale
Statutory accounts	Open	These are a matter of public record, and anyone can access them.
Balanced business scorecard	Internal	This shows employees how the organisation is performing, so it is open to anyone in the organisation. However, the information isn't public and shouldn't be shared outside the organisation.

Forecast sales	Confidential	Only authorised people should have access to the information in the organisation, as it is commercially sensitive.
Diversity data	Secret	Due to the sensitivity of the data, access and use of the data has additional controls.

Obfuscation and redaction

This is where parts of the data are made unclear (obfuscation) or blanked (redaction). Take, for example, a sales forecast. You might produce a report showing the employee names for the manager, but fictionalise or blank them out when shared with the team. The manager might have a legitimate interest knowing how everyone is performing, whereas the team members are unlikely to. In doing this, you need to be careful that people cannot work out who everyone is through other information people have available.

Aggregation

Aggregation is used where there is a small data set, and it can be easy to work out who data relates to even where it is obfuscated or redacted. This might be based on information in the data set or other information that people have access to. This could include inferences made about people and individuals' interaction with them. In these cases, you can aggregate the data set so that individuals are not represented. Aggregating the data means that it is displayed for a group rather than by individual. In many instances, small groups will be excluded from reporting, to protect privacy.

Anonymising data

In this case, you remove all personal data. However, you need to be aware that some of the data you retain might allow someone to be identified. For example, if you leave an Internet Protocol (IP) address and an age in a record, it is likely these data items could be used to identify an address and a particular person living there. Data could be changed for an ID or 'token', which removes personal data, for example each employee could be given a unique reference number. However, it should be remembered that the individual could be identified by someone who has the data set containing the unique reference numbers and names. You might recall that data where people can't be identified falls outside of GDPR. However, even where

data has been anonymised, it is often still possible to identify someone, for example combining anonymised mobile phone data with geolocation data from, say, a tweet can identify the user of the phone (Palmer, 2013). Therefore, this GDPR exemption should be used with caution.

Access controls

If you want to ensure someone doesn't misuse data, then a good way to stop it is to prevent them gaining access to the data in the first place. Access controls can be applied to both where data is stored and where it is reported. Many organisations have policies that set out who can access data and how permissions to access it are granted.

PRACTICAL TIPS

- Ethics and lawfulness are different, although they often align.
- Ethics are not defined and can change with context and time.
- Data regulation is based on principles that need interpretation, not on rules.
- Other laws, not just data protection, govern the use of data.
- Make sure you understand your organisation's policies on the use of data.

- Consider the ethical and lawful use of data at the start of the project.
- Keep the ethical and lawful use of data under review throughout the project.

DISCOVERY

When an organisation kicks off a data science project, one person is normally made accountable for its delivery: the sponsor. They might be the person who came up with the idea or someone to whom the responsibility has been delegated. One of their roles is to initiate the project.

The first step in the data analysis lifecycle is discovery.

Discovery is when the project is investigated to understand its objectives, scope, resource needs and constraints. Constraints might be time, financial, technology and, of course, data. An important part of discovery is determining whether the project is feasible. This includes not only whether it is worth doing, but also whether it has the resources (including expertise) to deliver it. Sometimes this will be referred to as a feasibility study.

The scope and scale of discovery will vary. Before starting a project, it is worth spending time to consider what it will take to deliver, the challenges you might encounter, whose help you'll need and ultimately whether it is worth doing. It could save you a lot of angst and heartache. This is true both for a simple sales forecast and pharmaceutical research, although in the latter case discovery is likely to be a significantly larger undertaking.

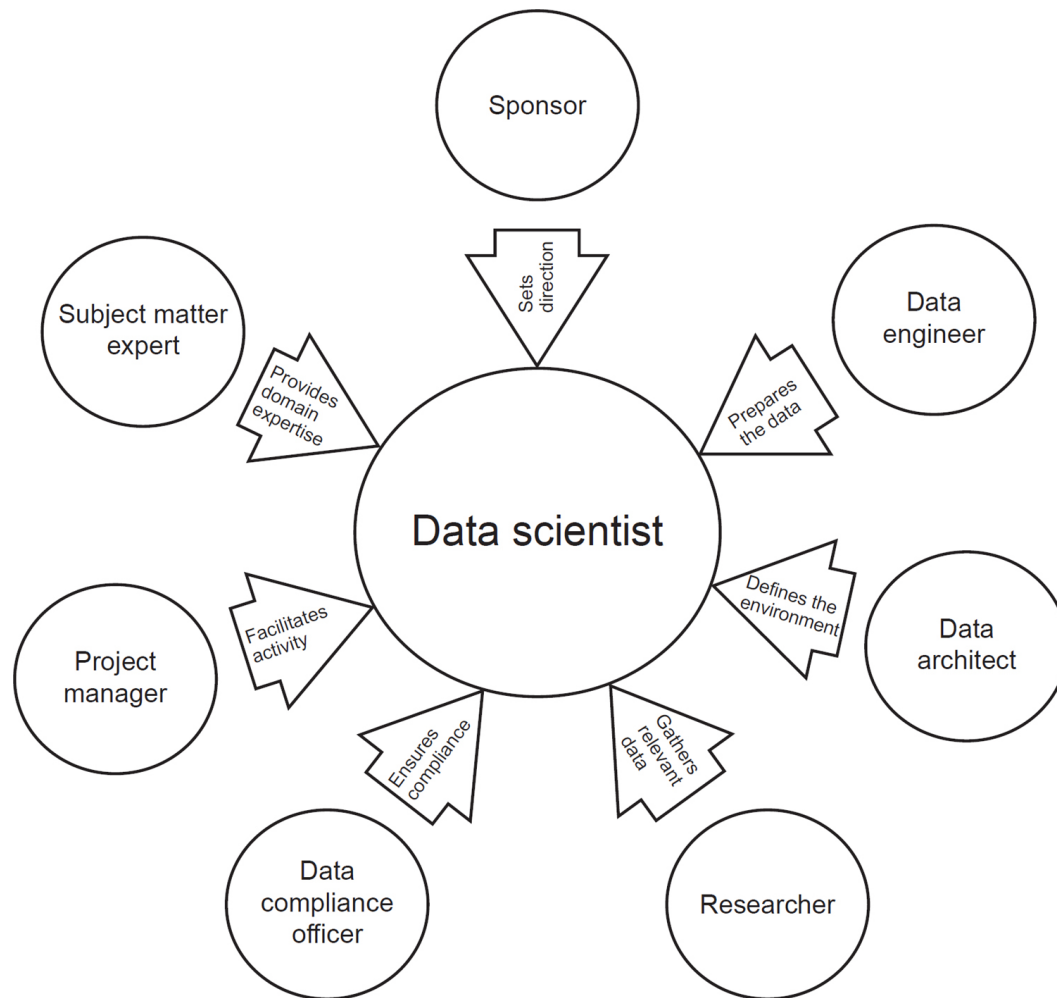
Discovery is the time to start looking at stakeholders. Important stakeholders in the project we haven't yet considered are the delivery team.

THE TEAM

To help identify who should be in the team, we think about roles involved in a data science project and the skills and expertise they bring. The descriptions we have used are generic, and different organisations might have their own role titles and definitions. In looking at this, we should remember that one person might play many, if not all, the roles. For larger projects, such as pharmaceutical trials, the roles might be broken down further with greater specialisation within them. When identifying the team, you might want to consider the data analysis lifecycle and who you will need help from in each step to make the project a success. Roles you might find on a project are illustrated in [Figure 5.1](#) and include:

- **Sponsor:** They set the direction of the project through defining its objectives and scope. This is likely to be a senior person, accountable for the project. In the setting of direction, the sponsor is likely to consult with others.

Figure 5.1 Generic data science project roles



- **Subject matter expert:** They provide the domain knowledge. They are key to the definition of the problem, understanding the data and analysing the results.
- **Researcher:** In some instances, work will need to be undertaken to gather the data needed for the project. This could be through studies, experiments or surveys. Often, the data has already been collected.
- **Data engineer:** They will transform the data and make it available in standard formats and to a certain quality. This is

often the extraction of raw data from its sources and then transforming and loading it into a warehouse for analysis. However, you will still need to prepare the data to ensure that it meets the needs of the project.

- **Data architect:** They define the environment (both technical and non-technical) in which the project takes place. For example, they might determine where the data can be stored to ensure it is secure. The creation of the environment might be undertaken by a technician.
- **Data compliance officer:** Compliance ensure that the data is used in line with regulations and the organisation's policies. This can be in the form of setting policies and operating procedures to ensure compliance.
- **Project manager:** They facilitate the activity required to deliver the project. Collaboration between the stakeholders is key to the success of most projects, and the project manager can play an important role creating an environment that encourages and supports this.
- **Data scientist:** They identify the data and methods required, build and analyse the models created and communicate the findings.

In putting the team together, it is worth considering who will provide a check and challenge on the models used. Having this

process reduces the risk of models being used inappropriately and invalid conclusions drawn.

Many organisations have the role of data analyst. A data analyst might be seen as focusing on the descriptive analytics (reporting what has happened), although many data analysts take the role of the data scientist in projects.

We should not forget that data science projects can take place as part of a wider programme of work and that other roles will be involved in delivering that programme. For example, change management and IT/development operations (DevOps) might be involved in operationalising the findings of the data science project.

UNDERSTANDING THE DOMAIN

It is said that nothing exists in a vacuum, and that is true of data science. The problem being looked at has a real-world context, so it is important to understand that context to properly understand the problem – this is often referred to as the domain context of the problem. The domain context helps to determine the scope and objective of a project. It provides a context in which to understand and interpret the data used and models created. It can also influence how the outcomes of the

project should be communicated, as it gives an indication of the audience. The PESTLE framework discussed in [Chapter 2](#) is a useful tool for understanding the external domain of a project.

Let's consider how the domain context for phone faults could differ if the organisation is an emergency response organisation, phones are used in extreme conditions and the reliability of phones could be a matter of life or death. This domain is very different from many other organisations that use phones. Likewise, the way the results will be communicated is likely to change if the managing director initiated the request rather than the phone manager. The managing director will have different drivers and knowledge from that of the phone manager.

RESOURCES

We have already talked about the people needed to deliver a project. Other resources that are required include the tools and infrastructure needed to undertake the analysis and, of course, the data itself.

In many instances, you will be constrained in relation to the tools and infrastructure available to you – you will need to work within the existing data architecture of your organisation.

However, it is beneficial to consider what is available and how well that meets the needs of the project. This might highlight challenges that will need to be addressed by the project. For example, you might need to resolve issues of data storage and computational power, or the tools might not provide the capabilities needed to easily prepare the data, or the methods they support might be limited.

It is also essential to understand the potential sources of data. This might throw up issues as to the availability and quality of data. It could also uncover issues with the right to use that data, or technical challenges of accessing or combining the data. At this point, initial consideration can be given to the models you might use in the project.

PROBLEM DEFINITION

When embarking on any project, it is always important to understand the scope and objective of the undertaking.

Agreeing and sharing these provides a common understanding and an anchor point for decisions. With data science projects, this is often done through defining the problem – this is also referred to as framing the problem.

In framing the problem, it is important to understand the context of the problem and why it is an important problem. This often requires input from several different stakeholders, as different people might have different perspectives.

Let's consider a music streaming company, we might have a problem with subscriptions. However, stakeholders might have different perceptions of that problem: the sales director might see it as one of attracting new subscribers, whereas the finance director could see it as one of retention because people don't use the service. Framing the problem helps to resolve these differences.

In this instance, what might the problem look like if the finance director prevails? The idea might be that the more a subscriber uses our services the less likely they are to cancel their subscription. Therefore, if we make recommendations of music that subscribers enjoy, they will listen more and retain their subscription. Recommendations could be based on artist or recording label. In this instance, there are several questions to explore:

- Does the amount of music streamed affect whether someone retains or cancels the service?
- Does recommending music lead to people listening more?

- If someone likes an artist, which other artists will they like?

It is also helpful to consider the point at which the project will be complete. For our music streaming service, it could be when we know what factors increase listening and hence retention. But what if it can't be demonstrated at what point you should stop working on the project? This could be defined in several ways, for example time-boxing the activity or setting resource limits. Time-boxing is where specific time is assigned to an activity, and that activity ends at the end of that time. Thinking about when the project should end before it starts is helpful to stop projects getting carried away and going past the point where they have no substantive value.

DEVELOPING THE HYPOTHESIS

Based on the problem, hypotheses can be developed. A hypothesis is a theory that explains a situation that has not been proven, and it provides starting points for investigating the problem. Exploring hypotheses allows the problem to be better understood, and decisions to be made in relation to it. The hypotheses are tested through the creation of models, created from methods and data. These models allow the hypotheses to be investigated, providing insight into the problem.

If we look at our subscriber retention problem, the hypothesis might be simple: people who stream more are more likely to retain their subscription. However, this problem is quite general; how much streaming are we talking about, and what time periods are we considering?

Our hypothesis could be:

- Streaming more than eight hours of music in a month affects whether a subscriber will cancel their subscription in the following month.

If this is proved, then further a hypothesis could be explored delving further into the problem. For example:

- The greater the number of music recommendations made in a month, the longer a subscriber listens to music.

And then

- Recommending an artist that the subscriber has previously listened to affects whether the subscriber listens to the recommendation.

A problem might have several hypotheses associated with it, which will be explored concurrently.

EFFECTIVE ENQUIRY

To define the problem and the resources needed to investigate it (along with any constraints), effective enquiry is required. This will involve both desk-based research and speaking to a range of stakeholders. Undertaking the desk-based research before speaking to stakeholders can be beneficial. It allows better questions to be asked of stakeholders and improves your credibility when talking to them, enhancing the quality of information gained and reducing the overall time and effort for everyone.

A good starting point is to understand the domain of the problem. A good source of information on this is external news, and sector-specific and technical websites. You can then start looking at the specifics of the problem space using your organisation's internal resources. These will vary across organisations: in some, it might be looking on the network to see what can be found, in others it will be consulting the intranet. Finding previous similar data science projects can be useful.

When tools and data sources are located, some initial exploratory work will be helpful. This can help in identifying constraints and challenges that will need to be addressed.

Understanding these can help to frame the problem and the plan to explore it. There is no point embarking on a project that it isn't feasible to deliver.

There are different methods to engage with stakeholders. Two of the most common methods are interviews and workshops. Interviews are one-on-one conversations. They can be useful in building relationships, and they provide the opportunity for in-depth questioning. Workshops are a group activity and effective when information needs to be shared and consensus built. A discovery stage is likely to make use of both of these techniques. However, whichever you use, it is important to ensure you prepare properly so that they are effective and positive events. For interviews, preparation often takes the form of developing a question set to use. This will often comprise open questions, with more specific follow up questions, allowing an effective and informed challenge. For workshops, structuring how the time will be used and the interactions you will need can help to keep the workshop on track and people focused on the desired outcomes.

PRACTICAL TIPS

- Take time to understand the domain of the project and the environment into which it will be delivered.

- Identify the skills and expertise needed to deliver the project.
- Understand who will provide the skills and expertise: ‘the team’.
- Determine the resources needed to deliver the project and whether they are available.
- Frame the problem clearly and identify the hypothesis to be tested.
- Think about potential sources of data and models that might be used.
- Draw on other people’s knowledge and expertise.

PROPERTIES OF DATA

THE IMPORTANCE OF THE RIGHT PROPERTIES

History is littered with examples of decisions being made based on people's gut feelings and things whispered in their ear. In the modern world we like to believe that we base our future on 'data' and can be more certain of what is likely to happen. Human psychology is a little beyond this book, but to make great decisions still requires the 'right' data.

When managers use figures and percentages from a spreadsheet, they look at numbers: this key performance indicator (KPI) has been met; this is how well we are doing against our target and so on. We expect that everyone knows (and most people do) what 100 per cent means, and that meeting this means something. What about the words that people use, though? 'That was a **great** outcome', '**Awful** customer service', 'I was really **disappointed**'. How do we use the thoughts and words of different customers and colleagues to get a view of what is happening? Many times, this information can prove far more useful than just numbers on their own.

Knowing what type of data we are using is important and will lead us in a particular direction. Many of us are brought up dealing with numerical data, but there are so many different ways of gathering information in the modern world. Sometimes we avoid a way of doing things if it is too 'difficult' for us, and sometimes we are led into a method that could be completely wrong. An accurate, detailed report showing lots of figures may just cloud the view of what is really happening when words and sentiment are far more useful.

Sears is an example of a company focused on their financial numbers, and for many years was a retail giant in the United States, reporting strong results and having a significant market presence. At its peak, it was one of the largest retailers in the country.

Despite its financial success, customer satisfaction began to decline sharply. Many consumers complained about poor customer service, outdated stores and a lack of product availability. The shopping experience was often viewed as uninviting and frustrating.

As customer loyalty waned, Sears struggled to adapt to the changing retail landscape, particularly with the rise of e-commerce. Their inability to innovate and improve the

customer experience contributed to a significant decline in sales.

In 2018 Sears filed for [Chapter 11](#) (reorganisation) bankruptcy, marking a dramatic fall from grace for a company that once thrived. Their financial results and traditional KPIs were overshadowed by mounting customer dissatisfaction and a failure to respond quickly to evolving consumer expectations (Riker, 2024).

The types of data we use can be given official-sounding names. Let's start at the top and look at the two main types of data which people tend to use:

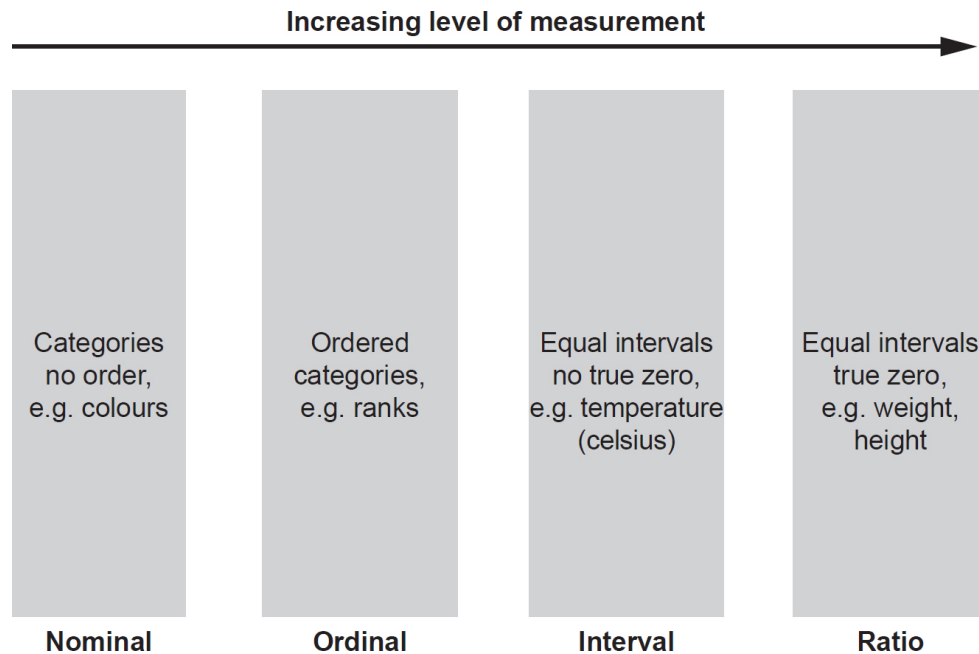
- **Quantitative** data defines an amount of something. It is almost always numbers and is the most common form of data that people tend to think of in their reports, scorecards, dashboards. Humans are trained to work with numbers so, even when we get data that doesn't fit this form, we try and make it quantitative just so that we can use traditional methods like percentages or totals (e.g. how many customers said we were 'good').
- **Qualitative** data is normally subjective and frequently takes the form of words (e.g. which of the cakes tastes the **best**? How do you **feel** about your broadband?). As we are getting

words in response to our questions, it is trickier to deal with the answers. Surveys are a common form of gathering people's feelings – and quite often try and turn this into quantitative data when it is being captured.

Quantitative data can come in different forms as well. It is critical to understand (and challenge) the source of the figures you are provided with, because numbers can have their drawbacks. When something comes in first place and something is in second place, we have a view of which one was 'better', but the data is missing details of how much better. Someone finishing the 100 metres sprint in first place but only one thousandth of a second better than second place is very different from them winning by five seconds.

To help us define the different formats of qualitative and quantitative data, there are four 'types' you need to be aware of: nominal, ordinal, interval and ratio (NOIR), set out in [Figure 6.1](#). Nominal and ordinal are usually seen as qualitative measures, with interval and ratio being quantitative.

Figure 6.1 NOIR data types



BIAS OR SKEWED DATA?

Mark Twain once used a phrase borrowed from earlier authors that there are 'lies, damn lies and statistics'. He was referring to the way that we can make numbers show a message that may not be the correct picture. There are lots of words used when describing data that 'appears' to have been influenced. Many people bandy about the terms 'skewed' and 'biased' – and also correlation – but we will talk about that later in the book.

Think of modern marketing phrases such as

‘You probs prefer Pepsi as 70% of the UK preferred it in blind taste tests’, Pepsi Co.

‘The world’s favourite airline’, British Airways

‘Nine out of ten doctors recommend Tylenol’, Johnson & Johnson

These rely on the use of figures and statements like ‘prefer’ or ‘favourite’ and do not give many details about the sample of people used for their tests, the questions asked, when the figures were gathered, removal of any outliers and so on.

As someone wanting to know their way around data, you should spend a few moments looking at what statements actually mean. The interpretation of data usually depends on the person that is reading it, and the data itself will have gone through a number of calculations and aggregations as it made its way onto the report or presentation.

Just as important as understanding the message that has been derived from the data is the ability to know whether the source of the data may have been influenced too. If you ask a group of young males if they want more football on the TV, there is a strong chance that more of them will say ‘Yes’. The person asking the question may want to show that there **should** be

more football on TV and can use the results from their survey to prove this. If you had asked the same question of a broader range of people, there is a strong chance you would get fewer people saying 'Yes'.

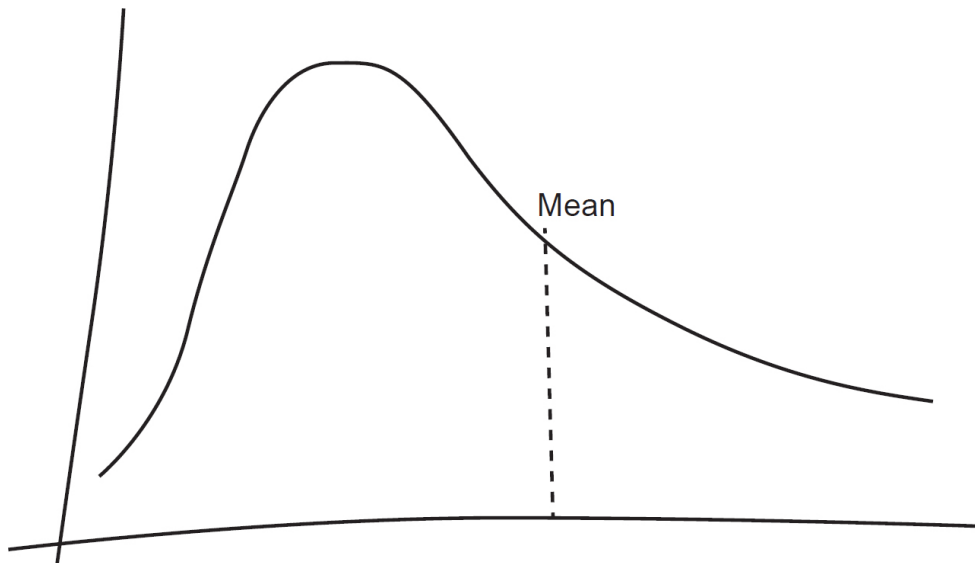
Let's explain a few of the key definitions used when we talk about the source of our data:

- **Population:** This means every person or thing that could be included in the scope of what you are doing (e.g. all customers that use your company, all people that watch a specific TV channel).
- **Sample:** This is a (usually) smaller group taken from the population (e.g. a group of 100 customers that you would send surveys to from the 100,000 customers that your company has).
- **Sampling:** This is how you pick the sample and as it has a potentially significant impact on the data, is a discussion that should involve relevant people. It should not be done by one person (their preferences may limit who is asked or included). At the very least, any results from the data should clearly state how the sample was decided upon.

There are a few ways that you can pick a sample:

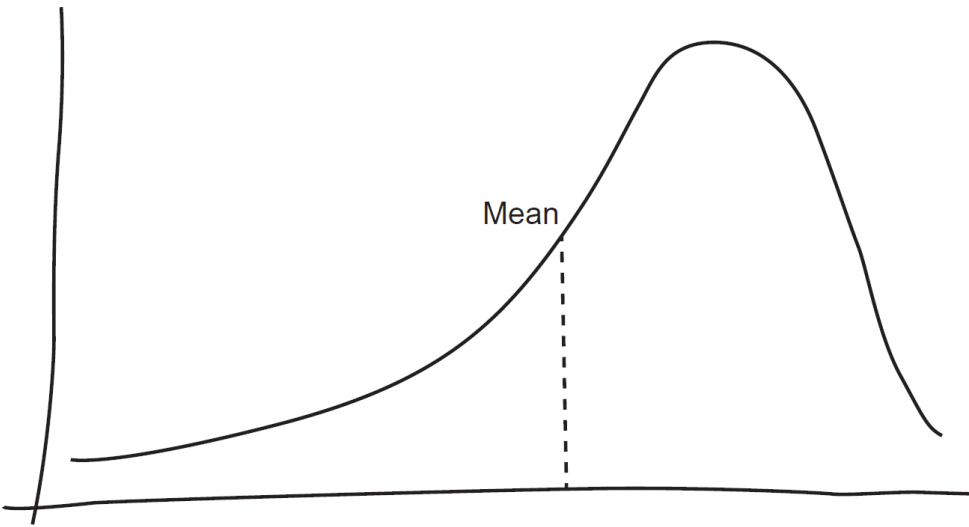
- **Random sampling:** Everyone in the population has an equal chance of being chosen, like drawing names out of a hat. Many would argue that this should always be used, but sometimes you want to make sure that all types of people will definitely be included in your work.
- **Stratified sampling:** You divide the population into groups and then pick a certain number from each group. An example of this would be ages, where you want to make sure that 60–69 year olds are always included in the sample.
- **Convenience sampling:** You pick the easiest people to reach, like your friends.
- **Skew** refers to how data is spread out or ‘leaning’ in one direction (see [Figures 6.2](#) and [6.3](#)). Imagine a set of employees. If there are more younger people working in a company, then the spread of people would be greater in the lower age groups. In data, if most of the values are higher or lower than the average (mean), the data is skewed.
- **Positive skew:** Most data is on the low side.
- **Negative skew:** Most data is on the high side.

Figure 6.2 Positively skewed data



- **Bias** is one of the most popular terms to use when people ‘feel’ that something is unfair or not accurate. Bias can happen when your sample or method of sampling causes the results to lean a certain way. For example, if you only ask customers who buy mobile phones if they would like to buy headphones. Your results might be biased because you’ve left out those who are less likely to buy headphones. Bias can lead to incorrect conclusions because your sample doesn’t truly represent the population, and making sure you know the best way of sampling is the easiest way of avoiding bias.

Figure 6.3 Negatively skewed data



When starting to work with data, you should look into using both quantitative and qualitative research to understand the problem you're addressing, the data you need and the context in which you'll be working.

QUANTITATIVE RESEARCH

Data collection and analysis:

- **Collect existing data:** Identify and gather relevant data sets that can help to answer your questions. This could be structured data such as numbers, statistics or measurements.
- **Statistical analysis:** Use descriptive statistics (mean, median, mode, standard deviation) to understand the basic features of the data. Perform exploratory data analysis (EDA) to

identify patterns, trends and relationships in the data. We will talk about this more in [Chapter 10](#).

- **Surveys and questionnaires:** If you're collecting your own data, design surveys or questionnaires with closed questions to gather numerical data from your target audience. Ensure you have a large enough sample size and are covering a range of different groups.

QUALITATIVE RESEARCH

Problem definition and user research:

- **Interviews:** Conduct in-depth interviews with stakeholders, domain experts or management to confirm what question needs to be answered by looking into data. This helps in defining the problem and setting expectations.
- **Focus groups:** Organise focus group discussions to gather diverse opinions and insights. This is particularly useful for understanding the broader context and user behaviour that might not be evident from data alone.
- **Observation:** Observe what happens in a 'normal' situation rather than asking specific questions. This may highlight unspoken needs and thoughts.

Learn the context:

- **Hands-on research:** Immerse yourself in the environment where the data will be used or where it originates. This helps you to understand the cultural, social or organisational context that might affect your work.
- **Case studies:** Study specific examples or cases that are relevant to your work. Analysing how similar problems have been addressed can provide valuable insights.

Literature review:

- **Academic papers and industry reports:** Read existing literature on similar projects, methodologies and technologies. This helps you to understand what has already been done by others and avoid reinventing the wheel.
- **Best practices:** Identify and understand best practices in using data and in your workplace. This can guide your approach and lead you to people that could help and advise.

Qualitative data analysis:

- **Thematic analysis:** Analyse qualitative data from interviews, focus groups or open-ended survey questions to identify common themes, patterns and insights.
- **Content analysis:** Categorise textual data to quantify patterns or themes, which can then be combined with

quantitative data for a richer analysis.

Integrating different research:

- **Cross-validate your findings:** Use both quantitative and qualitative methods to cross-validate your findings. For example, you might use qualitative insights from interviews to explain patterns you see in the quantitative data.
- **Iterative process:** Let your qualitative research inform your quantitative approach, and vice versa. For instance, initial qualitative findings might help to refine your data collection or analysis methods.

By combining both types of research, you'll gain a deeper, more comprehensive understanding of your problem, the data and the best approach to arrive at a solution. This holistic view increases the chances of your results being successful and effective.

THINGS TO NOTE

The world of data is rapidly changing the way we work. The 'traditional' methods of analysing and summarising data should be challenged. The sheer quantity of data available to us can be daunting, and knowing how to 'pull' the right parts from this will take some time and effort. Before you even think about

extracting data, you should look into what data you could use. Qualitative data is much easier to handle than it ever used to be, especially when used with tools that AI can support. These views of data can add to or change the perceptions of management and the direction of the company strategy. If you are thinking even further ahead, question what other types of data there could be. Could images or video add any useful information and, if so, how do you get at that data as part of your sample?

SOURCING

One of the key ingredients of data science is, as the name applies, data. As we have discussed in the previous chapter, the data used in any project can be the difference between a good outcome and a bad one. To be clear, a good outcome isn't just proving a hypothesis. Failing to prove a hypothesis can be equally valuable, potentially avoiding making a poor decision based on a false premise.

Sourcing data isn't just about finding some data, it is about ensuring the data set contains the data items needed and has the right data properties to allow the analysis. In some instances, the source data might need to be manipulated to derive data. In [Chapter 8](#) we have an example of this in data granularity, where daily data is used to calculate a monthly figure.

Remember, just because data is available, it doesn't mean that you can use it. There are both legal and ethical constraints that need to be considered. These are addressed in [Chapter 4](#).

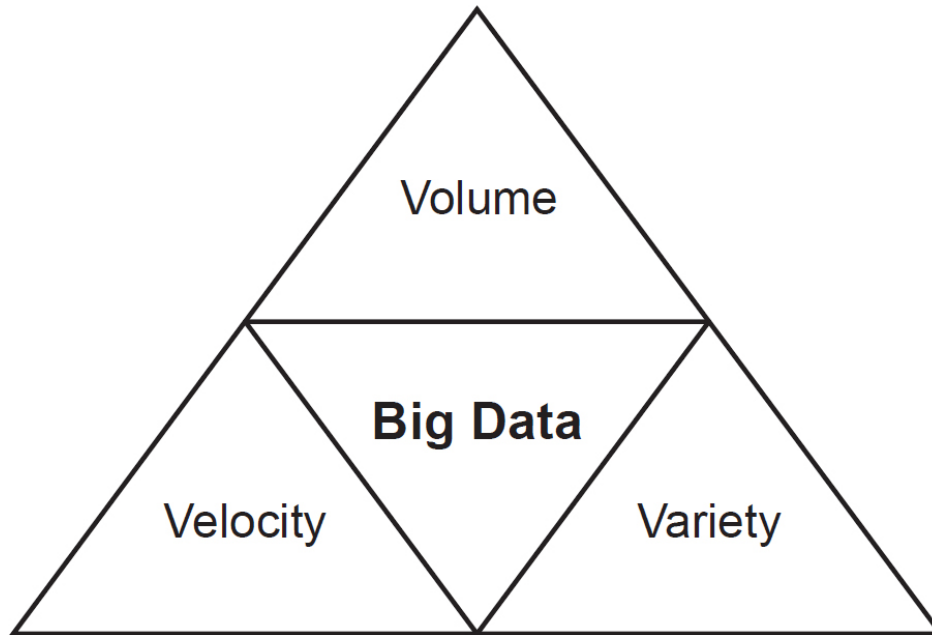
BIG DATA

Data is everywhere in the world around us. We are surrounded by the radio frequencies that carry data for the devices we use

every day. This data permeates all aspects of our lives, whether it is checking social media or tapping a payment card to travel on the Tube. The acceleration in data has led to the term Big Data and three concepts relating to it: volume, velocity and variety. These are known as the 3Vs ([Figure 7.1](#)), and can be useful in helping us to consider some of the challenges related to sourcing data.

- **Volume:** This considers the amount of data that is generated. Think about how much data a quick coffee stop can generate. As you walk through the door, the security camera will record you, the coffee machine will log its use, your payment will be recorded and that's all before you hit social media while you enjoy the caffeine hit. If you look at all the activities in a day that generate data, both human and automated, that's a lot of data.
- **Velocity:** Think about the speed at which the data is created and processed. The camera is a real-time recording, the payment is processed in the time it takes to say 'thanks', the coffee machine will instantaneously flag if it needs attention (after all, time is money) and your social interaction is co-temporal. As you can see, much of the data around us is moving fast – very fast.

Figure 7.1 The 3Vs of Big Data



- **Variety:** We can see the variety of data being created. We have the unstructured data of video and social media alongside the structured data of the payment and internet of things (the coffee machine sending its data). If we want to use this data, we need to be able to store it and, equally important, be able to access it.

Other Vs have been added to the 3Vs, and reference is often made to the 5Vs.

- **Veracity:** This relates to the quality and accuracy of the data. Data quality is covered in more detail later in this chapter.
- **Variability:** Organisations might have inconsistencies in their data. Take employees: in one system this might refer to

people with employment contracts, in another it might include agency staff.

For us as data scientists, this all creates challenges in identifying and accessing the data we need. It can also cause issues as we address the constraints of storing and processing the data.

DATA COLLECTION

We talked about several different sources of data when having our coffee break. These use different methods to collect the data:

- **Cameras** record digital images, which are data. These can be video or still images.
- **Sensors** automatically record data and frequently send it to a central data repository. This includes the coffee machine recording usage and flagging any issues.
- **Transferred data** occurs when you add your favourite snack to the coffee, because the barista might scan the barcode on it, transferring data from the item to the point-of-sale system. The same is true when you tap your card to pay for it, your details on the card are read by the system.

- **Human entry** via keyboards, mice, touch screens and other data entry devices. In the coffee shop, the barista might use a touch screen to record the sale of your coffee. You also enter data when you give the coffee shop a five-star review on the shop's app.
- **Derived data** is created when existing data is taken and manipulated to develop a new data set. The coffee shop might take all your coffee transactions and determine from this what customer persona you fit.
- **Synthetic data** is created when a data set is artificially generated. This can be beneficial where there is limited real-world data or concerns exist about its quality. In some instances, rather than using actual customer data, an organisation will synthesise customer data to overcome privacy concerns.

It is also worth considering how these different collection methods can be joined. Let's take our coffee machine. Its sensors record how hot the water is. This data is then transferred to head office. Here, an algorithm is applied using this and other data to determine whether the machine needs a service, resulting in a request to maintenance, where a service call is automatically scheduled. This is an example of the internet of things in action, where devices collect and share information with each other.

When we collect data from people, it is often in the form of having them enter data into a system or complete a survey. We consider the risks relating to this and how they can be mitigated further on in this chapter.

PRINCIPLES OF DATA: DATA TYPES

In sourcing data, different types of data can be considered. The data types we are considering here are the data types associated with the principles of data. The term ‘data type’ is used in different contexts, where it has a different meaning, an example being data types used by computers. We can think of the principles of data, data types, as indicating the availability of data and the reason for its creation ([Figure 7.2](#)). The difficulty with these labels is that they can have slightly different meanings depending on who you are talking to. If you come across these terms, it is always worth checking which are relevant and their meaning in the context you find them.

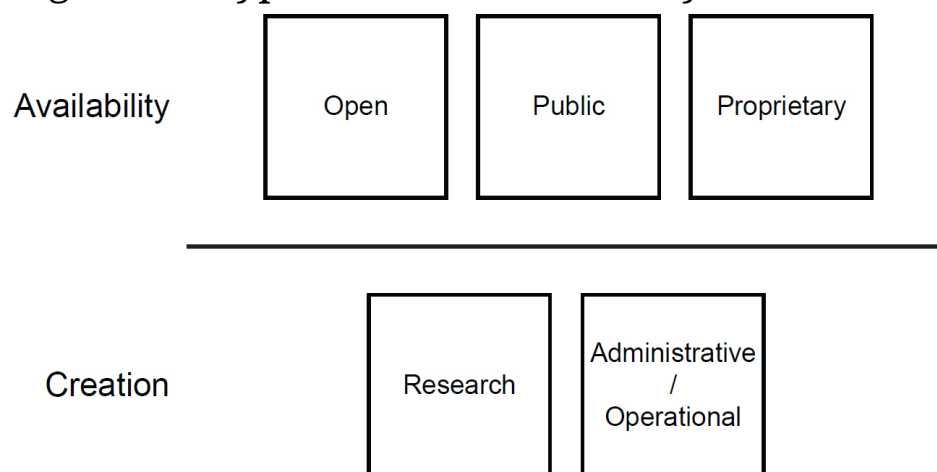
Availability refers to how accessible the data is, and has three broad categories:

- **Public data:** This is generally taken to mean any data that is publicly accessible, that is, anyone can access it. This includes data found on the internet and social media. It is worth

remembering that just because something can be accessed doesn't mean it can be used. If you are intending to go on holiday, then the flight times and prices presented to you online are public data. That doesn't necessarily give you a catch-all right to collect and use them.

- **Open data:** This is a subset of public data, normally released by a government or its agents. In this instance, the data is of a known quality, and it has a licence associated with it that determines how it can be used. The benefit of this is that the data is transparent and therefore can be used with confidence. This doesn't mean the data is free; in some instances, a fee might be payable for its use. Census data published by the Office of National Statistics (ONS) is an example of this type of data.

Figure 7.2 Types of data (availability and creation)



- **Proprietary data:** This is data that is not publicly available. The individual or organisation who owns it has not made it

public and they control how it is used.

Creation refers to how the data was generated. It can be split into two categories:

- **Administrative and operational data:** This is the data generated through the activities of an organisation, and the terms aren't mutually exclusive. The transaction details of our coffee purchase are administrative and operational data. Similarly, if we grew coffee and recorded the yield of the crops and the weight of beans roasted, these would also be administrative and operational data.
- **Research data:** Where data is specifically collected for a study or from an experiment, it is research data. Thinking back to the link between smoking and cancer, the data used to prove this would be research data. This data tends to be academic in nature, so if you respond to a survey about your morning coffee it might be better seen as administrative or operational data as it has been collected to run an organisation better.

FINDING DATA

When sourcing data there are various places we can find it. We might already have it in our organisation. If we haven't got it,

we might be able to obtain it from a third party. Failing that, we will need to collect it ourselves.

In a world of Big Data, organisations hold a lot of data. This is proprietary, administrative and operational data. It is not always clear what data is available nor its quality. Data governance, covered later in this chapter, can assist with this.

It is possible to source open and public data. In some instances, data is published online. This can provide a rich source of data (e.g. Census data). It is also possible to purchase data from third parties; an example of this is financial trading data. Data can also be obtained from social media and other websites using approaches such as screen scraping or via their application programming interfaces.

There is huge variation in the scope and scale of data collection activities. Consider the effort involved in collecting data in a pharmaceutical trial compared to a small customer survey. Thought must be given to the method of data collection to ensure the quality of the data. In all instances, don't underestimate the effort involved because there are often unforeseen challenges that need to be resolved, as can be seen in our case studies in [Chapter 15](#).

When we rely on people to collect data, we need to remember they are human, and humans make errors. This is particularly the case when typing in information. Therefore, it is helpful to ensure that the data people are entering is validated. This might be through using drop downs or tick boxes or checking the format of the information. Think about when you enter an address to have your latest online purchase sent to. You might type in the address and the system will check that the postcode is valid. It's even better if the postcode is used to look up the address, allowing you to select your postal address from a trusted source. In this instance, there is less chance of your parcel going astray. It also results in quality data for our algorithm to help us better profile our customers.

No doubt you are frequently asked to take part in a survey, normally on your customer experience or a product you have bought. The challenge with surveys is ensuring you have a representative set of data and that the questions you ask are clear. You might think that this is easy, but even the experts get it wrong, as can be seen the UK 2021 census. In this case, the statistics on gender identity were downgraded from official statistics, due to potential confusion over the question being asked (Melley, 2024).

STORING DATA

In many instances, organisations set up separate data stores, which allow data to be accessed without affecting operational systems. They also have the advantage of data reuse, reducing costs and increasing efficiency. Three types of stores are often referred to:

- **Data lake:** This is a store of data in its raw format, which can contain structured, unstructured and semi-structured data. The data can come from many different sources. For example, data from the enterprise resource planning (ERP), customer relationship management (CRM) and email systems might all be placed in the data lake.
- **Data warehouse:** This is a store of data used for reporting. It is normally at an enterprise level, bringing together data from different systems. In most cases, the data has undergone a level of data preparation, covered in [Chapter 8](#). Looking at our example, subsets of the ERP and CRM data might be brought together into the data warehouse, for example customer and supplier data.
- **Data mart:** This is a subset of a data warehouse focused on a single domain of an organisation. This could be in the form of customer and sales details being taken from the data warehouse and placed in a sales data mart.

It is important to note that not all organisations will have these data stores and data might also be found in other locations, for example on shared drives or in document repositories.

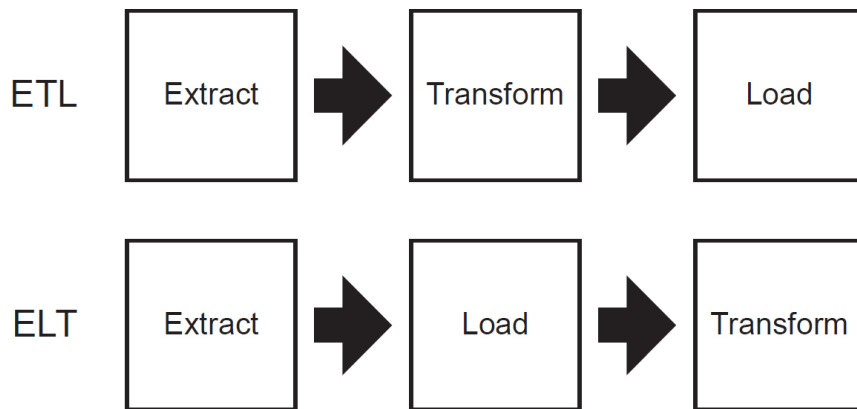
MOVING DATA

It is worth considering the process of moving data into these data stores from source systems. Similar processes are used to take data from the data stores to the tools you use. The process has three stages:

1. **Extract:** Accessing the source system and retrieving the data required.
2. **Transform:** Preparing the data from the source system. This might include bringing different data sets together, restructuring the data and addressing quality issues.
3. **Load:** Writing the data sets to the new data store.

These can be done in two different orders: ETL and ELT, as set out in [Figure 7.3](#).

Figure 7.3 ETL and ELT



The fundamental difference in the processes is where the transformation (data preparation) takes place. If we take a data lake, it is likely that ELT will be used. The raw data from the source will be placed in the data lake. For a data warehouse and data mart, it is likely that ETL will be used, as the data will be transformed to ensure the quality of the data loaded in the warehouse or mart. Many data scientists favour ELT, as they gain access to the raw data, which potentially presents more analytical possibilities.

DATA GOVERNANCE

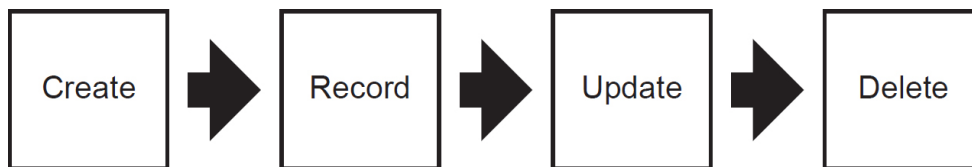
Data governance covers both the strategic decisions around how an organisation should use and manage data and the operationalisation of that strategy. For mature organisations, this will be set out in clear policies with an effective governance

structure. As a data scientist, you need to adhere to these policies and, if they are not in place, the benefits they deliver should be considered. Data governance not only ensures the safe use of data but also leads to the efficient use of data.

Data lifecycle

The data lifecycle is different from the data analysis lifecycle explored in [Chapter 3](#). In this case, it is the lifecycle of the data that is being looked at, not the lifecycle of the analysis of data. Unlike man who has seven ages, data has four ages.

Figure 7.4 CRUD, the four ages of data



The data lifecycle is often referred to as CRUD ([Figure 7.4](#)). This comes from the operations applied to data in a database: it is created, recorded, updated and then deleted. However, it is also useful to data scientists when thinking about how we govern our data.

We create a data set from our sourced data for any data science project. This data needs to be stored (recorded), and over the

course of the project it will be manipulated and updated. Once our project is finished, and the data is of no further use, it should be deleted.

Benefits of data governance

Data governance can ensure that we take appropriate action at each stage of the data lifecycle, resulting in quality data that is used safely and efficiently.

- **Safety:** Data governance can ensure that data is only used for proper purposes. This includes restricting access, mandating storage requirements and providing for the deletion of data no longer required.
- **Efficiency:** Although data is everywhere, we can forget the costs associated with its use. These relate to both the preparation of data and the storage of it. Good data governance can promote the efficient use of data through the reuse of data and deleting data no longer needed. This reduces both preparation and storage costs.

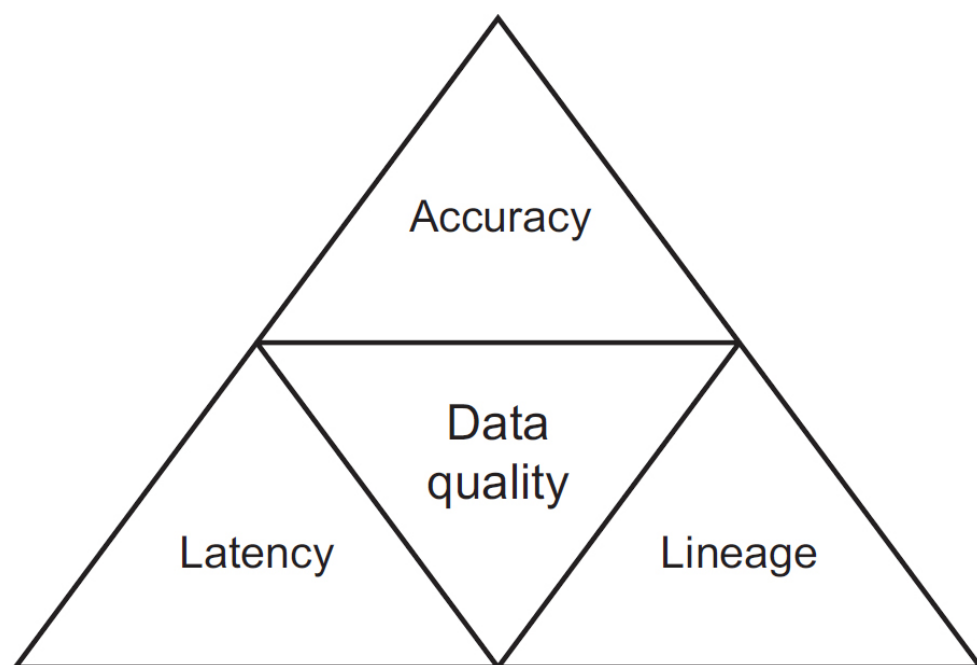
DATA QUALITY

You might recall our project looking at phone faults. How would you feel if the project used data from five years ago in the analysis? What about if the data from the field sales force was

missing? What if someone just had some data that they thought related to the organisation's phones? It is likely in all these cases that you would be sceptical, to say the least, about the findings of the project.

The quality of a data source can be seen through three lenses ([Figure 7.5](#)):

Figure 7.5 The three lenses of data quality



- **Accuracy:** Ask the questions: is the data set complete and has the data in it been properly recorded? To answer these questions, you really need to understand the domain context of the data.

- **Latency:** How old is the data set? Is it still relevant? This can vary with context: if you are working with census data, it might be valid for it to be years old; if you are looking at financial trading data, minutes or even seconds can count.
- **Lineage:** Do you know where the data came from? Can it be trusted? Many people trace their family tree to understand their origin and roots. Data lineage is about understanding the origin of the data and any changes that have been made to it over time. For example, if you took a data set on population from the ONS, you would know where it came from and its quality. If the ONS data was taken from your organisation's data warehouse would you be aware of any changes that had been made to it? Is there a possibility records have been removed or amended that could have an adverse impact on your project?

Good data governance can help in answering these questions for internal data. For external data, open data helps us to address these questions. For other sources, do your homework: ask commercial suppliers the questions.

In considering the quality of the data, it is also important to ask the questions: does this data relate to my problem, and will it help explore the hypothesis? It might be the most up-to-date,

accurate and timely data set, but if it isn't relevant, it hasn't value.

METADATA

To put it simply, metadata is data about data. It can help to govern the data we use and, where used effectively, help us to identify good sources of data. Going back to our phone analysis, the metadata associated with this could include information on when the data was created, last updated and its storage needs (size). It could also provide information on the content of the data (e.g. phone usage data) for a specified date range and whether it is classified as confidential. If we make the metadata accessible to people, then they can understand what data is available and its quality.

As you can see, thought needs to be given to the source of data we use in data science projects. There is a plethora of data, and the challenge is finding the right data set to use in a model. In some instances, we will need to be creative in how we source our data. We also need to recognise the effort that sourcing data requires, even if we don't collect the data ourselves. You will recall the iterative nature of the data analysis lifecycle, and sourcing the data can lead to both backward iterations to discovery and forward iterations to preparation.

PRACTICAL TIPS

- Remember: garbage in, garbage out.
- Consider if there are different sources of data, and their relative merits.
- Consider whether the data has the right properties to test the hypothesis.
- Consider whether the quality of the data is good enough.
- Consider how you will export, transform and load the data.
- Make use of and follow internal data governance.
- Check it is ethical and lawful to use the data.

PREPARATION

Preparation conjures up the image of a chef preparing a meal. For you, this might be the clinical precision in which they compose and combine the ingredients in an orderly and controlled manner, or it could be a more chaotic approach as the ingredients are thrown together in a creative maelstrom. Data preparation is not dissimilar. It is also referred to as data transformation, manipulation, conditioning and wrangling – the latter really emphasising the challenge in taming the data ready for analysis.

It is important that the data sets and the methods we use to build a model for a project are well suited to each other. If they are not aligned, then the method will fail or deliver a result with limited value. Even if the data sets and methods are well suited, the results are likely to be limited if the data is low quality. In preparing the data, we need to understand both the form and the quality of the data. We can then consider the risks and issues associated with using it and take steps to mitigate them. Spending time up front on this pays dividends. The allure of developing your models may be strong, but doing this too early, with sub-optimal data, will lead to longer timescales or worse: low-quality outcomes that (dangerously) might not be recognised as such.

In this chapter we will take a generic view of the things you might need to consider in preparing data. These will vary, depending on your data and the methods you are looking to use, and should be carefully considered in relation to each of them.

FORM

They say form follows function in design, so an object or building shape should follow its function. Most pens are round or hexagonal. Why? They are easier to grip.

With data, its form affects the function. The form of the data affects what you can and cannot do with it: its function. By form, we mean the inherent nature of the data set, rather than the content of the data set, which we often have limited control over. Form includes data granularity, structure, type, scale and properties. When sourcing and preparing the data, each of these must be considered.

Data granularity

Granularity refers to the level of detail in the data. If we think about recording data against time, we could record it at a daily or hourly rate. Hourly data is more granular than daily data. It is important that the granularity of the data matches the

question asked and the context of the question. Think about a contact centre, a question you might be asked is: how many employees are needed? This could be to support shift planning or to help shape the recruitment strategy. For shift planning, you are likely to need data on a daily or hourly basis; for recruitment, monthly data might be more appropriate.

The granularity of the data affects what you can do with it. So, if in our example you have monthly figures, the data would have very limited use in shift planning – daily or hourly data would be more appropriate. This might, however, have too much detail to be useful for recruitment planning. The daily or hourly data could be consolidated into monthly figures to aid recruitment planning, but the monthly data can't be broken down into daily or hourly figures.

We have previously considered faults in phones. If our data just records that the phone has a fault, it will not allow us to analyse faults in screens or faults with the operating system – more granular data would be required for this. This level of granularity might be in the data, but not in a form that can be used in a model. For example, the cause of the fault might be recorded in one field (see [Table 8.1](#), column 3), but the method we plan to use with the data requires a separate field for each of the faults. In this instance, we might create an additional

data column for each fault and record the fault in it. In the example in [Table 8.2](#), you can see that, in creating the additional columns, the fault type is marked as being present with a 1 and not present with a zero. The reason for this is explained further in Category of Data later in this chapter.

Table 8.1 Data with fault type in one column

Fault ID	Date fault logged	Fault type
101	12/5/2025	Screen
102	13/5/2025	Operating system
103	13/5/2025	Screen
104	14/5/2025	Screen
105	14/5/2025	Processor

Table 8.2 Data with fault types recorded in separate columns

Fault ID	Date fault logged	Screen fault	Operating system fault	Processor
101	12/5/2025	1	0	0
102	13/5/2025	0	1	0
103	13/5/2025	1	0	0
104	14/5/2025	1	0	0
105	15/5/2025	0	0	1

Creating additional columns is a one way of restructuring the data. However, in some instances, the structure of the tables the data is held in also needs to be considered.

Structure

Data can be held in many different structures. These data structures include lists, arrays, records, trees and tables.

- **Lists** are an ordered collection of data items. The items in a list can change.
- **Arrays** are a collection of items that can be accessed by an index. An array can have multiple dimensions – a two-dimensional array often being represented as a matrix.
- **Records** are collection of fields, much like a row in a table.
- **Trees** are used to hold hierarchical data. The data is held in nodes that have a parent–child relationship with other nodes.

- **Tables** consist of rows and columns. Each row represents an instance of data, and the columns hold the attributes relating to that instance. Each row is a record and each column is a field.

The data structures you encounter will depend on the data you work with and the tools you use. One of the most encountered data structures are tables. These can be found in both spreadsheets and relational databases. It is important that you have the right data structure to enable your analysis. This includes making sure your data is held in appropriate tables if it is in a relational database.

When we analyse our contact centre, we might want to look at the relationship between the number of contacts dealt with in a shift and the number of people working a shift. This data might be sourced from different systems and thus sit in two different tables ([Table 8.3](#)). For this to be used by our model, we are likely to need to combine the data into one table. In this instance, we will use the date as the common reference to do this – the key ([Table 8.4](#)).

Table 8.3 Data on contacts in two tables

Date	Contacts dealt with	Date	People working
12/5/2025	100	12/5/2025	4
13/5/2025	110	13/5/2025	5
14/5/2025	80	14/5/2025	4

Table 8.4 Data on contacts combined into one table

Date	Contacts dealt with	People working
12/5/2025	100	4
13/5/2025	110	5
14/5/2025	80	4

This is a simple example of how data often needs to be restructured as part of building a model. Structuring data to fit your model can become more complicated, as data can be

spread out over multiple tables with complex relationships and linking tables.

Types

The data type of each of the fields also needs to be considered. This isn't the data type we looked at in [Chapter 7](#). In this context, we are considering the type of data that can be held in a field that includes integers, floats, Booleans, date/time and strings.

- **Integers:** These are whole numbers, that is, no fractions or decimals. They can be positive, negative or zero.
- **Floats:** These are numbers that can include fractions and decimals.
- **Booleans:** This is a binary field that holds a true or false value.
- **Date/time:** This holds the date and/or time. The data held and its format varies across specific data types. For example, one type might just hold DD/MM/YYYY, while another might hold YYYY/MM/DD/Hour/Minute.
- **Strings:** These are made up of characters (letters, numbers, symbols).

To be used in our tools and models, the data might need to be of a particular data type. For example, if we are forecasting future values over time, the dates in our data set will need to have the data type date/time. However, they could have been stored as a string data type, or the data type might have been corrupted in our export or load. Another example is where an integer is stored as a float or a string. It is therefore important to check and, where necessary, change the data type.

When using dates, it is also worth considering their format. You might just be interested in the day, month and year, but the date may include a time as well. In this case, you might need to change the data type to address this or, in some instances, restructure the data to remove the parts that are not required.

Scale

When we measure things, we use different units of measurement, and these often have different ranges. For example, in measuring age we normally use years, and this range might be 0 to 120, if you are very lucky. In measuring daily mobile phone usage, the unit could be minutes and the range 0 to 1440, if you are an insomniac. Differences in the ranges of data can cause models to give undue weight to some attributes. In these circumstances, the data is scaled so that the

ranges across the attributes are the same. This then gives them equal weight in the model. There are different techniques to do this, referred to as feature scaling. Each attribute (referred to as a feature) is replaced with a value, often with a range between 0 and 1 or -1 to 1. In our example, using a range of 0 to 1, if we had an age of 120 and scaled it, it would result in a 1. If the age is 60, the value would be 0.5. A 1 would also be returned when someone spent 24 hours on the phone in a day. If they spent an hour, it would be a 0.042 (to three decimal places). This is often referred to as normalisation.

If we wanted to investigate phone faults using age and time the phone is in use, then feature scaling this data would ensure that the model is not influenced by time on the phone having a greater range than age.

Properties of data

In [Chapter 6](#) we looked at the properties of data. When preparing the data, it is important to understand the properties of the data set used and its suitability for the model being built. Rather than revisiting the concepts we will look at some practical examples.

Numeric data

Many models require the data in them to be numeric rather than textual. This means that data might need to be changed from a word into a number. If you look at the restructured table of phone faults in [Table 8.2](#), you can see that, rather than recording a yes/no in the field, we have used 1 and 0. This allows the data set to be used in models that do not accept alpha values.

Category of data

Creating additional columns and populating with 1s and 0s might seem a lot of hassle. If the model needs numeric values, why not just leave one field and give each of the faults a number ([Table 8.5](#))?

Table 8.5 Fault types

Fault type	Fault type value
Screen	1
Operating system	2
Processor	3

The difficulty with this is that the data is ordinal (i.e. it has a rank), and this might affect the model. Hence, the addition of columns, one for each of the fault types and the recording of the value as a binary (1 or 0), a type of nominal data. The technique to move categorical data in one field into binary data in many fields is called 'one-hot encoding'.

This example considers a model that requires nominal binary data. Other models might have different requirements as to the properties of the data used in them.

PROPERTIES OF DATA SETS

As well as the data having properties, data sets can also have properties. Two properties often encountered are patterns over time and skewness.

Patterns over time

When looking for patterns over time, the data might have a trend. It could be an upwards trend (values increasing over time) or a downwards trend (values decreasing over time). A pattern over time could also be cyclical, for example retail sales peaking in the run-up to Christmas every year. Data such as this isn't stationary, and some models require the data to be stationary. Stationary data is where its statistical properties,

such as mean and variance, don't vary over time. Where the data has a trend or cyclical pattern, the mean and variance do vary over time. Tests can be undertaken to determine if the data is stationary and, if it isn't, it might be possible to transform the data so that it is stationary, allowing it to be used as part of a model requiring stationary data.

Skewed data

We have seen in [Chapter 6](#) that skewed data is where the data is unevenly distributed around the mean. A model's output can be affected by how skewed the data is within it. If this is the case, steps can be taken to normalise the data, leading to an even distribution either side of the mean. This even distribution is referred to as normal distribution or Gaussian distribution. Statistical tests that assume normalised distribution are called parametric tests, those that don't are non-parametric tests.

QUALITY RISKS

As we have seen in [Chapter 7](#), when sourcing our data it is important to consider the data quality. Considering the data quality is equally important in preparing the data. At this stage, we are looking at the quality risks in the data set and the action that needs to be taken to mitigate them. These quality risks

might be inherent in the source data or have been introduced in the extraction, loading or subsequent manipulation of the data.

Missing values

Data sets can have missing values for many reasons. These could include data not being collected in the first instance or data sets not being complete. Where data is found to be missing, this should be investigated. Within our contact centre project, the record of the contacts might have been lost or corrupted, potentially due to an IT failure.

When combining data, it is important to consider the way the data is combined, as different techniques can produce different combined data sets. A missing value in a data set might lead to data in that table and corresponding tables being lost. For our contact centre, if there was no data for a particular day in either our contacts-dealt-with or people-working tables, we could lose the corresponding record in the other table, depending on the approach taken to combine the data.

Duplication

There is also a possibility that rows of data might be duplicated when the data is combined. Again, we need to take care when

selecting how we combine our data to avoid this. In doing so, consideration needs to be given to the data structure, particularly where the relationships between tables are not 1 to 1. By this, we mean that the unique key in one table has multiple records with the same key in another table.

Sample size

Your data within your data set might be perfect. However, the data set might be too small to be effectively used in a model. In most cases the more data you have, the more accurate the model you can build. Some models are better at working with small data sets than others.

Outliers

Where there are outliers in the data set, it is worth considering the cause of the outliers, to decide whether to keep or remove the data. For example, if we looked at our contact centre data and saw a spike in the number of people working due to new trainees being brought on board, we might decide to exclude that data as it is not the 'normal' operation of the contact centre.

Bias

We have considered some of the causes of bias in our data sets in [Chapter 6](#). Bias is an ethical issue, and bias might not be apparent. It is worth thinking about how bias might be introduced into the core data set. A classic example is if we look at the characteristics that made a person a good recruit for a job. If we look at the data in the curriculum vitae of successful applicants for similar roles, the data set might be biased due the biases within those recruiting to those roles.

Wrong data set

Our search for good candidates for a role also illustrates the challenge of ensuring we have the right data set to address the problem. It could be argued that the data from the curriculum vitae of successful applicants can only help to identify the characteristics of successful applicants, but it tells you nothing about whether they are good at the job. In this case, another data set would need to be found that reflects successful candidates who perform well in the role.

QUALITY CHECKS

Some quality checks require understanding the data set and the domain to which it relates, others can be done by looking at the quality of the data source. For example, in looking at data to

help find good candidates for a role, you need to consider whether the data sets have missing data. This could be done by looking at the data set for blank or null fields. However, it could be harder to determine if there was bias in the data set, for example because a recruiter had previously rejected candidates based on their age. In this instance, knowledge of the domain and comparing the data set to other similar data sets is likely to be helpful.

Visual inspection

A visual inspection of the data can be helpful to identify any anomalies that require investigation. Where the data set is large, a sample of the data can be inspected, and there are tools to assist in this. For example using filters to look for specific values and using conditional formatting to highlight them. Some tools are more sophisticated, for example they might help to identify duplicates. A visual inspection of the data is always a good idea.

Graphical inspection

The data can be presented graphically to help identify any anomalies in it. Box plots are effective in identifying outliers, and skew in data is apparent if plotted in a histogram.

Graphically presenting null values or duplicate records can also assist in identifying potential issues.

Cross checks

Where data has been transformed, it can be cross checked against the source data or data sets created during the transformation. Where the data structure hasn't changed, checking that the number of rows correspond can be helpful, as well as comparing statistics relating to the data (e.g. the mean value of a column). Check sums can still be used where the data had been manipulate, for example if we take a daily record of hours worked and derive the total hours worked per month, the total of both over any period should be the same. More detailed checks can be carried out by looking at fields and columns to check data isn't missing or corrupted.

Data quality dashboard

If a data set is updated on a regular basis, you might consider setting up a data quality dashboard. This would reduce the effort in quality checking the data and provide confidence in the data science and insights driven by it.

MITIGATION STRATEGIES

When a data quality issue is found, you need to decide how to mitigate the risk it poses. There are several strategies that can be adopted, and thought needs to be given to how the strategy will affect the data set and model of which it is part.

- **Accept:** This is where the issue identified is accepted and you proceed to use the data set with the quality issue in it.
- **Correct:** In some instances, it is possible to correct the data quality issue. This might be a manual or automated process.
- **Remove:** The poor-quality data is removed from the data set. This is often through deleting the row or column of data that contains the issue. Removing data can lead to bias in the data set, especially when the data that is missing is not random. For example, if everyone occasionally forgot to enter data into a system, this is more random than if one member of staff consistently failed to load data.
- **Imputation:** In this instance, a value is determined for the missing data item and that value is used as a substitute. The substitute value is normally an estimate based on other data in the data set.
- **Enrich:** If a data set lacks the data it needs, it can be enriched. The data set could be missing data, or the data set might be too small. In some instances, there might not be sufficient variation in the data for the model to be effective. Enrichment is where additional data is added to address the

issue. This data might be from an additional data source or it might be synthetic. When enriching data, the costs, the appropriateness of the data used and the ethical questions in relation to this need to be considered.

- **Discard:** In some cases, the data just won't be up to job and the whole data set will need to be discarded. In this case, it is worthwhile understanding why it was discarded to allow the earlier detection of those issues in other data sets.

As already mentioned, the preparation of data is closely related to the sourcing of data. Once a candidate data source has been found, then initial preparation work can take place. This might reveal issues with the source data that need to be addressed.

The methods used in the model also influence the data preparation. It is important that the properties of the data fit the method used and that the data is good quality. Both of these aspects can affect the output of the model and, where this isn't understood, can lead to decisions being made and actions taken on misleading insight.

PRACTICAL TIPS

- Allow lots of time for preparation.
- Identify the activities that need to be undertaken to ensure the data has the correct form.

- Consider whether any transformation undertaken could introduce data quality risks.
- Consider what activities need to be taken to mitigate the quality risks in the data.
- Determine how you will check the quality of the data.
- Ask yourself whether your data set is suitable to test your hypothesis.

BASIC CONCEPTS

In the world of data science, we often talk about data points, averages and trends. However, understanding the variability and uncertainty inherent in data is just as crucial. Measures of spread and probability provide us with the tools to do just that. This chapter will explore these fundamental concepts in an approachable way, helping you to build a solid foundation for more advanced data analysis.

AVERAGES

Mean, median and mode are measures of central tendency that summarise a data set with a single value representing the 'centre', or typical, value. Each measure has its own advantages and disadvantages, making them suitable for different types of data and analysis.

Mean (arithmetic average)

Definition: The mean is calculated by summing all the data points and dividing by the number of points. It is the most commonly used measure of central tendency.

Advantages:

- **Simplicity:** The mean is easy to calculate and understand, making it a straightforward choice for summarising data.
- **Mathematically tractable:** The mean is widely used in statistical analysis and mathematical modelling because it is algebraically manipulable.
- **Uses all data points:** The mean considers all data points in the data set, providing a comprehensive measure of central tendency.

Disadvantages:

- **Sensitivity to outliers:** The mean is highly sensitive to extreme values or outliers, which can skew the result and provide a misleading representation of the data set.
- **Not suitable for skewed distributions:** In skewed distributions, the mean may not accurately reflect the central location of the data.

Median

Definition: The median is the middle value of a data set when it is ordered from smallest to largest. If there is an even number of data points, the median is the average of the two middle values.

Advantages:

- **Robustness to outliers:** The median is not affected by outliers or extreme values, making it a more reliable measure of central tendency for skewed distributions.
- **Reflects the middle of the data:** The median provides a true central value, especially in data sets with significant skewness.

Disadvantages:

- **Ignoring data points:** The median does not consider all data points, potentially overlooking variations in the data set.
- **Less mathematically convenient:** The median is less amenable to algebraic manipulation and is not used as frequently in statistical modelling as the mean.

Mode

Definition: The mode is the value that occurs most frequently in a data set. A data set can be unimodal (one mode), bimodal (two modes) or multimodal (more than two modes).

Advantages:

- **Applicable to categorical data:** The mode is the only measure of central tendency that can be used with nominal data, such as categories or labels.

- **Identifies most common value:** The mode highlights the most common value in a data set, providing insight into the most frequent occurrence.

Disadvantages:

- **Multiplicity:** A data set can have multiple modes or no mode at all if no value repeats, making it less informative in such cases.
- **Less stable:** The mode is less stable than the mean or median, as small changes in the data set can lead to changes in the mode.

Choosing the right measure

- **Mean:** Best used with symmetrical distributions without outliers.
- **Median:** Preferred for skewed distributions or when outliers are present.
- **Mode:** Useful for categorical data or identifying the most common value in a data set.

In practice, it is often beneficial to consider all three measures of central tendency to gain a comprehensive understanding of the data's distribution and characteristics. Each measure

provides unique insights and can complement the others in data analysis.

MEASURES OF SPREAD

Imagine you have a data set of exam scores for a class of students. While the average score gives you an idea of how the class performed overall, it doesn't tell you how varied the scores are. Measures of spread help you to understand the diversity in your data.

Range

The range is the simplest measure of spread. It is calculated by subtracting the smallest value in the data set from the largest value. For example, if the highest exam score is 95 and the lowest is 65, the range is $95 - 65 = 30$. The advantage of using the range as a measure is that it is easy to compute and therefore understand. But be careful, the range can be misleading because it only considers the extreme values and these should be evaluated carefully.

Variance

Variance gives a more comprehensive measure of spread by considering how each data point differs from the mean. To

calculate variance, you take each data point, subtract the mean, square the result and then find the average of these squared differences. A high variance indicates that the data points are spread out over a wide range, while a low variance suggests that they are clustered closely around the mean.

Standard deviation

Standard deviation is simply the square root of the variance. It provides a measure of spread in the same units as the data, making it easier to interpret. In our exam score example, if the standard deviation is 10, you can say that most students' scores are within 10 points of the average.

Interquartile range (IQR)

The interquartile range focuses on the middle 50 per cent of the data, making it a robust measure of spread that is less affected by outliers. It is calculated by subtracting the first quartile (25th percentile) from the third quartile (75th percentile). The 'percentile' is just a technical term for how far through the data the values are, that is, the 25th percentile is 25 per cent of the way through the data (or a quarter of the way). In practice, the IQR tells you how much the central portion of the data varies, providing insight into the core distribution of the data set.

PROBABILITY

In the world of statistics and data science, there is very little we can be certain about. Most of the time we will be interpreting a view of data and using what we see to draw conclusions.

Unfortunately, these views may point towards a trend or pattern, but they will almost never 'prove' things are true (or false). Probability is the language of uncertainty and randomness. It helps us to quantify the likelihood of different outcomes and is a cornerstone of statistical inference in data science.

Basic probability concepts

- **Probability of an event:** Probability is a number between 0 and 1 that indicates how likely an event is to occur. An event with a probability of 0 will never happen, while an event with a probability of 1 is certain.
- **Complementary events:** If the probability of an event happening is $P(A)$, the probability of it not happening is $1 - P(A)$. This is the certainty of something happening (value of 1) minus the probability of it happening. These are complementary events.

Independent and dependent events

- **Independent events:** Two events are independent if the occurrence of one does not affect the probability of the other. For example, flipping a coin and rolling a die are independent events.
- **Dependent events:** Events are dependent if the outcome of one affects the probability of the other. For instance, drawing two cards from a deck without replacement makes the second draw dependent on the first.

Conditional probability

Conditional probability is the probability of an event occurring given that another event has already occurred. It is denoted as $P(A | B)$, meaning the probability of event A happening given that B has happened. This concept is vital in data science for understanding relationships between variables.

Probability distributions

- **Discrete distributions:** These describe the probability of outcomes for discrete random variables, such as the number of heads in coin tosses. Common examples include the binomial and Poisson distributions.
- **Continuous distributions:** These apply to continuous random variables and include distributions such as the

normal distribution, which is crucial in data science for modelling natural phenomena and statistical inference. Common examples of continuous variables are time, speed and temperature.

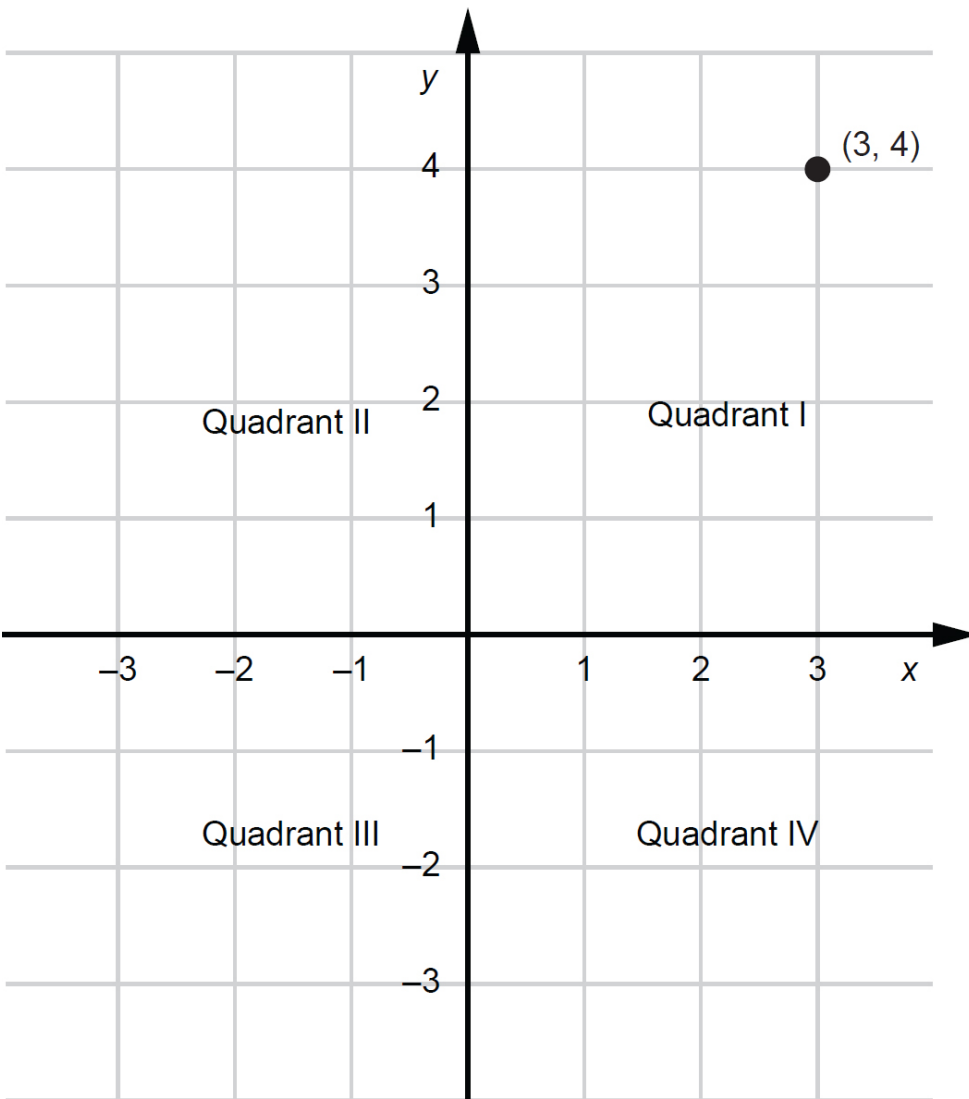
The law of large numbers

The law of large numbers states that as the number of trials increases, the empirical probability (observed frequency) of an event will converge to its theoretical probability. This principle underpins much of statistical reasoning, reinforcing the reliability of large data sets in making predictions. It is like saying if you flip a coin enough times, you will see half of the outcomes being heads and half being tails. The more times you flip the coin, the closer you will get to a perfect half-half. Being able to use vast quantities of data is a relatively new functionality for businesses. The tools now available to a wide range of people give us access to manipulate and analyse far greater amounts of data, and this in turn allows us to build better models that provide stronger pictures of the data.

NAVIGATING THE CARTESIAN PLANE AND MEASURING DISTANCES

In the realm of mathematics and data science, the Cartesian coordinate system serves as a fundamental framework for understanding spatial relationships and measuring distances. A number of statistical models use the concept of distance, so it is important to talk about the Cartesian plane ([Figure 9.1](#)). We will explain how to plot points and how to calculate distances between them using simple language and visual aids to ensure clarity and comprehension.

Figure 9.1 The Cartesian coordinate plane



Understanding the Cartesian coordinate system

The Cartesian coordinate system, named after the French mathematician René Descartes, is a two-dimensional plane for plotting points, lines and shapes. It consists of two perpendicular axes: the horizontal axis (x axis) and the vertical

axis (y axis). These axes intersect at a point called the origin, labelled as (0, 0).

In [Figure 9.1](#), the plane is divided into four quadrants:

- **Quadrant I:** Both x and y are positive.
- **Quadrant II:** x is negative, y is positive.
- **Quadrant III:** Both x and y are negative.
- **Quadrant IV:** x is positive, y is negative.

Each point on the plane is represented by a pair of coordinates (x, y) , indicating its position relative to the origin.

Plotting points

Plotting a point involves identifying its x coordinate and y coordinate and locating its position on the plane. For example, the point (3, 4) is three units to the right of the origin and four units up.

Steps to plot a point:

1. **Locate the x coordinate:** Start at the origin and move horizontally to the x coordinate. Positive x values move right; negative x values move left.

2. **Locate the y coordinate:** From the x coordinate position, move vertically to the y coordinate. Positive y values move up; negative y values move down.
3. **Mark the point:** Place a dot at the intersection of the horizontal and vertical positions. Label the point with its coordinates for clarity.

Measuring distance between points

The distance between two points on the Cartesian plane can be calculated using the distance formula, derived from the Pythagorean theorem. This formula provides a straightforward way to find the straight-line distance (or Euclidean distance) between two points.

Distance formula

For two points, (x_1, y_1) and (x_2, y_2) , the distance d between them is given by:

$$\text{distance } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: calculating distance

Let's calculate the distance between the points A(1, 2) and B(4, 6).

Identify coordinates:

$$x_1 = 1, y_1 = 2$$

$$x_2 = 4, y_2 = 6$$

Apply the formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(4 - 1)^2 + (6 - 2)^2}$$

$$d = \sqrt{(3)^2 + (4)^2}$$

$$d = \sqrt{9 + 16}$$

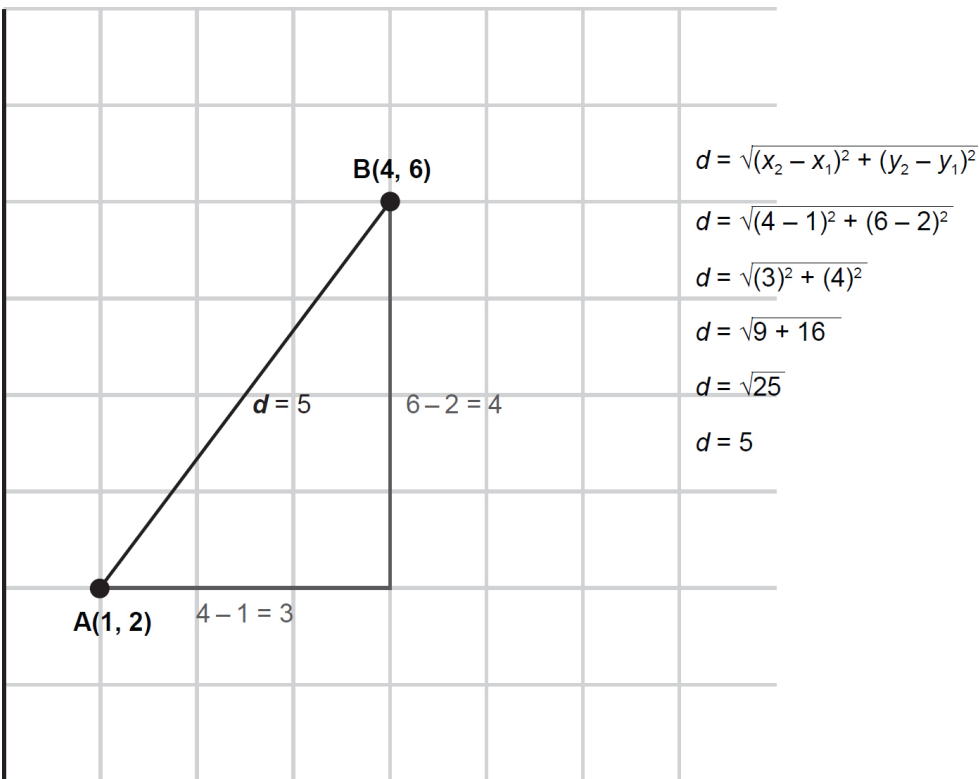
$$d = \sqrt{25}$$

$$d = 5$$

Therefore, the distance between points A(1, 2) and B(4, 6) is five units.

In [Figure 9.2](#), the line connecting points A and B represent the distance we calculated using the distance formula.

Figure 9.2 Visualising distance calculation



CONCLUSION

In summary, measures of spread and probability are essential tools in the data scientist's toolkit. They help us to understand variability and uncertainty, allowing us to make informed decisions based on data. As you continue your journey in data science, these concepts will serve as the building blocks for more advanced analyses, guiding you in uncovering insights and making predictions with confidence.

The Cartesian coordinate system provides a powerful and intuitive way to visualise and analyse spatial relationships. By understanding how to plot points and calculate distances, you can solve a wide range of mathematical and real-world problems. This foundational knowledge will serve you well in your journey through data science and beyond.

MODEL SELECTION

WHICH MODEL TO USE?

Data science is a transformative field that combines statistics, computer science and domain expertise to derive actionable insights from data. This chapter delves into the core concepts and methodologies of data science, such as hypothesis testing, statistical measures, predictive modelling and various analytical techniques, enriched with practical business examples to illustrate their applications. There are lots of potentially confusing terms to decipher when we start talking about statistical techniques, so we will attempt to give you a ‘fact sheet’ on each one, linking into examples of how it can be used.

This book is not intended to teach you the ins and outs of statistical formulae – it is much more important that you can understand how appropriate a model is to your situation and that you have an awareness of the potential pitfalls of using particular data sets with a model. If you can quiz a data scientist on why they have used specific data with a specific model, then you are in a very good place to apply the important fundamentals of data science.

We are therefore going to look at some of the key models, but focus on what they look like and important factors to consider in your data. Each of the following models will contain the basic theory of the statistics involved, but we will add examples of what it looks like in real-life:



Look out for the ‘What does it look like?’ icon.



There will also be know your data pointers, ‘KYD yourself’, to help avoid potential issues.

Choosing the right model depends on the hypothesis and the type of data you have. Some models are designed for prediction, while others are for classification, clustering or association.

You should also be wary of the human influence on model selection – it’s natural to stick to models that you have run before and your tools allow you to build, but this doesn’t always mean they are the best option. A good data scientist will question the model selection every time, but beware the

temptation to take the easiest path of running models that are already in place.

We will start with what data science can be used for and its approaches, before talking about some common terms that are often poorly, or even mistakenly, used. We can then move on to the core categories of modelling that are increasingly used by budding data scientists.

Approaches to data science

To better understand what data science does, we can split it into three main types of analytics: descriptive, predictive and prescriptive analytics. Each type represents a different stage of understanding and decision-making, and together they provide a powerful framework for making informed choices. In the earlier chapters we mentioned the different approaches in data analysis. Let's revisit these terms.

DESCRIPTIVE ANALYTICS: UNDERSTANDING THE PAST

Descriptive analytics focuses on summarising historical data to understand past events. Techniques such as data aggregation, visualisation and basic statistical measures provide insights into trends and patterns. Descriptive analytics is similar to being a detective looking at what has happened: it involves

analysing historical data to understand what has already occurred. The goal is to summarise and interpret data in a way that makes it easy to see trends, patterns and key insights.

Applications:

- **Business reporting:** Creating dashboards to visualise sales performance over time.
- **Customer segmentation:** Analysing demographic data to identify key customer segments for targeted marketing.

Examples of descriptive analytics:

- A retail store analysing last year's sales data to see which products sold the most.
- A website tracking the number of visitors and their behaviour over the past month.
- A hospital examining patient admission rates over the last year to identify seasonal trends.

Tools and techniques:

- Metrics such as averages, totals, percentages and trends.
- Data visualisation tools such as bar charts, pie charts and line graphs.
- Dashboards that summarise KPIs.

Descriptive analytics helps you to paint a clear picture of the past, but it doesn't explain why things happened or what might happen next.

PREDICTIVE ANALYTICS: PEEKING INTO THE FUTURE

Predictive analytics takes things a step further – it's like using a crystal ball powered by statistics and machine learning to predict **what might happen** in the future. By analysing historical data, predictive models identify patterns and relationships that can help to forecast future outcomes.

Think of predictive analytics as answering the question: what's likely to happen next?

Applications:

- **Demand forecasting:** Anticipating product demand based on historical sales and market trends.
- **Risk assessment:** Predicting credit default risks for loan applicants using financial history and behavioural data.

Examples of predictive analytics:

- An e-commerce platform predicting which customers are likely to make a purchase, based on their browsing history.

- A weather app predicting tomorrow's temperature and chance of rain.

Tools and techniques:

- Statistical models such as linear regression and time series forecasting.
- Machine learning algorithms such as decision trees, random forests and neural networks.
- Risk assessments and probability scores.

Predictive analytics gives you a glimpse into the future, but it doesn't tell you what to do with that information.

PRESCRIPTIVE ANALYTICS: DECIDING WHAT TO DO

Prescriptive analytics is the final and most advanced stage. It's not just about understanding the past or predicting the future – it helps you to decide **what actions to take** to achieve the best outcome. Prescriptive models combine data, predictions and optimisations to recommend the best course of action.

Think of prescriptive analytics as answering the question: what should we do about it?

Applications:

- **Pricing strategy:** Optimising prices for maximum profit while considering competitor pricing and customer price sensitivity.
- **Supply chain optimisation:** Recommending inventory levels and logistics strategies to minimise costs and improve efficiency.

Examples of prescriptive analytics:

- A delivery company optimising routes to minimise fuel costs and delivery times.
- A marketing team deciding which promotional emails to send to different customer groups to maximise conversions.
- A hospital determining how to allocate resources during a busy flu season to ensure patient care.

Tools and techniques:

- Optimisation algorithms and simulations.
- Decision trees and what-if analysis.
- Advanced AI systems such as recommendation engines.

Prescriptive analytics empowers decision-makers by providing actionable insights, often using real-time data.

HOW THEY WORK TOGETHER

To see how these three analytic types work together, imagine you're managing a coffee shop:

- **Descriptive analytics:** You analyse last month's sales data and learn that lattes are your best-selling product on weekdays.
- **Predictive analytics:** You use this data to predict that latte sales will likely increase in the winter months when it's colder.
- **Prescriptive analytics:** Based on this prediction, you decide to stock up on latte ingredients, introduce winter-themed latte flavours and adjust staff schedules to handle higher customer volumes.

When combined, descriptive, predictive and prescriptive analytics create a full-circle approach to data-driven decision-making.

UNDERSTANDING THE DATA FOR YOUR MODEL

The very first step in creating a model is to fully understand the data you have (and would like to have). As with many technical processes, this has been given a definition – exploratory data analysis (EDA), as discussed in [Chapter 6](#). During this stage of work, the explore stage of the data analysis lifecycle, it is always

recommended that you visualise your data set – this could take the form of different charts and graphs that will help you see what the data looks like. We will talk more about visualisations in [Chapter 11](#).

There are also some figures that can be calculated to show how the data is spread out, describe the values using an average view and highlight how consistent they are. We mentioned these earlier in the book, but here we will go into how they are calculated. **Central tendency** (mean, median and mode) gives you a single figure that describes the group of numbers, but just as important is understanding how your data is spread out, which is where variance and standard deviation come into play.

Variance: Variance measures how far your data points are from the mean (average). A high variance means your data points are very spread out, while a low variance means they're tightly clustered around the mean. Variance helps to assess data variability, informs risk assessment and forms a basis for other statistical concepts such as standard deviation and covariance.

For a data set with n observations and a mean $\bar{x}$, variance σ^2 is calculated as:

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

We use a squared value to take account of distance away rather than just above or below the mean. It also helps to emphasise the effect of extreme values.

Standard deviation: Standard deviation is just the square root of variance. It's easier to interpret because it's in the same units as your data. A small standard deviation means your data points are close to the mean, while a large standard deviation means they're more spread out. The value of the standard deviation can be directly compared to the data that you are using, for example if the standard deviation of house prices is 2450, then this will be in £s and you can immediately see how close values are to the mean house price.

Standard deviation is calculated using the formula

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

Why these matter

Variance and standard deviation help you to understand the 'spread' of your data. You can see how a distribution with a standard deviation of 9.79 looks in [Figure 10.1](#).

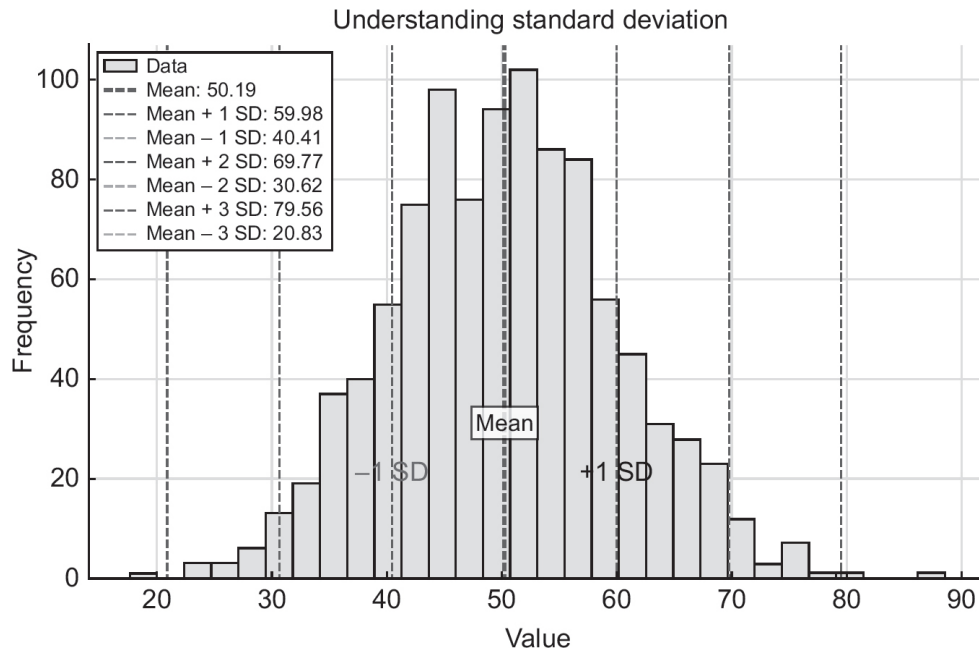
- As an example, if you're analysing house prices and the standard deviation is very high, it means house prices vary a lot in your data set.
- If standard deviation is low, most houses in your data set are similarly priced.

Knowing this can help you to decide whether you need to apply transformations or focus on outliers.

Business example

An investment firm evaluates the standard deviation of portfolio returns to assess volatility and guide asset allocation decisions. If the standard deviation is a large number, they will need to focus on how different the portfolio returns are – do some customers make a lot while others are losing out? If the standard deviation is low, then the investment firm can have some confidence that the returns are reasonably consistent among the portfolios and if they change something then they could consider it may affect a majority of portfolios.

Figure 10.1 Example of standard deviation measures from the mean in a data set



UNDERSTANDING THE HYPOTHESIS

What is a hypothesis?

In data science, a hypothesis is a formal statement proposing a potential explanation or relationship between variables. It serves as a foundation for testing assumptions and drawing conclusions based on data-driven evidence. As a science, there are clear rules on how we should propose – and test – our hypothesis.

Types of hypotheses

- **Null hypothesis (H_0):** This assumes no effect or relationship between variables. It acts as the default assumption tested against the alternative hypothesis.
- **Alternative hypothesis (H_A):** This suggests that an effect or relationship exists. Researchers aim to support this hypothesis through statistical testing.

Formulating a hypothesis

Crafting a hypothesis involves identifying a research question and developing a testable statement. For example, a retail chain might investigate the effect of a new store layout on sales:

- H_0 : The new store layout does not affect sales.
- H_A : The new store layout increases sales.

Hypothesis testing

Hypothesis testing involves statistical methods to determine whether to reject the null hypothesis:

1. **Selecting a significance level (α):** Commonly set at 0.05 (5 per cent), this represents the probability of rejecting the null hypothesis when true (Type I error).
2. **Calculating a test statistic:** This quantifies the difference between observed data and the null hypothesis, using

metrics such as t -score or z -score.

3. **Making a decision:** A critical value is calculated based on a number of factors including the significance level and test type. This is then compared to the test statistic to determine whether the null hypothesis should be rejected or accepted.

Business example

A pharmaceutical company tests whether a new drug reduces blood pressure more effectively than an existing drug. The hypothesis test determines if the observed difference is statistically significant, guiding product development decisions.

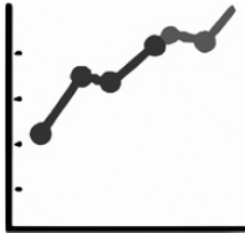
HOW TO CHOOSE BETWEEN DIFFERENT MODELS

When working on a data science project, one of the most important decisions you'll make is choosing the right method to analyse your data. With so many techniques available – clustering, classification, correlation, regression, time series and association – it's easy to feel overwhelmed, but don't worry. Each method serves a specific purpose and is best suited for certain types of problems. By understanding the basics of these techniques, you can confidently decide which one to use.

Let's start with the fundamental differences in the modelling techniques. Choosing between clustering, classification,

correlation, regression, time series and association is not just about the data you are using but also depends on what type of answer you are after.

Correlation: understanding relationships between variables

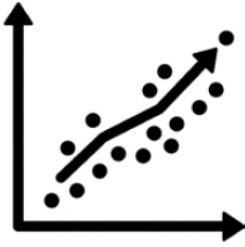


What is it? Correlation measures the strength and direction of the relationship between two variables. It tells you whether two variables move together (positive correlation), move in opposite directions (negative correlation) or have no relationship at all.

Why choose correlation?

Correlation is ideal for exploring relationships between variables. However, remember that correlation does not imply causation – just because two things are related doesn't mean one causes the other.

Regression: predicting numerical values

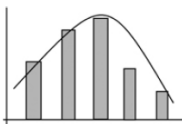


What is it? Regression is a method used to predict a continuous numerical value based on one or more input variables. It's like drawing a best-fit line through your data to make predictions. Use regression when your goal is to predict a number or understand the relationship between variables.

Why choose regression?

Regression is the right choice when your output is a continuous number and you want to quantify relationships between variables.

Time series: analysing data over time



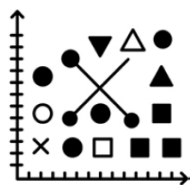
What is it? Time series analysis focuses on data that is collected over time, such as daily stock prices, monthly sales or hourly temperature readings. It's used to identify trends and seasonality and make forecasts. There are several algorithms

that can be used, but when your data is time-dependent and you want to predict future values, one of the time series methods will be the best approach.

Why choose time series?

Time series is perfect for problems where understanding trends over time is crucial. It's especially useful for forecasting future outcomes based on past data.

Classification: sorting data into categories

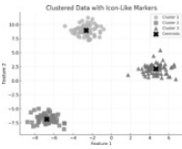


What is it? Classification is a supervised learning technique where you assign data points to predefined categories or classes. It's like teaching a model to recognise patterns based on labelled examples. Use classification when you have a data set with categories or labelled outcomes and you want to predict which category new data points belong to.

Why choose classification?

Classification is your go-to method when you need to make decisions (yes/no, A/B/C) or split things into distinct groups.

Clustering: grouping similar data together



What is it? Use clustering when you want to find natural groupings or segments in your data. You're not looking to make predictions here; instead, you want to explore how your data is organised. It's particularly useful when you don't have predefined labels or categories for your data.

Why choose clustering?

Clustering is perfect for exploratory analysis when you're trying to uncover patterns or hidden structures in your data.

Association: discovering relationships between items



What is it? Association analysis identifies rules or patterns in your data, often used to find relationships between items in transactional data. It's the technique behind market basket analysis used by retailing companies. It will help find patterns

or rules about how items co-occur in a data set, and can be particularly good at looking through large amounts of data to spot these patterns.

Why choose association?

Association is great for uncovering hidden relationships and making recommendations, particularly in huge amounts of retail or e-commerce transactions.

STATISTICAL MODEL: CORRELATION

Understanding correlation

When you want to understand if one thing happens because of another – for example, whether people buy shorts when the temperature goes up – correlation can give you a view of how ‘related’ one thing is to another. There are some questions to ask that require domain understanding, though. Which of the variables depends on the other? Do people buy more knives because there is more knife-crime or is there more knife-crime because more knives are bought?

The output

Correlation quantifies the strength and direction of a linear relationship between two variables, expressed as a correlation coefficient between -1 and $+1$.

- **Positive correlation:** Both variables move in the same direction. As one goes up, so does the other one.
- **Negative correlation:** Variables move in opposite directions. As the temperature drops, more gas is used for heating.
- **Zero correlation:** No linear relationship exists.

The formula

Pearson's correlation coefficient, r , is calculated as:

$$r = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{(n \sum x_i^2 - (\sum x_i)^2) (n \sum y_i^2 - (\sum y_i)^2)}$$

where x_i and y_i are data points for variables x and y , respectively.



What does it look like?

Correlation is one of the most over-used terms when talking about data and statistics. Journalists will quite often use the correlation of two data sets to build a story when it helps to justify that it is someone’s fault for something happening.

Let’s try an example of looking at how time spent on computer games affects student grades.

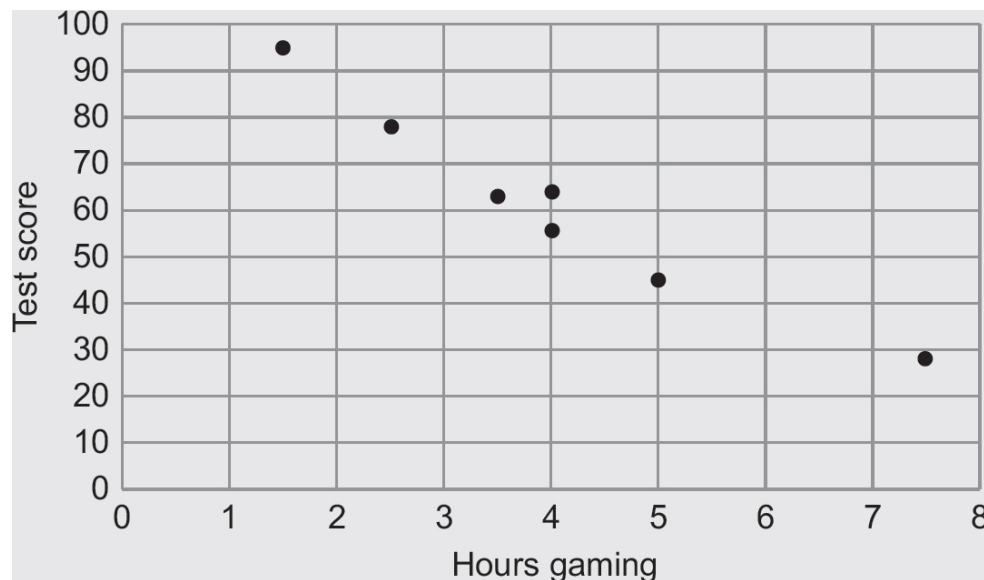
We can gather data on students by asking for their test grades at school and also asking how many hours a week they spend gaming. The data may look like this:

Student	Test score	Hours gaming
A	45	5
B	95	1.5
C	63	3.5
D	28	7.5
E	78	2.5

F	56	4
G	64	4

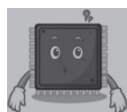
If we plot the data, the chart would give us a very clear picture of a trend, as shown in [Figure 10.2](#).

Figure 10.2 Chart of hours gaming trend



Using the formula for the correlation coefficient, the value for these two sets of data gives us an answer of -0.97 . As the value can only go between $+1$ and -1 , this shows there is a very strong **negative** correlation – the more hours spent gaming, the lower the test score.

It is worth noting that 'Test score' is shown on the vertical, or y , axis and 'Hours gaming' is shown on the horizontal, or x , axis. It is normal practice to use the y axis for the variable that you think changes based on the other one. In this case, we are saying that the test scores are aligned to gaming hours. (You could also argue that 'Hours gaming' is not the cause of the test scores, but that is a debate to be held.)



KYD yourself

Correlation can happen by chance between two completely independent things. For examples, the number of people streaming TV programmes is rising steadily, as is the Earth's average surface temperature. You are likely to be able to show a positive 'correlation' figure between these two things but that does not necessarily mean there is a relationship between the two things. Understanding the context of the data and whether there could be a relationship between them is absolutely critical.

You should also be aware of people using the word 'correlated' without any backup information. Don't be afraid to ask about

the data that has been used and exactly how strong (or weak) the correlation is.

Business example

A marketing team uses correlation analysis to identify the relationship between advertising spend and sales revenue, guiding budget allocation decisions. Are sales impacted by the amount of television adverts broadcast?

STATISTICAL MODEL: LINEAR REGRESSION

What is linear regression?

Imagine you're trying to predict how much your plant will grow based on how much water you give it. Linear regression is like drawing a straight line through a bunch of data points to show the relationship between these two things. It helps us to understand how one thing changes when another thing changes.

Linear regression models the relationship between a continuous target variable and one or more predictor variables by fitting a linear equation to the data.

When to use linear regression:

- **Predicting outcomes:** If you want to forecast what might happen in the future based on past data, for example predicting how many ice creams you'll sell based on the temperature.
- **Understanding relationships:** If you want to see how two things are connected, for example seeing if there's a link between hours of study and exam scores.
- **Finding trends:** If you want to spot patterns in data, for example seeing if sales increase steadily over time.

The output

Linear regression will produce an equation that shows how a variable is affected by one or more other variables. It will give weightings to each of the variables, and you can use this equation to predict other outcomes by inputting different values.

An example is:

$$y = 2.2x + 0.5$$

We can put a value in for the x part of the equation, say 10, and the calculation would give us an answer of the y value being $2.2 \times 10 + 0.5 = 22.5$. We can tell that, when x is 10, y is likely to be 22.5.

The formula

For a single predictor variable, the linear regression equation is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where:

- y is the predicted value.
- β_0 is the intercept.
- β_1 is the slope (coefficient) of the predictor variable x .
- ε is the error term.

Now, what does this actually mean? Well, linear regression models one (or more) things against another. For example, we could be looking at how many mobile phones fail as they get older. If the number of phones failing increases as their age increases, we may be able to predict how many phones will fail after a number of years. In this equation, y is the number of phones, and x the number of years. If we can come up with a pattern, then the other parts of the equation just tell us the scale of the problem and what the starting point is.



What does it look like?

Let's try an example of looking at how mobile phones fail over time.

If we had the following data, we could plot this to see if there are any patterns.

Phone age (years)	Battery capacity (%)
0	100
1	97
2	93
3	90
4	84
5	79

6

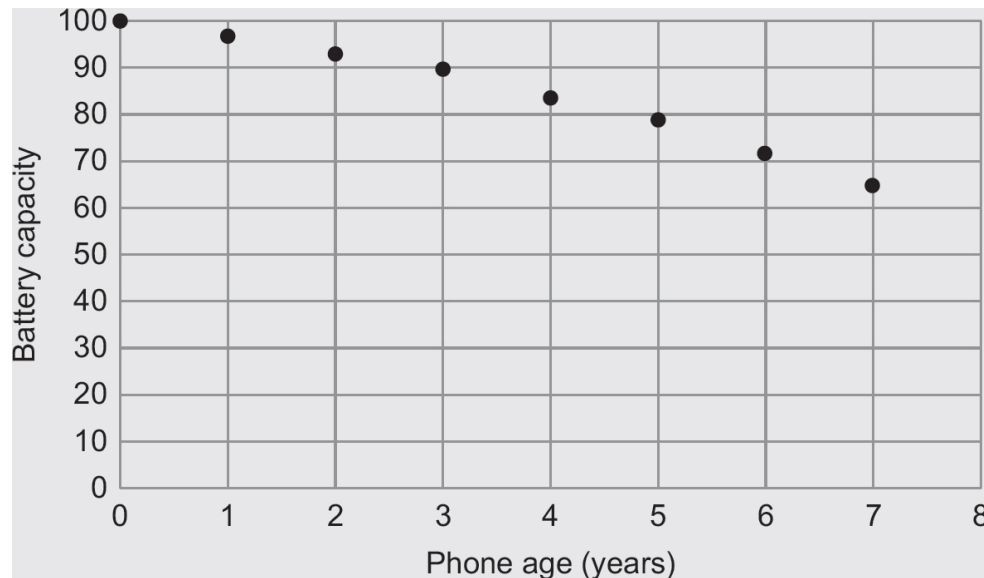
72

7

65

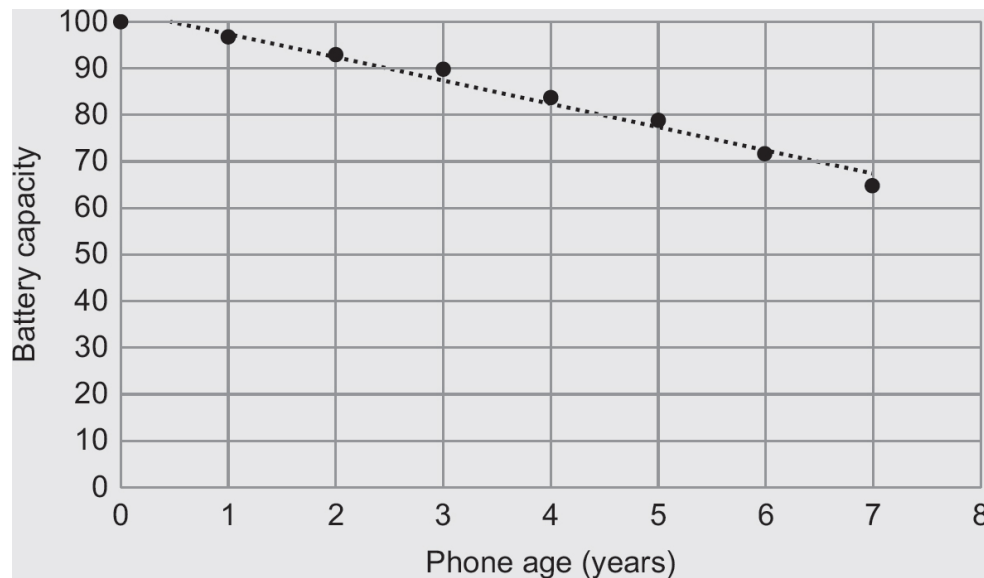
The plot would show a trend for battery capacity reducing with age, as shown in [Figure 10.3](#).

Figure 10.3 Plot chart of battery capacity



A natural next step is to add a line of best fit, showing the direction that the data is headed in. This line of best fit is actually calculated using linear regression ([Figure 10.4](#)).

Figure 10.4 Linear regression of battery capacity



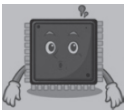
The line has an equation; in this case it is $y = -4.9762x + 102.42$. We can use this equation to ‘try-out’ different values of x (the age of the phone). This could lead to a prediction on what the capacity is after 10 years.

Evaluating model performance

There are some technical measures that can quantify how well the model is performing. It can be evaluated using metrics such as mean absolute error (MAE), mean squared error (MSE) and R-squared. These metrics assess how well the model’s predictions align with actual values.

Example:

- **MAE:** This is the average absolute difference between predicted and actual values.
- **MSE:** This is the average squared difference, highlighting larger errors.
- **R-squared:** This is the proportion of variance in the target variable explained by the model.



KYD yourself

Linear regression is quite a straightforward technique to apply. The key warnings when using it with your data are related to the 'linear' approach. A lot of real-life data does not follow a nice, simple straight-line approach. Imagine the battery capacity prediction: do we really think that it will take 20 years for the battery to be 0 per cent capacity? Or do we think that after 10 years things are going to be behaving quite differently?

Business examples

- **Weather forecasting:** Meteorologists use linear regression to predict tomorrow's temperature based on today's weather patterns.
- **Business decisions:** Companies use it to forecast sales, plan inventory and set prices.

- **Medical research:** Doctors use it to understand the relationship between risk factors and diseases.

PREDICTING VALUE OVER TIME: TIME SERIES ANALYSIS

Time series analysis involves forecasting future values based on historical data collected over time, used in various domains to predict trends and patterns. When we look at data across time, things might be more complicated than a straight line (as we saw in linear regression). Often, we can see movements across different times of year – think about sales around Christmas. Many time series analysis methods take account of this ‘seasonality’ and use mathematical techniques to model them. The equations used in these methods are more complex and beyond the scope of this book, but the modern data scientist would use tools to run the models, so the important priority is understanding what the model shows rather than the calculations themselves.

Imagine you’re tracking your weight over a year. Time series analysis is like looking at how your weight changes month by month, spotting any patterns (like gaining weight during the holidays) and predicting what your weight might be next month. It deals with data that happens over time, like stock prices, weather patterns or website traffic.

When to use time series analysis:

- **Forecasting:** Use for predicting future values, for example predicting next year's sales for a business.
- **Trend spotting:** Use for identifying long-term patterns, for example seeing if the number of tourists visiting a city is increasing or decreasing over time.
- **Seasonality detection:** Use for finding repeating patterns within a certain time period, for example noticing that ice cream sales are always higher in the summer.
- **Anomaly detection:** Use for finding unusual or unexpected events, for example spotting a sudden drop in website traffic.

Components of time series:

- **Trend:** The long-term movement in the data.
- **Seasonality:** Regular, repeating patterns or cycles.
- **Noise:** Random variability not attributed to trend or seasonality.

Techniques for time series forecasting:

- **Exponential smoothing:** This applies exponentially decreasing weights to past observations for forecasting, which means it gives less weight to older values.

- **Autoregressive integrated moving average (ARIMA):** This is a model capturing trend and seasonality in time series data. It uses elements of regression and moving averages.

STATISTICAL MODEL – AUTOREGRESSIVE INTEGRATED MOVING AVERAGE

Understanding time series analysis – ARIMA

ARIMA breaks down into three main components:

- **AR (auto regressive):** This means the model uses past values to predict future ones. You may remember that we used the word ‘regression’ in our linear regression model – it is a similar principle here.
- **I (integrated):** This means the model looks at changes between values instead of the raw data. This helps to deal with trends or data that changes over time.
- **MA (moving average):** This uses a mean average of the data, but, instead of looking at all of the data items, it will use a certain number of past and future values. This forces the model to keep re-evaluating the ‘average’ based on nearby values.

When you want to understand how data moves with time, you have to take account of changes in the trend. The ARIMA model

is adept at using relevant values to predict the next changes by focusing on a moving picture of the data.

The output

The ARIMA model will predict future values, and these tend to be plotted on a graph.

The formula

The simplest ARIMA method is known as the Box–Jenkins model. It is a combination of the linear trend and the moving average.

$$Y_t = \delta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p}$$

$$+ \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$



What does it look like?

Applying ARIMA takes a number of steps, and it will certainly be necessary to use tools to calculate results on any reasonable-

size data set. Let's work through the steps that are taken as an example of what happens during the process.

Using some basic mobile phone sales data, let's try and predict what will happen in Week 8:

Week	Mobile phones sold
1	500
2	520
3	530
4	570
5	580
6	600
7	630

Step 1: Check if the data is stationary

ARIMA works best when your data does not have any trends (also known as seasonality). We are therefore aiming to make the data **stationary**.

To make data stationary (i.e. remove the trend), we calculate the **difference** between consecutive values. This is called ‘differencing’.

Week	Mobile phones sold	First difference (change)
1	500	–
2	520	$520 - 500 = 20$
3	530	$530 - 520 = 10$
4	570	$570 - 530 = 40$
5	580	$580 - 570 = 10$
6	600	$600 - 580 = 20$
7	630	$630 - 600 = 30$

We can see that the differences (20, 10, 40, 10, 20, 30) don't have a clear trend – they are not obviously increasing or decreasing.

Step 2: Identify the ARIMA parameters (p , d , q)

ARIMA has three main parameters:

1. **p (auto regressive order):** This is the number of past values (lags) the model looks at. For example, if $p = 1$, the model uses the previous one week's sales to make predictions.
2. **d (differencing order):** This is the number of times the data is differenced to make it stationary. From Step 1, we differenced the data **once**, so $d = 1$.
3. **q (moving average order):** This is the number of past errors the model uses to make predictions. For example, if $q = 1$, the model uses the error from the last week.

We'll use $p = 1$, $d = 1$ and $q = 1$ for this example (this is a common starting point).

Step 3: Create the ARIMA model

The ARIMA model combines the three components (AR, I, MA) to forecast future values. Here's how it works:

1. **AR:** The model predicts the next value using the previous value(s), for example if sales increased by 20 phones last week, they might increase by 20 again this week.
2. **I:** The model works with the differenced data – changes in sales, not the raw sales.
3. **MA:** The model adjusts its prediction using past errors.

Step 4: Fit the model to the data

Let's fit the ARIMA(1, 1, 1) model to our data:

Formula for ARIMA(1, 1, 1):

Forecast = (AR part)+(MA part)+(differencing part)

AR part: This looks at the previous value.

MA part: This adjusts for past errors.

Differencing part: This accounts for the change between values.

The ARIMA model predicts the next week's sales using the next steps:

Step 5: Forecast for Week 8

1. **Start with the differenced data:** From Week 7, the difference was +30 (change in sales from Week 6 to Week 7).
2. **Use AR:** Look at the previous difference (+30) to predict the next difference. If $p = 1$, the model assumes the next difference might be similar (i.e. close to +30).
3. **Add MA:** Adjust for past errors. Let's say the model underestimated Week 7's sales by 10 phones. The MA component corrects this by taking 10 from the prediction.
4. **Combine everything:** The model predicts the next difference as

$$\text{Next Difference} = (\text{Previous Difference}) + (\text{Error Adjustment})$$

For our example:

$$\text{Next Difference} = 30 + (-10) = 20$$

Undo the differencing: To get the actual forecasted sales for Week 8, we add the predicted difference (+20) to the sales from Week 7 (630):

$$\text{Week 8 Sales} = 630 + 20 = 650$$

Step 6: Forecast for future weeks

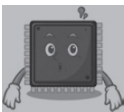
You can repeat this process for additional weeks. For Week 9:

1. Use the predicted difference from Week 8 (+20).
2. Add it to the predicted sales for Week 8 (650):

Week 9 Sales = $650 + 20 = 670$

Final forecast:

- **Week 8:** 650 phones.
- **Week 9:** 670 phones.



KYD yourself

When you are picking the AR order (p), differencing order (d) and the moving average order (q), you will need to know the context of your data. If there is something ‘special’ that happened as a one-off, this can affect how you look at the pattern or trend. A common problem for many time series analyses over the past few years has been the effect of the Covid pandemic – trends that would have followed through the years have been radically changed during a very different time. Knowing whether to include, exclude or deal with this data is important to the potential predictions.

Evaluating time series models

Metrics such as mean absolute percentage error (MAPE) and root mean squared error (RMSE) assess forecast accuracy.

Business example

An airline uses time series analysis to forecast passenger demand, thereby optimising flight schedules and pricing strategies.

CLASSIFICATION

Understanding classification

Classification is a supervised learning technique used to categorise data points into predefined classes or labels. It is widely applied in applications such as spam detection, image recognition and medical diagnosis. Imagine you have a box of mixed fruits. Classification is like sorting those fruits into piles – apples in one pile, oranges in another and so on. In data science, it's about categorising data points into different groups or classes based on their characteristics.

Common classification algorithms

- **Logistic regression:** A simple and interpretable model for binary classification.

- **Decision trees:** A tree-based model that splits data into branches based on feature values.
- **Support vector machines:** A model that finds the optimal hyperplane to separate classes.
- **Random forests:** An ensemble method that combines multiple decision trees for improved accuracy.

Evaluating classification models

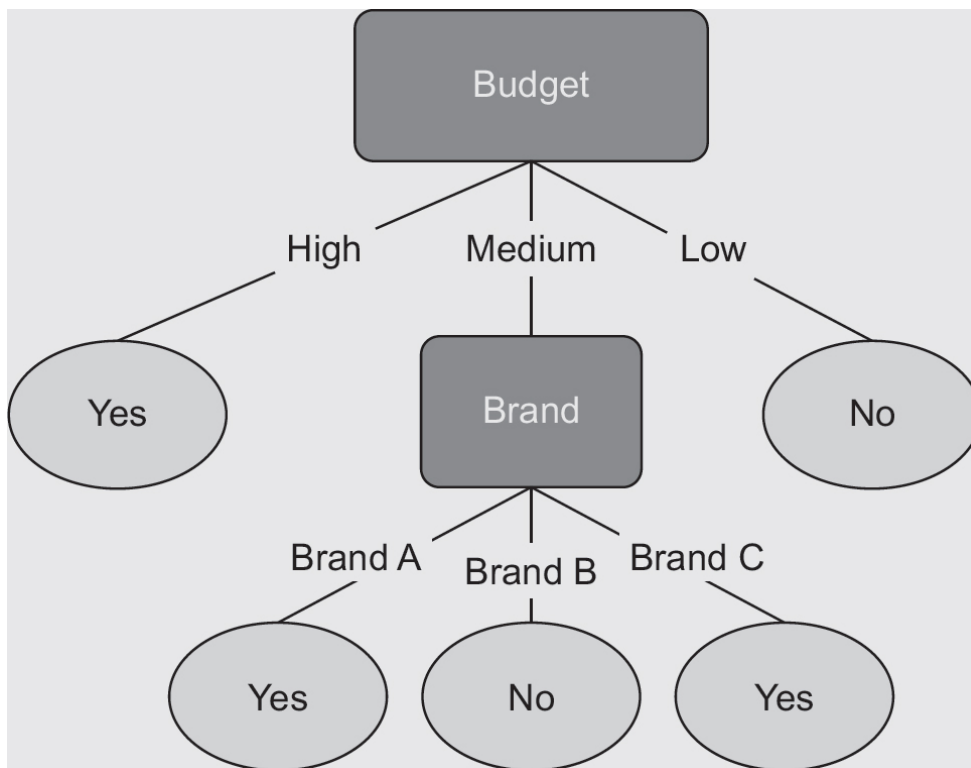
Classification models are evaluated using metrics such as accuracy, precision, recall and F1-score. Confusion matrices provide insights into model performance by showing the true positive, false positive, true negative and false negative rates.

With such a range of different algorithms available, it is worth understanding the nuances of each one. For the sake of an example, we will work through one of the more common approaches: decision trees.

The output

STATISTICAL MODEL – DECISION TREES

Figure 10.5 Decision tree for budget and brand



A decision tree model will show outcomes from various ‘decisions’ in the form of a tree. The decision tree in [Figure 10.5](#) show whether someone will buy our new phone based on their current phone brand and budget. The nodes give the outcome based on these factors.

The formula

At first glance, decision trees might seem like they’re just branching diagrams to make decisions, but behind the scenes, there’s some maths involved to figure out **how to split the data**

at each step. A decision tree works by splitting the data into smaller groups at each step (called **nodes**) to make the groups as 'pure' as possible. **Pure** means that most of the items in a group belong to the same category (e.g. all 'Yes' or all 'No'). There are different ways of calculating how pure things are; one of the common methods is using **entropy** and **information gain**. Entropy measures uncertainty, and information gain tells us how much we reduce uncertainty by splitting:

$$H = - \sum_c p(c) \log_2 p(c)$$

To determine the best split, decision trees rely on a mathematical concept called information gain.

H_{attr} is the entropy

$$H_{\text{attr}} = - \sum_v p(v) \sum_c p(c|v) \log_2 p(c|v)$$

where c is the outcome, v is the attribute and p is the probability of it occurring.

$$\text{InfoGain}_{\text{attr}} = H - H_{\text{attr}}$$

$\text{InfoGain} = H$: This means that the attribute and outcome are correlated and H is on a scale of 0 to 1, so InfoGain can be up to

1 too.

When InfoGain = 0, it means that there is no link between the attribute and outcome.



What does it look like?

We can gather data on a customer's previous phone and see if they then bought our phone.

Customer	Budget	Brand preference	Screen size	Camera quality	Buy phone?
1	High	A	Large	High	Yes
2	Low	B	Small	Low	No
3	Medium	A	Medium	Medium	Yes
4	High	C	Large	High	Yes
5	Low	A	Small	Low	No
6	Medium	B	Medium	Medium	No
7	High	A	Large	Low	Yes
8	Low	C	Small	High	No
9	Medium	C	Medium	Medium	Yes
10	Low	B	Small	Low	No

To create a decision tree, we'll decide which feature splits the data best at each step. Investigating the data set we can see:

Feature 1: Budget

- **High budget:** Mostly 'Yes' outcomes.
- **Low budget:** Mostly 'No' outcomes.
- **Medium budget:** Mixed outcomes.

Splitting by **Budget** seems promising because it separates 'Yes' and 'No' outcomes quite well. Budget can be used as our **root node**.

We can then branch out based on the values of the Budget feature:

Branch 1: Budget = high

The data for High Budget customers looks like this:

Brand preference	Screen size	Camera quality	Buy phone?
A	Large	High	Yes
C	Large	High	Yes
A	Large	Low	Yes

All outcomes are 'Yes', so this branch ends with **Buy phone = Yes**.

Branch 2: Budget = low

The data for Low Budget customers looks like this:

Brand preference	Screen size	Camera quality	Buy phone?
B	Small	Low	No
A	Small	Low	No
C	Small	High	No
B	Small	Low	No

All outcomes are 'No', so this branch ends with **Buy phone = No**.

Branch 3: Budget = medium

The data for Medium Budget customers looks like this:

Brand preference	Screen size	Camera quality	Buy phone?
A	Medium	Medium	Yes
B	Medium	Medium	No
C	Medium	Medium	Yes

For Medium Budget customers, the best split comes from Brand preference:

- **Brand A and Brand C** → Mostly 'Yes'.
- **Brand B** → Mostly 'No'.

This gives us enough information to create a visualisation of the decision tree, as shown in [Figure 10.5](#).



KYD yourself

Decision trees use a training set of data to work out the most likely outcomes. Anytime you use a set of data to train a model, you should know how good the data is. Any bias or trends in the data will be a fundamental inclusion in the predictions that a decision tree produces.

Business examples

- **Online shopping:** Websites use classification to recommend products to you based on your past purchases and browsing history.
- **Fraud detection:** Banks use it to identify fraudulent transactions in credit card statements.
- **Spam filters:** Email providers use it to filter out unwanted emails.

CLUSTERING

Understanding clustering

Clustering is an unsupervised learning technique that groups similar data points together based on their features, used for EDA, pattern recognition and data compression. Imagine you have a box of mixed chocolates. Clustering is like sorting those chocolates into piles based on their similarities, such as colour, shape or filling. In data science, it's about grouping data points together that are like each other.

Common clustering algorithms

- **k-means:** A partitioning method that assigns data points to clusters based on their proximity to cluster centroids.
- **Hierarchical clustering:** A method that builds a hierarchy of clusters using either an agglomerative or divisive approach.
- **Density-based spatial clustering of applications with noise:** A method that identifies clusters based on the density of data points, allowing for the detection of arbitrarily shaped clusters (sometimes abbreviated to DBSCAN).

When to use clustering:

- **Customer segmentation:** Grouping customers with similar buying habits to tailor marketing campaigns.
- **Image analysis:** Grouping similar images together, like photos of cats or landscapes.

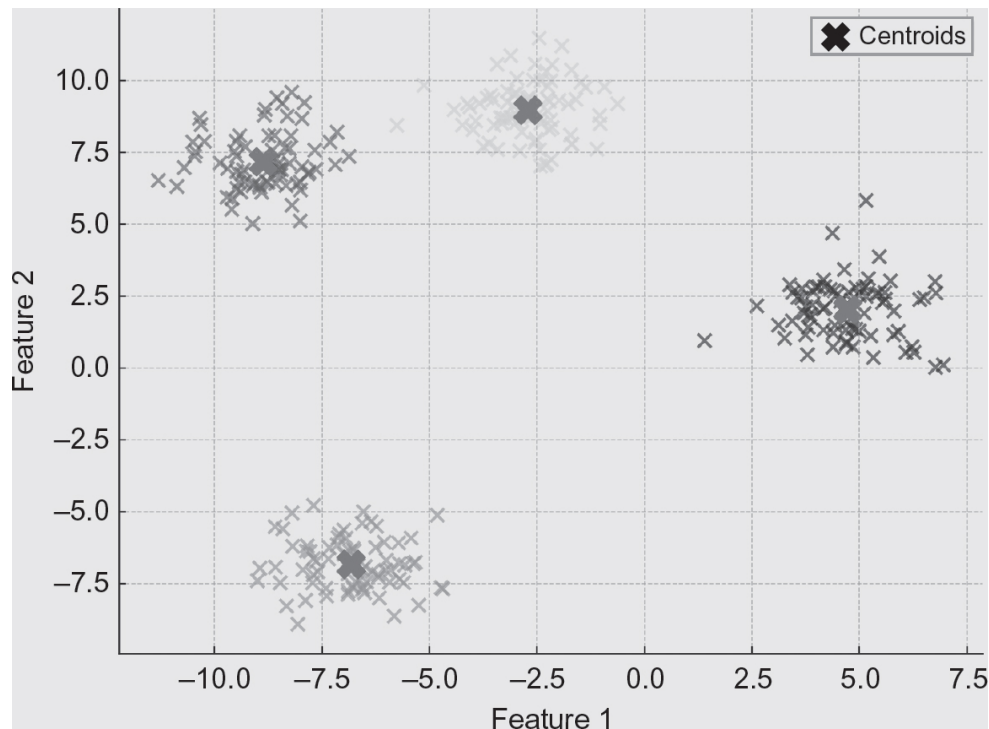
- **Anomaly detection:** Finding unusual data points that don't fit into any group, which could indicate fraud or system failures.
- **Document clustering:** Grouping similar documents together, such as news articles on the same topic.

Evaluating clustering results

Clustering results are evaluated using metrics such as silhouette score, Davies-Bouldin index and within-cluster sum of squares. Visualisation techniques such as scatter plots and dendrograms also aid in assessing cluster quality.

The output

Figure 10.6 *k*-means clustering example



k-means clustering will produce a graphical visualisation showing data points grouped around a given number of central points (centroids). The example in [Figure 10.6](#) has four clusters with four centroids, showing that the data can be clearly grouped into four separate groups.

The formula

The purpose of *k*-means clustering is to minimise the total of **all** of the distances between a data point and a centroid. We do this by changing the centroids around, and keep on trying until we get to a point that gives us the smallest total. Therefore, the

formula we use, related to adding up all the distance between points and centroids, is:

$$J = \sum_{j=1}^n \sum_{i=1}^c \left\| x_i - d_j \right\|^2$$

where n is the number of clusters, c the number of data points and the right half is measuring the distance between each data point and a centroid.



What does it look like?

k -means clustering involves a lot of calculations, as we have to check all data points against different centroids. To show a very simple example of how this works, let's try a small data set.

Using the following data about six students for their **height** (in cm) and **weight** (in kg), we can try and group them into two clusters:

Student	Height (cm)	Weight (kg)
---------	-------------	-------------

A	160	55
B	170	65
C	165	60
D	180	80
E	175	75
F	190	90

Step 1: What is the '*k*' in *k*-means?

The *k* in *k*-means is the number of clusters (groups) you want to create. Here, we will use $k = 2$ for two groups of students based on their height and weight.

Step 2: Initialise cluster centres

To begin, we randomly pick two points as 'starting cluster centres'. Let's randomly choose:

- **Cluster centre 1 (C_1):** (160, 55) → Same as Student A.
- **Cluster centre 2 (C_2):** (190, 90) → Same as Student F.

These centres are just starting points. They will move as we calculate better cluster groupings.

Step 3: Assign each student to the nearest cluster

Now, for each student, we calculate which cluster centre is closer – C_1 or C_2 – based on the distance formula we described in [Chapter 9](#).

Formula for Euclidean distance

Let's calculate the distance for each student to both cluster centres:

Student	Height (cm)	Weight (kg)	Distance to C_1 (160, 55)	Distance to C_2 (190, 90)	Closest cluster
A	160	55	0	50.00	C_1
B	170	65	14.14	36.06	C_1
C	165	60	7.07	42.72	C_1
D	180	80	36.06	14.14	C_2
E	175	75	29.15	18.03	C_2
F	190	90	50.00	0	C_2

This gives us the results:

Cluster 1 (C_1): Students A, B, C

Cluster 2 (C_2): Students D, E, F

Step 4: Recalculate cluster centres

After assigning students to clusters, we recalculate the centre of each cluster (**centroid**). To find the new centroid:

- Take the average of the **Height** and **Weight** for all the students in the cluster.

New centroid for Cluster 1 (C_1):

Students in Cluster 1: A (160, 55), B (170, 65), C (165, 60)

$$\text{New Height} = \frac{160+170+165}{3} = 165$$

$$\text{New Weight} = \frac{55+65+60}{3} = 60$$

New $C_1 = (165, 60)$

New centroid for Cluster 2 (C_2):

Students in Cluster 2: D (180, 80), E (175, 75), F (190, 90)

$$\text{New Height} = \frac{180+175+190}{3} = 181.67$$

$$\text{New Weight} = \frac{80+75+90}{3} = 81.67$$

New $C_2 = (181.67, 81.67)$

Step 5: Repeat assignment and recalculation

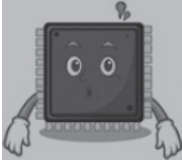
Now we repeat **Step 3** (assign students to the nearest cluster) and **Step 4** (recalculate cluster centres) until the cluster centres stop moving significantly (this is called **convergence**).

Let's go through one more iteration of assigning students to clusters:

Student	Height (cm)	Weight (kg)	Distance to new C_1 (165, 60)	Distance to new C_2 (181.67, 81.67)	Closest cluster
A	160	55	7.07	36.53	C_1
B	170	65	7.07	23.28	C_1
C	165	60	0	27.21	C_1
D	180	80	25.0	2.36	C_2
E	175	75	18.03	8.33	C_2
F	190	90	36.06	12.5	C_2

- **Cluster 1 (C_1):** Students A, B, C
- **Cluster 2 (C_2):** Students D, E, F

Since the clusters haven't changed, we've reached **convergence**, and the algorithm stops.



KYD yourself

When you are clustering data, the first thing to consider is how many groups/clusters would be appropriate. There may be cases where the data will say that five groups would make the most sense, but in real life this number of groups would not be workable.

As we are looking at ‘distances’, you should also be aware of clustering using geographical data. An example would be looking into opening a new shop and trying to choose the best spot – there are going to be many factors apart from distance that can affect the clustering of customers.

Business examples

Customer segmentation: A telecommunications company uses clustering to segment customers based on usage patterns, enabling personalised marketing campaigns and service offerings.

- **Social media:** Social media platforms use clustering to recommend friends or groups to you based on your interests

and connections.

- **Search engines:** Search engines use clustering to group similar webpages together in search results.
- **Market research:** Companies use clustering to identify different customer segments and understand their needs.

Association

Understanding association rule mining

Association rule mining identifies relationships and patterns within data sets, often used in market basket analysis to discover associations between products purchased together. Imagine you're at a grocery store – association rule mining is like noticing that people who buy water often also buy crackers. It helps us to discover interesting relationships and patterns between different items in a data set.

Apriori algorithm

The Apriori algorithm is a popular method for mining frequent item sets and discovering association rules. It uses a bottom-up approach, generating candidate item sets and pruning those that do not meet a minimum **support** threshold.

STATISTICAL MODEL – APRIORI ASSOCIATION RULES

When to use association rule mining:

- **Market basket analysis:** Understanding which products are frequently bought together to optimise product placement and create targeted promotions.
- **Web usage mining:** Analysing website traffic to understand user behaviour and improve website design.
- **Medical research:** Identifying potential risk factors for diseases by analysing patient data.
- **Fraud detection:** Discovering patterns of fraudulent transactions.

The output

Association rules do not **predict** outputs, but are very good at spotting patterns in very large data sets that a human would struggle to see just by looking.

The model outputs a few values that quantify associations between items.

Key concepts:

- **Support:** This is the proportion of transactions that contain a particular item set.

- **Confidence:** This is the likelihood that a rule is true for a transaction (two or more items) that already contains the first item.
- **Lift:** This is the ratio of the observed support to the expected support if the items were independent.

The formula

Support = the percentage of transactions that contain an item (e.g. item 1).

Confidence = the percentage of transactions that contain item 1 and that also contain item 2.

$$\text{Lift} = \frac{\text{Support for Item 1 and Item 2}}{\text{Support for Item 1} \times \text{Support for Item 2}}$$



What does it look like?

Imagine you own a mobile phone shop and want to understand what products customers typically buy together. This could help you to place different products together and increase sales even more.

Let's look at five customer transactions:

Transaction 1: {battery pack, case, screen protector}

Transaction 2: {battery pack, case}

Transaction 3: {case, screen protector}

Transaction 4: {battery pack, case, screen protector}

Transaction 5: {battery pack, screen protector}

- **Support:** How often items appear together as a percentage of total transactions.
- **Confidence:** How likely it is that item 2 is purchased when item 1 is purchased.

Let's find some rules with minimum support of 40 per cent (appears in two out of five transactions):

Step 1: Find frequent item sets

- {battery pack} appears in 4/5 transactions = 80 per cent.
- {case} appears in 4/5 transactions = 80 per cent.
- {screen protector} appears in 4/5 transactions = 80 per cent.
- {battery pack, case} appears in 3/5 transactions = 60 per cent.

- {battery pack, screen protector} appears in 3/5 transactions = 60 per cent.
- {case, screen protector} appears in 3/5 transactions = 60 per cent.
- {battery pack, case, screen protector} appears in 2/5 transactions = 40 per cent.

Step 2: Generate rules and calculate confidence

For {battery pack, case}:

- If customer buys battery pack, they buy a case:
- Confidence = $3/4 = 75$ per cent.
- If customer buys a case, they buy a battery pack:
- Confidence = $3/4 = 75$ per cent.

Insight: If someone buys battery pack, there's a 75 per cent chance they'll also buy a case, so you might want to place these items near each other.

Lift measures how much more likely items are to be bought together compared to if they were bought independently. A lift > 1 means items complement each other, while < 1 suggests they substitute each other.

Let's calculate everything for the rule 'battery pack → case':

1. Support

1. $\text{Support}(\text{battery pack}) = 4/5 = 0.8$.
2. $\text{Support}(\text{case}) = 4/5 = 0.8$.
3. $\text{Support}(\text{battery pack, case}) = 3/5 = 0.6$.

2. Confidence

1. $\text{Confidence} = \text{Support}(\text{battery pack, case}) / \text{Support}(\text{battery pack})$.
2. $\text{Confidence} = 0.6 / 0.8 = 0.75$ (75 per cent).
3. Interpretation: When people buy battery pack, 75 per cent also buy case.

3. Lift

1. $\text{Lift} = \text{Support}(\text{battery pack, case}) / \text{Support}(\text{case}) \times \text{Support}(\text{battery pack})$
2. $\text{Lift} = 0.6 / 0.64 = 0.9375$

Key points:

1. **Lift = 1:** Items are independent.

2. **Lift > 1:** Items are complementary.

Example: chips and salsa are quite commonly bought together and might have lift = 3.0.

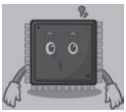
3. **Lift < 1:** Items are substitutes.

Example: Coke and Pepsi might have lift = 0.5.

Some things that you can assume from the results you generate (known as pruning in Apriori):

- If {Item 1} isn't frequent, then {Item 1, Item 2} can't be frequent,
- If {Item 1, Item 2} isn't frequent, then {Item 1, Item 2, Item 3} can't be frequent.

These properties help to reduce the number of combinations to check.



KYD yourself

A common mistake in association rule mining is ignoring the domain knowledge and context of the data. Experts in what is actually happening in a business can help to select relevant items for the analysis and highlight rules that are redundant, allowing you to focus on the more meaningful patterns. For example, if you know that everyone always buys a case with a new phone, then it could be worth ignoring this to focus on patterns that have not been realised. You should also take

warnings for other factors that might affect results – seasonality, location and demographics of the customers can change buying habits significantly.

Business examples

Product combinations: A grocery store uses association rule mining to identify frequently purchased product combinations, informing cross-promotional strategies and store layout designs.

- **Retail:** A grocery store might place bread and milk next to each other on the shelves based on the discovery that they are often bought together.
- **E-commerce:** Online stores might recommend products to you based on what other customers who bought the same items as you also purchased.
- **Web design:** Website designers might place related content or advertisements together based on user browsing behaviour.

VISUALISATIONS

CONSIDERATIONS

Telling your story is a well-proven tactic when trying to get a message across to others, but using the power of statistics can also enable you to create visualisations that leave lasting impressions on your audience. Knowing what people will ‘enjoy’ looking at, and ensuring they ‘see’ the right conclusions, are a critical part of working with data – numbers can transform into stories, and insights leap off the page. In this chapter, we’ll explore the art and science of creating compelling visualisations that not only convey information but also engage your audience. We’ll cover key considerations, discuss user experience (UX) and accessibility and stress the importance of clear labelling.

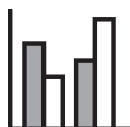
Know your audience

Before you even start choosing or designing your visualisation, it’s crucial to understand who will be viewing it. Are they data-savvy analysts or business executives looking for quick insights? Knowing your audience helps you to tailor the complexity and style of your visualisations to meet their needs. Analysts might appreciate detailed visualisations with

interactive elements. Executives often prefer high-level overviews with clear, concise messages.

Choose the right chart type

Selecting the appropriate chart type is like choosing the right tool for a job. Different types of data and messages call for different visual representations.



Bar charts: These are great for comparing quantities across categories.



Line charts: They are ideal for showing trends over time.



Pie charts: These are useful for displaying percentages or the share of a whole. Be careful, though, because pie charts are often over-relied upon and used for the wrong purposes. Use sparingly.



Scatter plots: These are perfect for illustrating relationships between variables.

Simplify, simplify, simplify

When it comes to visualisations, less is often more. Avoid clutter by focusing on the most important data points and removing unnecessary elements. Simplicity not only enhances clarity but also makes your visualisations more aesthetically pleasing.

Use colour wisely

Colour can be a powerful tool in your visualisation arsenal, but it must be used thoughtfully. Stick to a cohesive colour scheme and use colour to highlight key information rather than overwhelm the viewer.

- **Consistent colours:** Maintain consistency across visualisations for easy interpretation. Lots of organisations have a ‘theme’ or ‘palette’ of colours to use.
- **Contrasting colours:** Use contrast to draw attention to important data points.

USER EXPERIENCE IN DATA VISUALISATION

Intuitive design

An effective visualisation should be intuitive, allowing users to grasp the message quickly. Arrange elements logically and ensure that the flow of information is natural.

- **Hierarchy:** Organise elements to guide the viewer's eye to the most important information first.
- **White space:** Use white space strategically to separate elements and improve readability.

Interactivity

Interactive elements can enhance user experience by allowing viewers to explore data at their own pace. Consider incorporating features like tooltips, filters and zooming to provide a richer experience ([Figures 11.1](#) and [11.2](#)).

Figure 11.1 Tooltips: offer additional information when hovering over data points

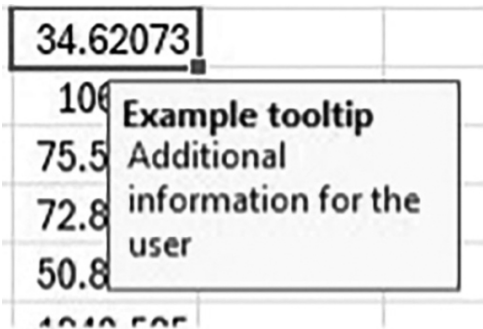
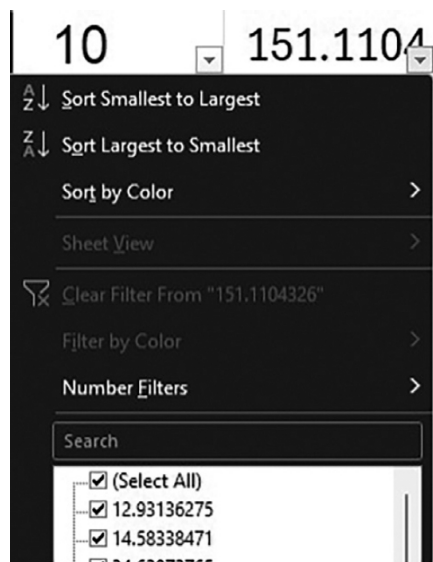


Figure 11.2 Filters: allow users to customise views and focus on specific data subsets



Responsive design

In today's multi-device world, ensure that your visualisations are responsive and look great on all screen sizes, from desktops

to mobile phones.

- **Scalable graphics:** Use vector graphics or responsive frameworks to ensure clarity on any device.
- **Mobile-friendly layouts:** Design with touch interactions in mind for smaller screens.

Accessibility in data visualisation

- **Colour blindness considerations:** Approximately 8 per cent of men and 0.5 per cent of women have colour vision deficiency (American Optometric Association, 2024). To make your visualisations accessible, consider using colour-blind friendly palettes and avoid relying solely on colour to convey information.
- **Colour-blind palettes:** There are many tools available in modern applications that include palettes designed for accessibility.
- **Alternative markings:** Use patterns or shapes in addition to colour to differentiate data points.

Text and font size

Ensure that all text in your visualisations is legible. Use clear sans-serif fonts and avoid text sizes that are too small.

- **Minimum font size:** Keep font sizes above 12 points for readability.
- **Contrast:** Ensure sufficient contrast between text and background colours.

Descriptive text and alt text

Incorporate descriptive text and alternative text (alt text) for users who rely on screen readers. This ensures that your visualisations are accessible to individuals with visual impairments.

- **Descriptive labels:** Provide context and explanations for charts and graphs.

Labelling in data visuals

Use clear and concise labels. Labels are the unsung heroes of data visualisations. They provide essential context and help users to understand what they're looking at.

- **Axis labels:** Clearly label axes to indicate what data is being represented.
- **Data labels:** Add labels to data points when necessary for clarity.

- **Avoiding overlapping labels:** Ensure that labels do not overlap or clutter the visualisation. This can make the data difficult to interpret and detract from the overall design.
- **Smart placement:** Use algorithms to position labels intelligently, avoiding overlaps.
- **Legend use:** When space is limited, use a legend to explain colours or symbols.

Consistency and formatting

Maintain consistent formatting for labels throughout your visualisations. This includes font choice, size and alignment.

- **Uniform style:** Apply a consistent style to all labels for a cohesive look.
- **Alignment:** Align labels neatly to enhance readability and aesthetic appeal.

BAR CHART

What is a bar chart?

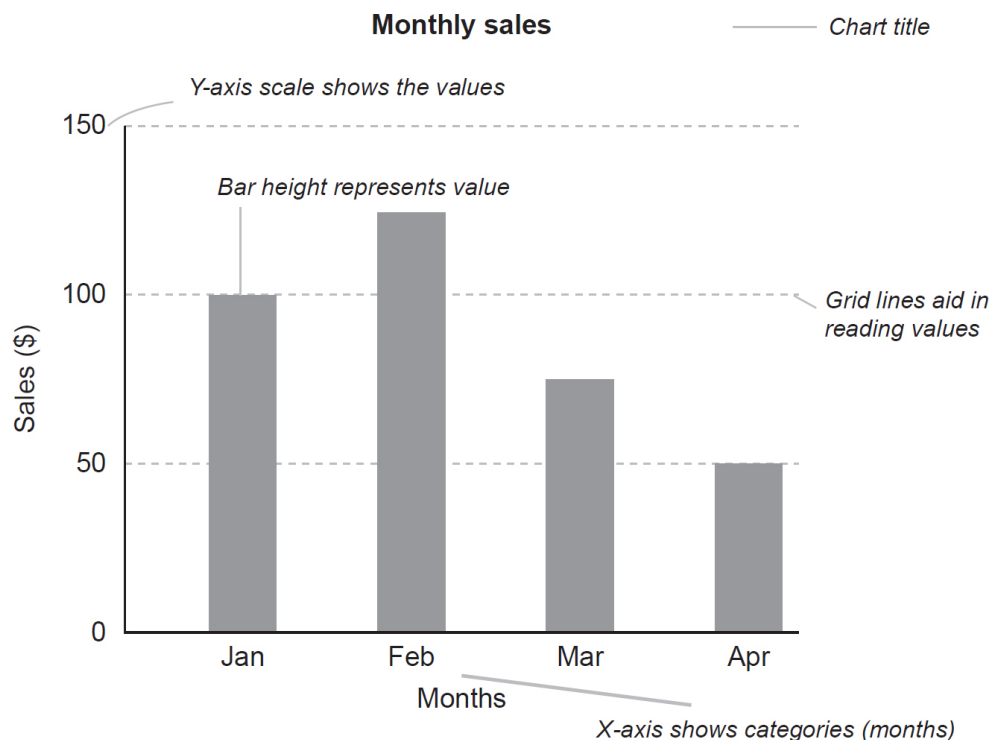
Bar charts are one of the most common and versatile types of visualisations, used to compare quantities across categories. They display rectangular bars with lengths proportional to the values they represent ([Figure 11.3](#)).

When to use bar charts

- **Comparing categories:** They are ideal for showing comparisons between discrete categories, such as sales across different regions.
- **Displaying counts:** They are useful when illustrating the frequency of different values in a data set.

Remember: Keep your bar chart simple by limiting the number of categories and using consistent colour schemes to avoid overwhelming your audience.

Figure 11.3 Example monthly sales bar chart



HISTOGRAM

What is a histogram?

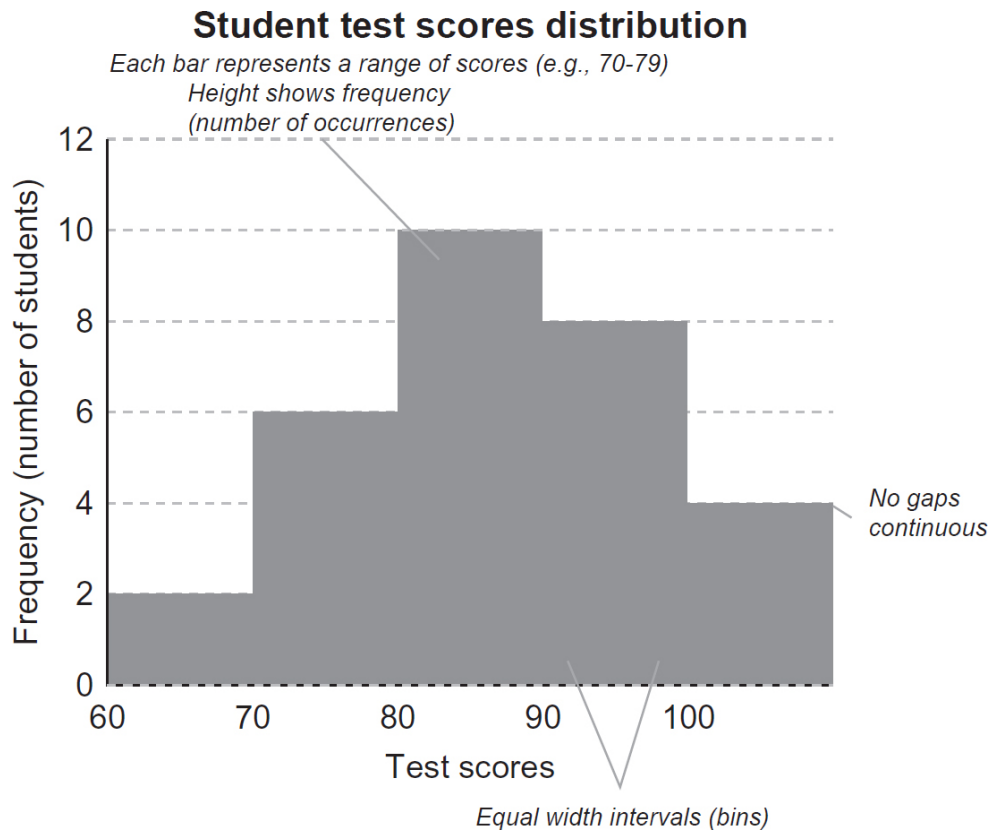
Histograms are similar to bar charts, but are specifically used to display the distribution of a continuous data set. They divide data into bins or intervals, showing how often data points fall within each interval ([Figure 11.4](#)). The shape of the histogram gives a view of how the data is distributed across the range of values.

When to use histograms

- **Understanding distribution:** They are perfect for visualising the distribution of a continuous variable, such as exam scores or ages.
- **Identifying skewness:** They are useful for spotting skewness or outliers in your data.

Remember: Choose an appropriate bin size to ensure your histogram accurately represents the data distribution without losing detail.

Figure 11.4 Example histogram

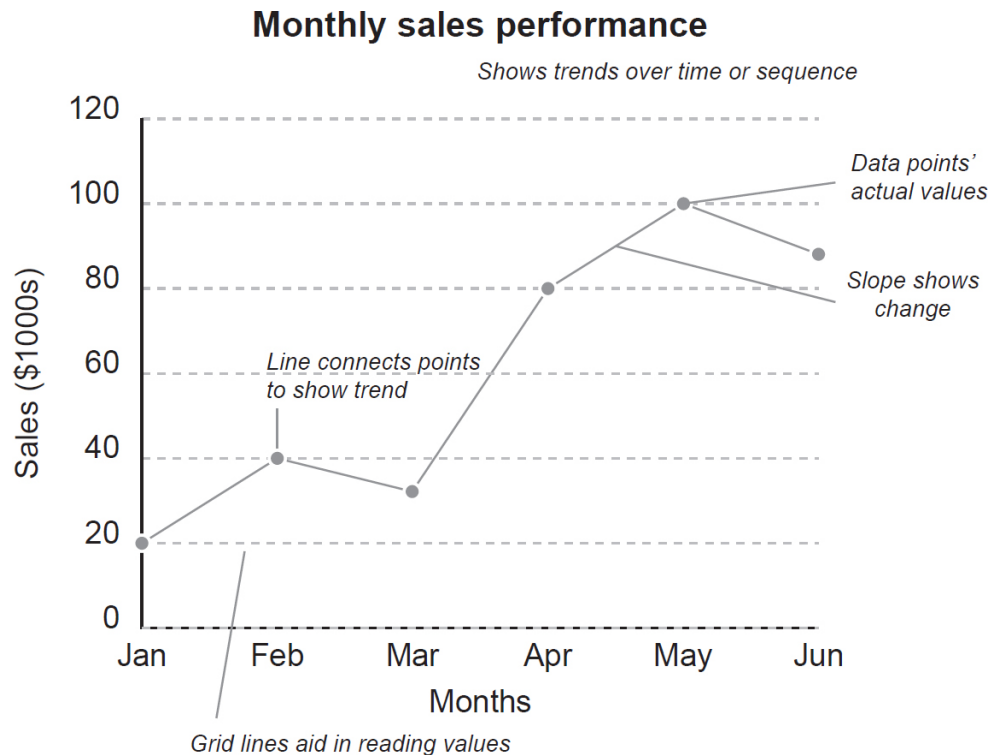


LINE GRAPH

What is a line graph?

Line graphs use points connected by lines to represent data points over time. They're perfect for visualising trends in continuous data ([Figure 11.5](#)).

Figure 11.5 Example line graph



When to use line graphs

- **Tracking changes over time:** They are ideal for showing how a variable changes over time, such as stock prices or temperature.
- **Comparing multiple series:** They are useful for comparing multiple data series on the same graph.

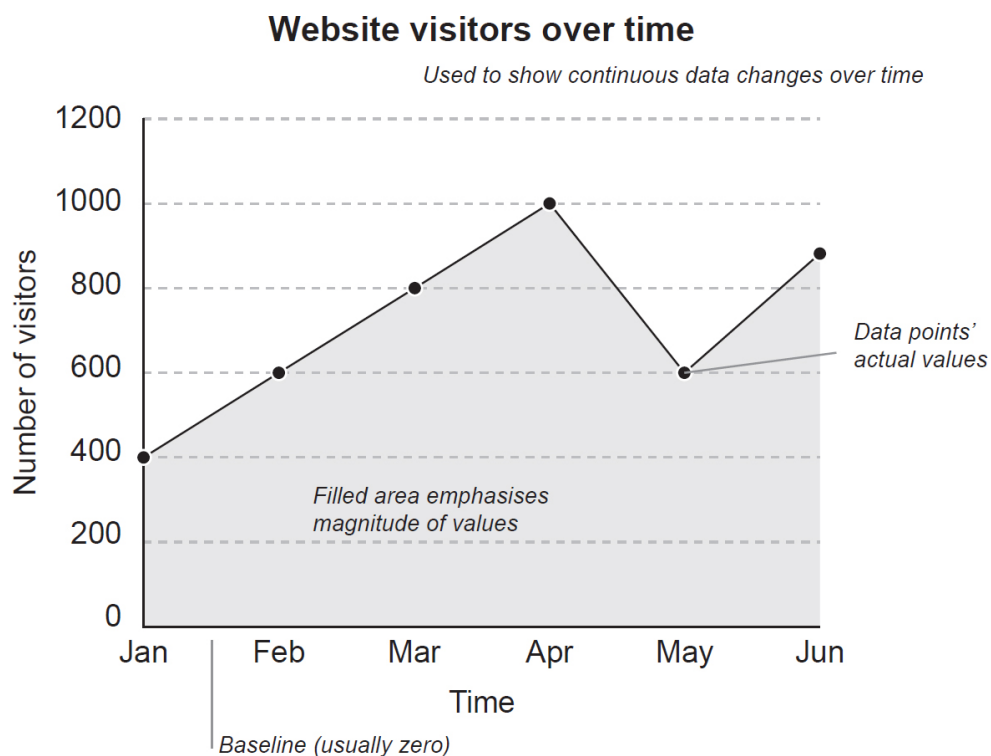
Remember: Use different line styles or colours to distinguish between multiple lines on the same graph, ensuring clarity.

AREA CHART

What is an area chart?

Area charts are like line charts, but the area beneath the line is filled in, emphasising the magnitude of change over time. They're great for visualising cumulative data ([Figure 11.6](#)).

Figure 11.6 Example area chart



When to use area charts

- **Showing trends over time:** They are useful for illustrating trends and comparing multiple quantities over time.
- **Visualising part-to-whole relationships:** They are ideal for showing how individual parts contribute to a whole over

time.

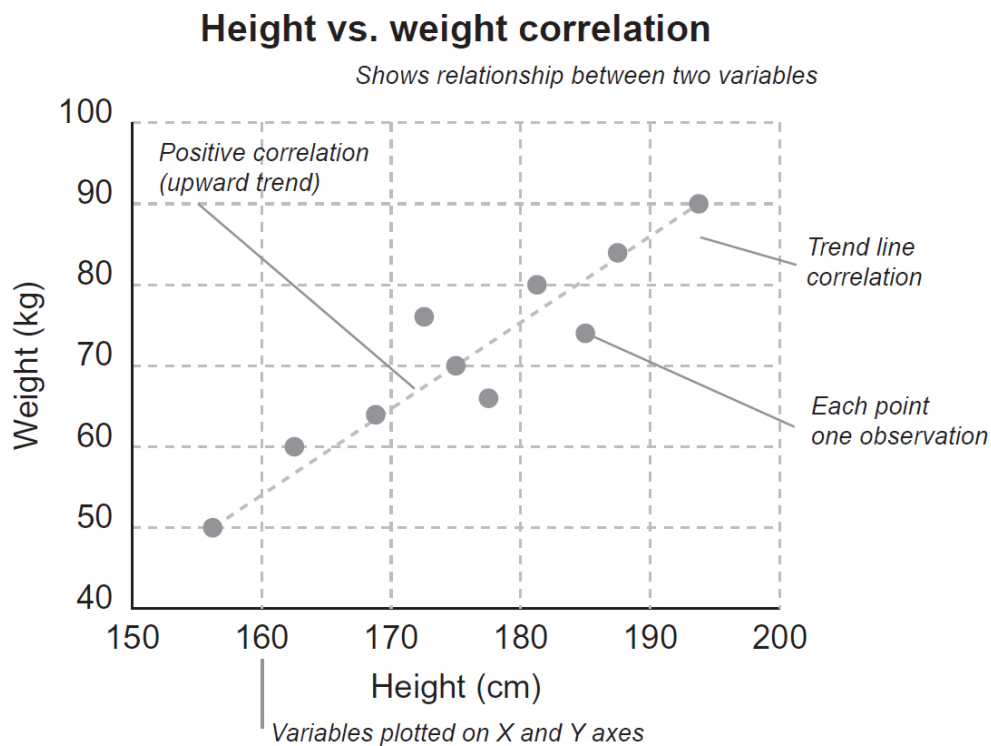
Remember: Use transparency in overlapping area charts to ensure all data is visible and easily interpretable.

SCATTER PLOT

What is a scatter plot?

Scatter plots use dots to represent the values of two numerical variables, allowing you to visualise relationships or correlations between them ([Figure 11.7](#)).

Figure 11.7 Example scatter plot



When to use scatter plots

- **Exploring relationships:** They are great for identifying relationships, trends or clusters between two variables.
- **Spotting outliers:** They are ideal for detecting outliers that deviate from the overall pattern.

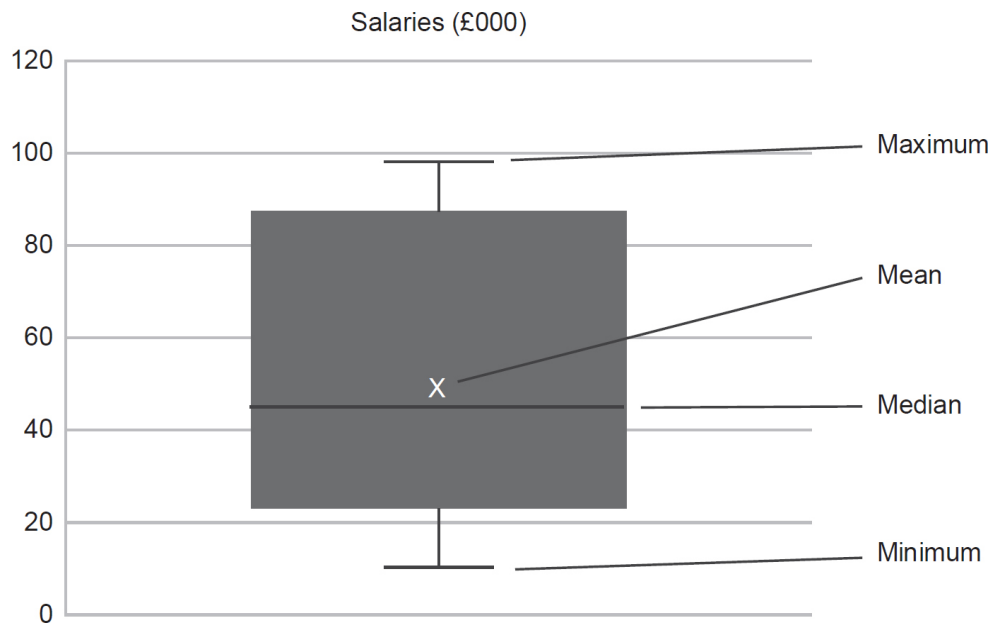
Remember: Consider adding a trendline to your scatter plot to highlight the overall direction of the relationship between variables.

BOX AND WHISKER PLOT

What is a box and whisker plot?

Box and whisker plots, or box plots, summarise data by displaying the minimum, first quartile, median, third quartile and maximum values. They're great for visualising the spread and skewness of data. They also offer an easy way to interpret potential outliers in the data. The size of the box quickly shows the 'interquartile range' ([Figure 11.8](#)).

Figure 11.8 Example box and whisker plot



When to use box and whisker plots

- **Comparing distributions:** They are useful for comparing the distribution of data across different categories.
- **Identifying outliers:** They are ideal for spotting outliers in your data set.

Remember: Box plots are especially helpful when you have multiple data sets to compare, as they provide a clear summary of each distribution.

HEAT MAP

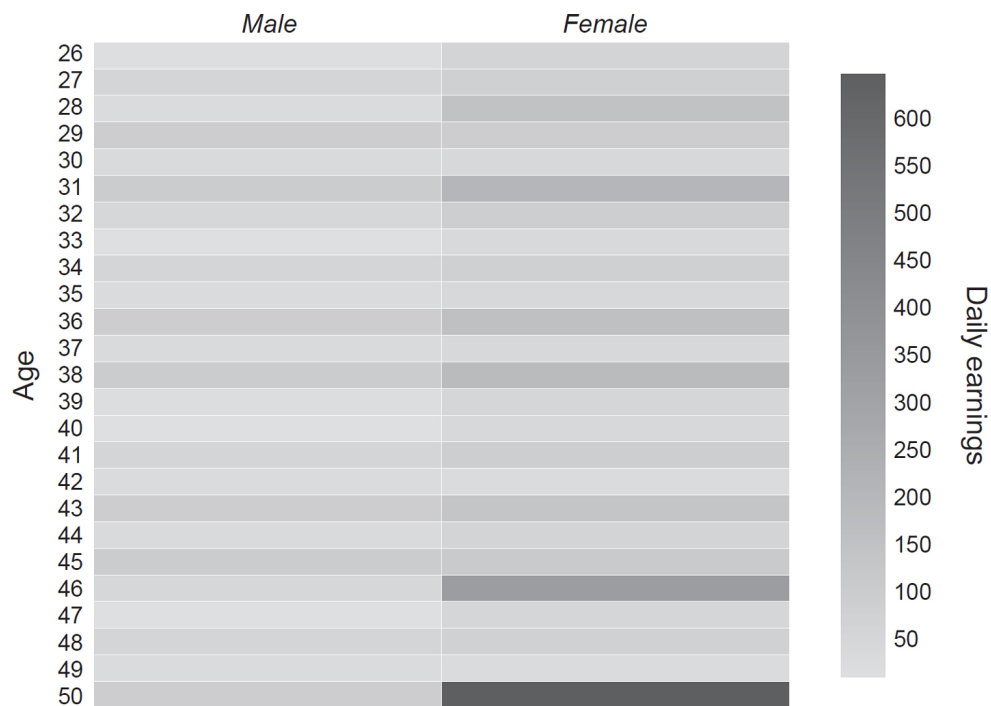
What is a heat map?

Heat maps use colour to represent data values in a matrix format, making it easy to visualise complex data relationships. You can easily see where ‘hot spots’ are in your data ([Figure 11.9](#)).

When to use heat maps

- **Visualising correlations:** They are perfect for displaying correlation matrices or relationships between multiple variables.
- **Highlighting patterns:** They are useful for spotting patterns, trends and anomalies in large data sets.

Figure 11.9 Example heat map



Remember: Select a colour gradient that clearly distinguishes between low and high values to enhance interpretability.

STEM-AND-LEAF PLOT

What is a stem-and-leaf plot?

Stem-and-leaf plots display quantitative data in a semi-tabular format, preserving individual data points while organising them into stems and leaves. They are particularly useful for small data sets. The stem is the first part of the data value, with the leaf showing individual final parts of the value, for example in [Figure 11.10](#), Stem 1 and Leaf 0 show the data point 10, Stem 8 and Leaf 7 indicate the data point 87 (and we can quickly see there are four data points of 87).

When to use stem-and-leaf plots

- **Exploring small data sets:** They are ideal for visualising small data sets while retaining the original data values.
- **Understanding distribution:** They are useful for quickly visualising the shape and distribution of data.

Figure 11.10 Example stem-and-leaf plot

Stem	Leaf			
1	0	0	0	2
2	3	3	3	3
3	4	4	4	
4	5	5	5	
5	6	6	6	6
6				
7				
8	7	7	7	7
9	8	8	8	

Remember: Stem-and-leaf plots are often used in educational settings to teach data organisation and distribution concepts. They are part of the core maths curriculum in the UK.

COMBINING CHARTS

Using more than one chart, representing multiple data sets, can create a deeper view of the data, and there is a particular pairing that is common, but powerful: scatter plots and line graphs. Together, these charts can provide a richer and more insightful view of your data.

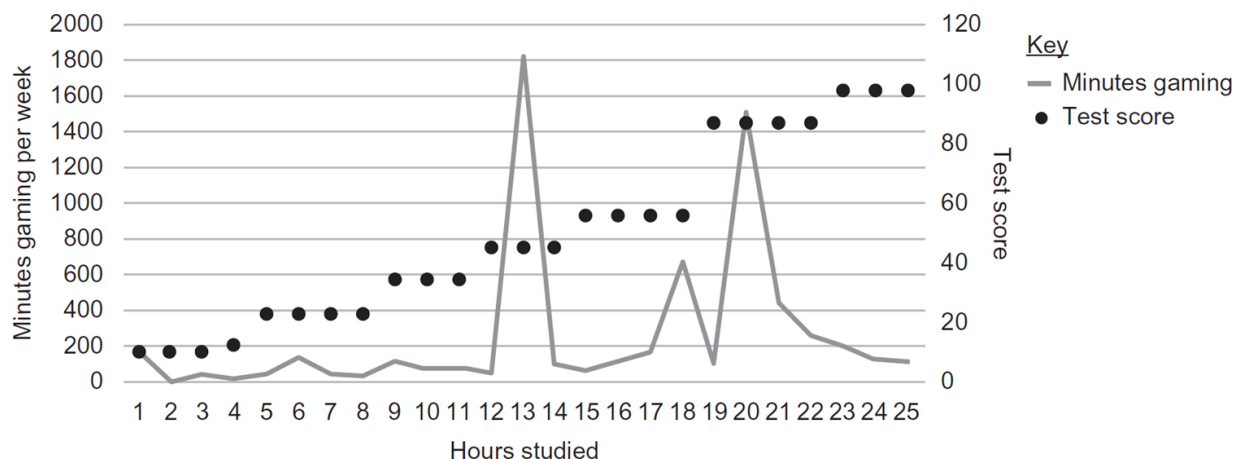
Why combine scatter plots and line graphs?

Telling a complete story

While single charts can convey specific aspects of your data, combining them allows you to tell a more complete story. Scatter plots and line graphs complement each other perfectly by highlighting individual data points and overall trends simultaneously ([Figure 11.11](#)).

- **Scatter plots:** These showcase the relationship between two variables by plotting individual data points.
- **Line graphs:** These emphasise trends and changes over time by connecting data points with a line.

Figure 11.11 Example scatter plot and line graph combined



Uncovering insights

Combining scatter plots and line graphs can reveal insights that might be missed when using either chart alone. This combination allows you to see both the granular details and the big picture, making it easier to identify patterns, correlations and outliers.

Creating the perfect combo

Step 1: Start with a scatter plot

A scatter plot serves as the foundation for your combined chart. By plotting individual data points, you can explore relationships and spot patterns between two variables.

Example: Imagine you're analysing the relationship between study hours (x axis) and test scores (y axis) for a group of students. A scatter plot will help you to see how each student's study habits relate to their performance.

Step 2: Add a line graph

Overlaying a line graph on your scatter plot provides a visual representation allowing the relationship between data sets to be explored.

Example: Adding a line graph plotting hours studied against minutes gaming allows you to explore the relationship between the three variables: hours studied, test score and minutes gaming

Step 3: Enhance with colour and style

Use colour and style to differentiate between the scatter plot and line graph, ensuring clarity and visual appeal. Consistent colour schemes and line styles can highlight key insights and make your combined chart easy to interpret.

Example: Use distinct colours for the scatter points and line to avoid confusion. You might opt for blue dots for the scatter plot and a bold red line for the line.

Real-world applications

Sales performance analysis

Businesses often use combined scatter plots and line graphs to analyse sales performance over time. By plotting individual sales transactions as scatter points and profit as a line graph, companies can identify seasonal trends between sales and profit.

- **Scatter plot:** Individual sales transactions plotted over time.
- **Line graph:** A line showing profit over time.

Scientific research

In scientific research, combined scatter plots and line graphs are used to explore the relationship between variables, such as dosage and drug efficacy, temperature and reaction rate, or any other two related phenomena.

- **Scatter plot:** Experimental data points showing dosage over time.
- **Line graph:** A line showing the efficacy of a drug over time.

Tips for effective combined charts

Keep it simple

While combining charts can enhance insights, avoid overloading the visualisation with too much information. Keep your design clean and focus on the most relevant data points and trends.

Label clearly

Ensure that both the scatter plot and line graph are labelled clearly, with axes, titles and legends that describe the data

accurately. This helps viewers to understand the context and significance of the visualisation.

Interactive elements

Consider adding some of the interactive elements we discussed earlier, such as tooltips or filters.

Visualisations are often subjective and sometimes even quite emotive. People can feel very strongly about what they 'like' to see and you should always bear this in mind – knowing your audience is crucial. Try not to fall into the trap of making visualisations look incredibly technical; the message from the data is often lost when someone does not quite understand what they are seeing.

While we do not want to offer direct advice on what you should choose, there are certainly some styles to avoid – 3D graphs are almost never a good idea.

PRACTICAL TIPS

- Gather audience feedback to know what works for different people (but remember you will probably not please everyone, just aim for the important majority).

- Choose the right chart type for your message. Starting with the wrong chart type will make it very difficult to have a clear message.
- Keep it simple; prioritise clarity. Start with the least possible contents and justify adding more, rather than having everything and taking things away.
- Label everything clearly.
- Maintain a consistent colour scheme (and if you have to meet corporate standards – make sure you do).

MODEL EVALUATION

In this chapter, we delve into the essential process of evaluating data science models. Evaluation is crucial to ensure that the models you build are not only accurate but also meaningful and reliable. We will cover key concepts and techniques to assess model performance, interpret results and ensure they generalise well beyond the data used to train them. This activity relates to the analyse stage of the data analysis lifecycle.

Let's start at the beginning. When we talk about the outputs from models, there is going to be mention of likelihood, significance, confidence and so on, so let's start with probability, which we mentioned in [Chapter 9](#), but will go into more detail now.

WHAT IS PROBABILITY?

Definition: Probability is the measure of how likely an event is to happen. It's a number between 0 and 1, where:

A probability of 0 means the event will definitely not happen.

A probability of 1 means the event will definitely happen.

A probability in between (e.g. 0.5) means there's some chance the event will occur.

Key formula for probability:

$P(\text{Event}) = \text{Number of favourable outcomes} \div \text{Total number of possible outcomes}$

Example 1: rolling a die:

Imagine you roll a standard six-sided die. What's the probability of rolling a 3?

Favourable outcome: Rolling a 3 (just 1 outcome).

Total possible outcomes: Rolling any number from 1 to 6 (6 outcomes).

$$P(\text{Rolling a 3}) = 1 \div 6 \approx 0.167$$

So, the probability of rolling a 3 is roughly 16.7 per cent.

Example 2: flipping a coin:

If you flip a coin, what's the probability of getting heads?

Favourable outcome: Heads (1 outcome).

Total possible outcomes: Heads or tails (2 outcomes).

$$P(\text{Heads}) = 1 \div 2 = 0.5$$

So, there's a 50 per cent chance of flipping heads.

WHAT IS A *P*-VALUE?

Definition: A *p*-value tells us how likely it is to observe the data we have (or something more extreme) if the null hypothesis is true. It is a measure of 'surprise' in our data.

A small *p*-value (e.g. 0.05 or less) suggests that our data is unlikely to happen unless something unexpected is happening.

A large *p*-value (e.g. > 0.05) means our data is consistent and as we expected.

Think of the *p*-value as answering this question:

What is the probability of observing data like this if there is nothing going on?

It is commonly accepted that a 0.05, or 5 per cent, chance of something happening is a good level to work at. If there is a 5 per cent chance, or less, of an event occurring, and it does happen, we can say in the social sciences this is unlikely enough

to surprise us. Sometimes the value we choose can be different, for example 0.01.

HYPOTHESIS TESTING: A STEP-BY-STEP GUIDE

Hypothesis testing is a formal process we use to determine whether a hypothesis about a population is supported by sample data. Here are the steps to conduct a hypothesis test:

Step 1: State the hypotheses

There are always two hypotheses in hypothesis testing:

Null hypothesis (H_0): This is the default assumption. It states that there is no effect, no difference or no relationship.

Example: A coin is fair, and the probability of heads is 0.5.

Alternative hypothesis (H_a): This is what you're trying to prove. It states that there is an effect, a difference or a relationship.

Example: The coin is biased, and the probability of heads is not 0.5.

Step 2: Choose a significance level (α)

The significance level is the threshold for deciding when to reject the null hypothesis. When would we think that there is a bias?

Common choices: $\alpha = 0.05$ (5 per cent) or $\alpha = 0.01$ (1 per cent).

If the p -value is less than α , we reject the null hypothesis, and this means that there is some bias. Be aware that you should choose the level before we run our model and not risk being influenced by our answers.

Step 3: Collect data and calculate the test statistic

The test statistic is a number that summarises your data and helps you to compare it with what the null hypothesis predicts. The type of test statistic depends on the test you're performing (e.g. z -test, t -test, etc.).

Step 4: Calculate the p -value

Using the test statistic and a statistical table (or software), find the p -value (also referred to as the critical value). This tells you how unusual your data is under the null hypothesis.

Step 5: Make a decision

If $p \leq \alpha$: Reject the null hypothesis (H_0). This means there's enough evidence to support the alternative hypothesis (H_a).

If $p > \alpha$: Fail to reject the null hypothesis. This means there's not enough evidence to support H_a .

Example of hypothesis testing: Is a coin fair? Let's say you flip a coin 20 times and get 17 heads. You suspect the coin might be biased.

State the hypotheses:

H_0 : The coin is fair ($P(\text{Heads}) = 0.5$).

H_a : The coin is biased ($P(\text{Heads}) \neq 0.5$).

Choose a significance level:

$\alpha = 0.05$.

Collect data and calculate the test statistic:

Number of heads = 17.

Total flips = 20.

Use a binomial test to calculate the p -value. Note: the binomial test is a method of calculating the probability of something

happening or not happening in many independent events.

Calculate the p -value:

Using software or a binomial probability table, find the p -value for getting 17 or more heads (or 17 or fewer heads for a two-tailed test). The p -value might be, for example, 0.003.

Make a decision:

Since $p = 0.003 < \alpha = 0.05$, we reject the null hypothesis. There's strong evidence that the coin is biased.

IS THIS CHANCE? UNDERSTANDING SIGNIFICANCE

When evaluating a model, it's vital to determine whether its performance is genuine or merely a result of random chance. **Significance testing** helps in this regard, telling you how likely it would be to happen so that you can make a decision on whether you have spotted a pattern or trend. Apart from p -values and null hypotheses, you may come across the term confidence intervals.

- **Confidence intervals:** These provide a range within which we expect the true performance metric (e.g. accuracy,

precision) to lie with a certain level of confidence, usually 95 per cent.

By using these measures, you can judge whether your model's predictions are meaningful or just flukes.

HOW MUCH DOES THE MODEL EXPLAIN? COMPARING TRAIN AND TEST

A critical aspect of model evaluation is understanding how well the model generalises to new, unseen data. This involves comparing its performance on the **training set** (the data used to build the model) with the **test set** (new data not seen by the model during training and therefore a way of checking the results).

- **Overfitting:** This occurs when a model performs exceptionally well on the training set but poorly on the test set. It means that the model has learned the noise and specifics of the training data rather than the underlying patterns. It has been overly designed to match the test set.
- **Underfitting:** Conversely, underfitting happens when the model performs poorly on both training and test sets, indicating it hasn't captured the underlying trend in the data.

- **Performance metrics:** Common metrics include accuracy, precision, recall and F1-score. However, it's essential to monitor these across both the training and test sets. Significant disparities between the two may indicate overfitting or underfitting.

Striking the right balance between these sets ensures that your model explains the data well and can generalise effectively.

False positives and false negatives

In many real-world applications, not all errors are created equal. If you build a model that predicts something will happen and it doesn't, then this is an error. The impact of this error depends on the context of a problem. Imagine predicting that someone will catch the flu, but they don't. You may have given them an unnecessary vaccine, which is not likely to be the end of the world. Being able to label the errors in your model is key to understanding its impact.

- **False positive (Type I error):** This occurs when the model predicts a positive outcome incorrectly. For example, diagnosing someone as having a disease when they don't.
- **False negative (Type II error):** This happens when the model misses a positive outcome, such as failing to detect a

disease in a patient who actually has it.

The consequences of these errors vary by context. In medical diagnosis, a false negative might be more critical than a false positive. However, in spam detection, false positives (marking legitimate emails as spam) might be more problematic.

The numbers of errors are often shown in a simple table called a **confusion matrix**. It shows how well your model's predictions align with the actual outcomes in your data. If we use the spam emails example, when detecting an email as spam we can have one of four outcomes:

- | | | |
|----|--------------------------------------|---|
| 1. | We mark it spam, and it is | – <i>got it right.</i> |
| 2. | We mark it spam, and it isn't | – <i>got it wrong (false positive).</i> |
| 3. | We do not mark it spam, and it is | – <i>got it wrong (false negative).</i> |
| 4. | We do not mark it spam, and it isn't | – <i>got it right.</i> |

After testing the model on 100 emails, you compare the model's predictions with the actual classifications. Here is what a confusion matrix would look like:

	Predicted: spam	Predicted: not spam
Actual: spam	40 (true positives)	10 (false negatives)
Actual: not spam	15 (false positives)	35 (true negatives)

To evaluate these errors, metrics such as **precision** (how many predicted positives are actual positives) and **recall** (how many actual positives are correctly identified) are utilised.

The F1 score is a commonly used metric that combines **precision** and **recall** into a single score. The F1 score is particularly useful when you want to balance precision and recall, such as in cases where false positives or false negatives are equally important.

HOW TO INTERPRET THE MODEL

Understanding and interpreting the model's outcomes is just as critical as building it. Here's how you can approach it:

- **R-squared:** This indicates how well the data fits a regression model. R-squared is expressed as a percentage between 0 and 100. A higher R-squared value generally means a better fit, but it's important to consider other factors as well, such as the complexity of the model and the sample size.
- **Feature importance:** Many models, especially decision trees and ensembles, provide insights into which features are most influential in making predictions. Analysing feature importance can help to explain the model's decisions and identify the key drivers of outcomes.
- **Partial dependence plots:** These visualise the relationship between a particular feature and the predicted outcome, holding other features constant. This helps in understanding how changes in a feature affect the model's predictions.
- **Residual analysis:** By examining residuals (the differences between observed and predicted values), you can assess whether the model's predictions are unbiased and identify any patterns it failed to capture.
- **Model explainability tools:** Advanced techniques such as SHAP (SHapley Additive exPlanations) and Local Interpretable Model-agnostic Explanations (LIME) can break down complex models to explain individual predictions,

enhancing transparency and trust in the model. These are a little beyond the scope of this book, but worth looking into when you are further into your data science journey.

Support, confidence and lift: evaluating rules

In association rules and other recommendation systems, three key metrics help to evaluate the strength of rules derived from data:

- **Support:** This indicates how frequently an item set appears in the data set. For example, if 30 per cent of transactions include bread, the support for bread is 0.30.
- **Confidence:** This measures the likelihood that a rule holds true. If 80 per cent of transactions that include bread also include butter, the confidence for the rule 'bread $\rightarrow$ butter' is 0.80.
- **Lift:** This evaluates the strength of an association rule relative to random chance. A lift value greater than 1 indicates a positive association, suggesting that the occurrence of one item increases the likelihood of the other.

These metrics help to determine the usefulness and relevance of the discovered patterns or rules. By thoughtfully interpreting

the model, you can ensure its predictions are not only accurate but also understandable and actionable.

COMMUNICATION

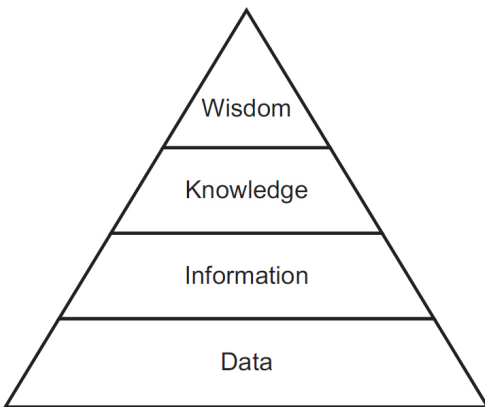
You might recall that we talked about framing the problem and developing a hypothesis back at the beginning of the data analysis lifecycle, in discovery. Having completed all the hard work in investigating the problem and drawing insight from it, we now need to communicate our findings. If we communicate them well, people will have knowledge that can inform and influence their decisions and actions. However, sub-optimal communication might mean that people don't properly understand, and result in poor decisions. It is therefore important to think carefully about and invest time in how we communicate our findings. We need to ensure that it is meaningful to and resonates with our audience.

The data, information, knowledge, wisdom (DIKW) pyramid (Wikipedia, n.d.) helps us to think about the differences between the data, our tested hypotheses, the insight from these and the actions that result.

In our version of this model ([Figure 13.1](#)), data represents the raw data that we sourced, information derives from our testing of our hypotheses, knowledge is what we will communicate and wisdom is what we hope our audience will gain from our communication. To derive knowledge from the information, we

will need to combine our information with other information derived from the domain in which we are working.

Figure 13.1 The DIKW model

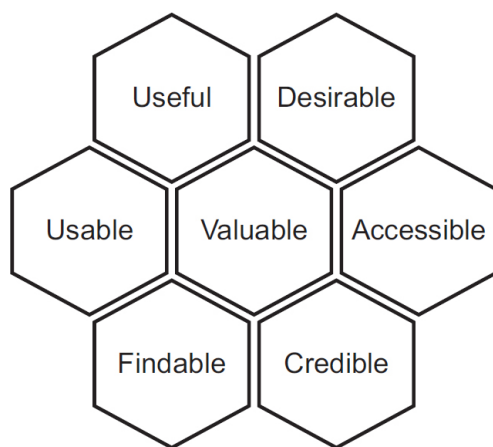


PRINCIPLES OF USER EXPERIENCE

Communication can be made in many forms, including written, visual and oral. Often, it is a combination of these – for example, a presentation. In this case, the presentation materials will normally be made up of words and images, with a voice-over by the presenter. In this instance, there is an opportunity to explain and react to an audience. Where a written or visual report is produced and made available to the audience, the opportunity to explain further does not exist. It is important to understand the communication channel to ensure that you tailor your communication to deliver a good user experience.

When considering user experience, the user experience honeycomb can be a useful guide (Morville, 2004). Although developed for user interfaces, it sets out seven principles of user experience that can help when thinking about how and what to communicate ([Figure 13.2](#)).

Figure 13.2 The user experience honeycomb



Let's consider what this means in terms of communicating our findings, in relation to our problem and having tested our hypothesis.

- **Usable:** Is it easy for the audience to engage with?
- **Useful:** Does it meet the audience's needs?
- **Desirable:** Does the audience want to use it?
- **Findable:** Can the audience locate the information they need? Both within the communication and where to find the communication?

- **Accessible:** Can those with disabilities access it?
- **Credible:** Do and should the audience trust it?
- **Valuable:** If all the above are met, then do the audience value it?

If you think about it, the whole data analysis lifecycle contributes to a good user experience. The usefulness, desirability and credibility of the communication are dependent upon the framing of the problem, the data sourcing and preparation and the methods applied. The design of the communication also plays a part in whether these principles are met. Design is similarly an important factor in whether the knowledge shared is useful, findable and accessible. How the communication is stored and shared also plays a role in whether the knowledge is findable and accessible.

Let's look at resource planning and how we might communicate our findings. Rather than consider this from the data scientist's perspective, let's step into the shoes of the resource planner. Their need is simple: a forecast of customer contact volumes for the next month, updated each week. It might be possible to manually pull together a report each week in Excel – consisting of a multitude of different rainbow-coloured charts with a full range of models covered. This could be sent out by email around the start of the week. Sounds OK? Perhaps not if you

compare it with a live report that the resource planner can access at any time by clicking a link on your intranet home page, and which clearly shows the forecast number of contacts for the next month.

IMPROVING USER EXPERIENCE

In many instances, the audience might have a preference as to how they are communicated with. Often, there are opportunities to improve on this; however, in the first instance changing the way you communicate with an audience can have a negative effect on the user experience. This doesn't mean to say that the communication shouldn't change, but it needs to be handled sensitively. The issue is how to educate your audience. Consider:

- Who will act as an advocate?
- Can small changes be made?
- Are you able to build a consensus in your audience over time?

KNOW YOUR AUDIENCE

In the previous section we stepped into our audience's shoes. Our audience will be made up of the project's stakeholders or a subset of those stakeholders. These stakeholders might have

different needs from the communication. Where this is the case, the audience should be segmented and appropriate information shared with each segment.

Even for a simple data science project, the audience can often be segmented. It is worth considering four potential segments, consisting of:

- **Technical:** This audience will want technical detail of the models used, and understands the technical language.
- **Managerial:** This audience wants the key insights expressed in simple non-technical terms, with the ability to drill into detail that is relevant to them.
- **Executive:** This audience wants the key insights expressed in simple non-technical terms.
- **Team:** Like the executive audience, this one wants key insights expressed in simple non-technical terms. However, the scope of information provided might be more limited.

This audience segmentation is a generalisation, providing a useful starting point. How might this play out in practice? Let's take the example of forecasting sales for a company with regional sales teams. The executives might want to know what the expected sales are going to be. Members of the regional teams are also likely to be interested in this. The managers of

each regional team could benefit from more details (e.g. the forecast for each of their sales representatives). Individual forecasts are probably too much detail for the executives and information that should not be shared with team members. Colleagues in the data science team are likely to want to know about the model used to create the forecast.

When looking at the different audience segments, it is worth considering their level of knowledge and whether they have preconceived ideas that the findings might reinforce or challenge. This can require sensitive handling. When communicating to different audiences, it's also important that the fundamental messages are consistent, otherwise credibility is undermined and difficult questions are likely to be raised.

DOMAIN CONTEXT

We talked about the importance of the domain context when looking at the discovery stage of the data analysis lifecycle. Domain context is important both in translating the information we have as the result of our hypothesis testing into knowledge, and in determining how to communicate that knowledge.

Looking at the outcome of our hypothesis testing in the context of the domain to which it relates can help us to understand it. A simple example could be our sales forecast: if the model appears weak, could that be due to changed behaviours during Covid? Another Covid example is the predicted A-level grades discussed in [Chapter 4](#). The domain context allows us to see that the schools with too small a cohort to predict grades are private schools. It also lets us recognise that some pupils might be prejudiced by using the model, as they would not be awarded the grades their teachers believed they were worthy of.

The domain context also helps us to understand our audience. How well do they understand the domain and how do they perceive it? These are factors to be considered in how we communicate our findings. How might we change the way we communicate if our sales forecasts show that sales will be significantly lower than the executive expect, or one influential region is forecast to perform poorly? Our audience is human, and we need to recognise that they will have an emotional reaction to our communication.

INFORMATION TO KNOWLEDGE

What should you consider when transforming the information you have gleaned from your hypothesis testing into knowledge?

We have already considered the domain context, but what about the results of the hypothesis testing themselves?

Most models will return a set of performance metrics to assist in determining the value of your model. Make sure you understand them and correctly interpret them. It is likely that most of your audience will rely on you to do this as they will not have the skills and expertise to do it themselves. Where possible, it is advisable to have someone else check or confirm your interpretation.

Once you have interpreted the data, you need to consider how you communicate the uncertainty that will inevitably be associated with the patterns found and predictions made. We also need to add unpredictable outside factors to the uncertainty in our models. There are two points worth considering here:

- **The strength of the model:** If it is a strong model, that should give confidence. If it is a weak model, should it be used?
- **External information:** Is there other information that supports or challenges the model?

It is also worth noting that, just because a hypothesis isn't proved on testing, it does not mean that the work hasn't been worthwhile. It has value in demonstrating that something thought to have importance doesn't. This means it can be excluded or downgraded in decision-making and effort, and resources can be put into finding those things that are important, for example if your hypothesis is that increasing the spend on advertising on a particular platform will increase sales and you are unable to prove it, you can redirect that spend to other platforms where you do get a return.

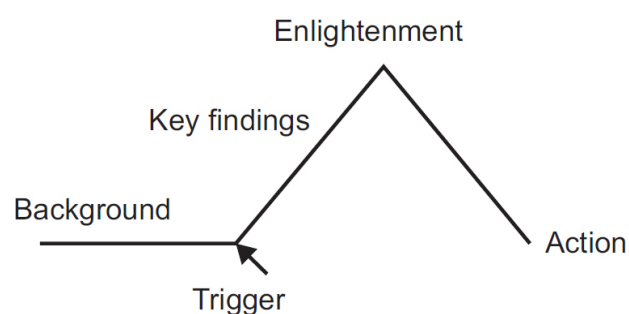
You might also need to be conscious of limitations in the data sets used and, where appropriate, communicate them and the potential impact on the findings.

STORYTELLING

The idea of telling stories around data has its roots in data journalism. In data journalism, data is used to both find and evaluate stories. The story is then told based on the data. The story brings a narrative to the data that in turn helps the audience to understand the knowledge imparted. It creates an emotive response in the audience and acts as a call to action.

In the 19th century, Gustav Freytag put forward a pyramid model for narrative that has been adapted to data storytelling (Dykes, 2019). In this model, the narrative consists of four stages and a hook (which we have called a trigger), a version of which is shown in [Figure 13.3](#).

Figure 13.3 Data storytelling pyramid



The stages are:

- **Background:** This is explaining the background to the problem.
- **Key findings:** These are what the key findings were, the information.
- **Enlightenment:** This is the insight and knowledge gained.
- **Action:** This is the options and suggested actions.

The **Trigger** is what caused the problem being investigated.

In preparing your narrative, think about the subject of the narrative. We often base the narrative around the data scientist,

the actions they perform and what they find. Changing the subject of the narrative can create a stronger connection with the audience. That subject could be a person, place, object or even a concept. Building the backstory not only provides context, but is an opportunity to build empathy and trust.

We can compare the narrative for an investigation into contact centre performance: one as a data scientist might present it, the other with the customer as the subject following the narrative model. This is set out in [Table 13.1](#).

Table 13.1 Narrative comparison for contact centre performance investigation

Data scientist as the subject	Customer as the subject
Took contact data from the source.	Customer interactions are important for our brand, especially when the customer has a problem.
Analysed the trend of customer satisfaction.	How customers experience our contact centre is important to their brand

perception and our net promoter score (NPS).

Downward trend in NPS.
Correlation between reduced staff contact time and NPS.

Customers' willingness to recommend us is falling, and is projected to continue to fall as we reduce staff contact handling times.

Customers are becoming less satisfied, and this looks to be an ongoing trend. This is correlated to shorter contact time with each customer. Qualitative feedback from customer comments indicates the lack of time contributes to a poor experience.

If the trend in NPS continues, we will be in the bottom quartile for customer experience compared to our major competitors within six months.

Increase contact centre staffing.

Our customer experience could be improved by allowing staff more time to

handle each customer contact, resulting in an increase in brand value and NPS.

Recommend A/B testing of contact handling times to determine impact on NPS.

Note the use of qualitative as well as quantitative data within the story. The qualitative feedback, potentially via a survey, provides evidence of shorter contact time causing reduced customer satisfaction. In thinking about your narrative, don't lose sight of the fact that the detail in the data science can still be drilled into if needed when telling the customer story.

It is often said that conflict makes a good story. It can be helpful to think of this in terms of competing interests. If you take our contact centre example, the competing interests could be between keeping costs down and keeping customer satisfaction high.

Just in case you haven't come across net promoter score, it is a measure of how likely a customer is to recommend an organisation to others. A/B testing is where you trial two competing services to determine which performs best. In this

case, the services are distinguished by the amount of time allowed for each contact.

RISKS IN STORYTELLING

Within any set of data, many stories can be found. Back in [Chapter 5](#) we discussed how the problem should be framed and the hypothesis developed. This determines to an extent the stories we will find in our data. Therefore, these are important steps in determining the story our data will tell. However, once we have gained the knowledge from testing the hypotheses, we still have choices around the story we tell. If you take our contact centre example, the story might have been one of how reducing contact time was expected to result in cost efficiencies.

Selecting the right stories to tell can raise ethical questions. Have the stories people might not want to hear been buried? In our contact centre, do people want to hear the story of declining staff welfare being correlated to shorter customer contact times?

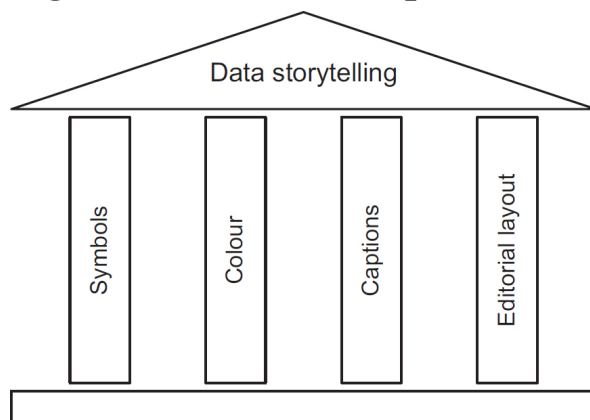
We also need to consider whether we are spinning a story to meet a particular end. If this is the case, then we are in the territory of propaganda: using a misleading data story to promote a particular point of view.

VISUALISING A STORY

We have discussed how a story can aid communication and how this requires good narrative. This can be supported by effective visualisation. We have already explored how to visualise our data in [Chapter 11](#), but how do we visualise our story? The four pillars for data storytelling set out design principles that can assist with this (Feigenbaum and Alamalhodaie, 2020). These pillars look at how the story is visualised, which goes beyond the charts that might be used in the telling of the story ([Figure 13.4](#)).

- **Symbols:** These are frequently used to aid communication. Drive down a road and you will see them, whether it is a sign warning of horses or a school crossing. To be effective, the symbols need to resonate with your audience and be relevant to them. Beware of picking a symbol just because it seems a good idea; ask yourself how your audience would view it.

Figure 13.4 The four pillars of data storytelling

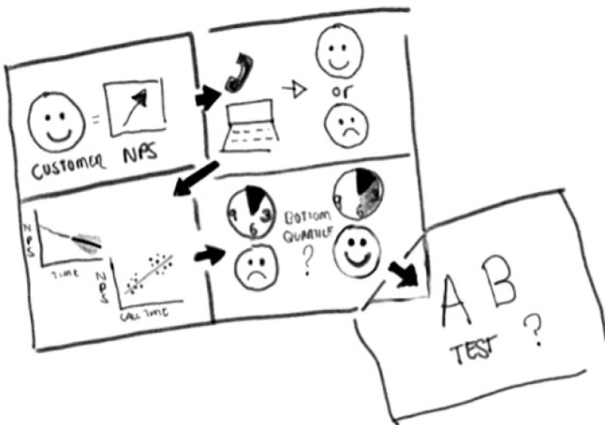


- **Colour:** This can be used to draw attention to important information. For example, in a chart looking at forecast sales figures for a team, those team members who had received training might be shown in a different colour. In this case, it might illustrate the effect additional training has had on expected performance. In using colour, give thought to the use of contrasts to aid accessibility. People attach meaning to colour – for example, green for good and red for bad, or blue for cold and red for hot. Using these associations can aid communication and cause confusion if ignored. It is also important to consider any cultural differences in colour association that might cause confusion. If we created clusters of sales regions, we might choose to show the best performing cluster in green and the worst performing in red.
- **Captions:** The symbols used to aid storytelling can have different meanings, and captions help to guide people in interpreting the symbols and thus aid their comprehension

of the story being told. Using an unhappy face in telling our contact centre story helps to convey feeling; however, we would need a caption to distinguish the customer from the contact handler.

- **Editorial layout:** This is the arrangement of the visual elements of the story in a way that supports the narrative. This might be the layout of a dashboard, the order of a slide presentation or even the flow of a comic strip.

Figure 13.5 Draft storyboard



The use of the four pillars is shown in the draft storyboard in [Figure 13.5](#). Symbols are used to illustrate the subject and themes within the narrative; captions are used to signpost the meaning of the symbols; contrast is used to show the forecast NPS and additional contact time that could make a difference to customer satisfaction. Finally, the editorial layout supports telling the story in five steps.

RECOMMENDATIONS

Recommendations will often flow from investigating our problem and testing our hypothesis. However, it is very easy to make recommendations to address something our investigation has revealed that are not backed up by the hypothesis testing or other evidence.

Going back to our contact centre example, we recommend that contact time is increased to improve customer satisfaction. However, we don't know that this is the case from the analysis we have undertaken. Yes, satisfaction is forecast to continue to fall, and yes there is correlation between contact time and satisfaction. That doesn't mean that shorter contact time is the cause of lower customer satisfaction. It is a theory, hence the proposal of A/B testing to develop understanding.

When putting forward recommendations, be clear as to which can be evidenced and which are only theories.

OPERATIONALISE

Where there is an ongoing need for the knowledge, then there is a need to operationalise the production of a report. This might be through repeating the process you have used to

investigate the problem and test your hypotheses. However, this process might need to change to deliver a good user experience.

At this point it is worth revisiting the stakeholder needs. They are likely to be different from a one-off investigation. A key question is how frequently they need the report and when it needs to be received by. Also, the scope of the information required might change. In this instance, the problem will need to be reframed. There might also be benefit in automating the process, both to speed up its production and to increase trust by reducing errors.

PRACTICAL TIPS

- Understand your audience and their needs.
- Consider all aspects of the user experience.
- Ask what knowledge can be derived from the information obtained from your models.
- Thinking about the domain should assist in generating knowledge from information.
- Consider who or what the subject of the story should be.
- Plan the story you want to tell and how you will visualise it.
- Beware of telling a story people want to hear, rather than the one that should be told.

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

In [Chapter 1](#) we recalled the dystopian future of *Terminator*: a future where machines have taken over the world and determine to eradicate humans. This has been a reference point for many of those writing about the risks of artificial intelligence. But what is artificial intelligence? How does machine learning relate to it? Where does data science fit into this?

As with so many other terms we have come across, there is no consensus on the term artificial intelligence. The UK government define it by its adaptivity and its autonomy (Department for Science, Innovation and Technology, 2023). This is looking at AI from a technology and risk perspective, the adaptative and autonomous characteristics of the technology making it difficult to explain, predict and control. A similar approach is taken to defining AI in Section 3 of the EU's Artificial Intelligence Act, which also refers to adaptiveness and autonomy (European Union, 2025), these characteristics allowing the AI system to infer and act on implicit as well as explicit objectives. Both of these definitions have been drafted in relation to regulating AI. A step back from these takes us to a wider definition, which NASA has succinctly stated as:

Artificial intelligence refers to computer systems that can perform complex tasks normally done by human-reasoning, decision making, creating, etc. (NASA, 2024)

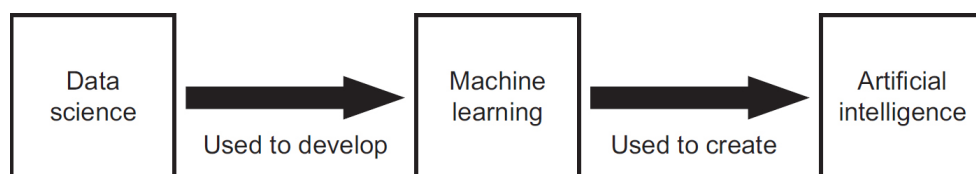
Therefore, we can see AI as adaptive and autonomous technology that performs tasks that have previously been the preserve of humans. This includes a wide range of applications, from dealing with a customer query to creating visual images – taking on the human roles in both call centres and creative endeavours.

AI is currently built on digital technology, and a key component of this is machine learning (ML). What is machine learning? Well, it is a machine that can learn. We can think of this as software that is designed to perform a task or tasks, whose performance of the task improves the more it undertakes the task (Mitchell, 1997). AI is built on complex ML techniques that give it the ability to adapt and act autonomously.

That's great, but where does data science fit into this? Data science is fundamental to ML. Machine learning is based on the application of algorithms (based on algebra, statistics and probability) to data, which – as you know – is data science.

To summarise, data science methods are used to develop ML. ML can be applied to data science problems and to create AI ([Figure 14.1](#)). This means that, to an extent, the terminology can be used interchangeably, for example image recognition technology could be the application of data science, the use of ML or an example of AI.

Figure 14.1 The relationship between data science, machine learning and artificial intelligence



MACHINE LEARNING

We have defined ML as software that is designed to perform a task or tasks, and the performance of the task improves the more it undertakes the task. Let's take image recognition, a technology that uses ML: the more images you show it, the better it will be at identifying the images. How does it do this?

Training and predicting

For ML to work, the models need to be trained. Only when it has been trained can it be used to predict.

Training involves exposing the ML to data from which it can learn. What if you were asked to develop image recognition to determine whether there were potholes in the road? How would you train it? In this instance, you could train it by allowing the ML to consume many images of roads with potholes and those without.

Once the ML has been trained, it can be used to predict. In our example, our image recognition system could be shown an image of a road and it would predict whether the road has potholes in it or not. A classification model.

Learning approaches

There are two common methods of learning: supervised and unsupervised.

Supervised learning

In supervised learning, the data used is labelled or tagged. This allows the ML to identify the patterns in the data associated with labels. The more data it consumes, the better the data patterns associated with the labels, and thus the better the ML is at applying labels based on the data presented.

How would this work with our pothole images? In this instance, you might expose lots of images of roads to the ML. Those with potholes in them would be labelled 'pothole', and those without wouldn't. The ML would identify the patterns in the data associated with the 'pothole' label and, when shown an image with those patterns, would identify them as being associated with the label 'pothole', thus allowing potholed roads to be identified.

Unsupervised learning

In the case of unsupervised learning, the data used is not labelled or tagged. Instead, the ML looks for patterns within the data. It can then cluster the data based on those patterns.

How would this work for pothole detection? Simply put, the ML would consume images of roads with and without potholes. It would detect patterns in the data that distinguished the pothole images from those without, in effect creating two clusters: those with and those without potholes.

If you think about it, supervised learning is great when you know what you are looking for, in our case potholes.

Unsupervised learning might not be as effective in these cases. For example, when looking at the images of roads, it might

detect patterns relating to whether there are pavements or the time of day the images were captured. However, unsupervised learning comes into its own when you are unsure what you are predicting, for example if you are looking for patterns to help find faults and don't know what those faults look like.

These methods can be combined to enable the benefits of both supervised and unsupervised learning to be realised.

DATA SCIENCE AND MACHINE LEARNING

Data science is used to develop ML. If you look at our image recognition example, you can see some of the data science methods we have explored being used. In our supervised ML, classification is used to identify the potholes. In our unsupervised ML, we are using clustering to determine the state of the roads.

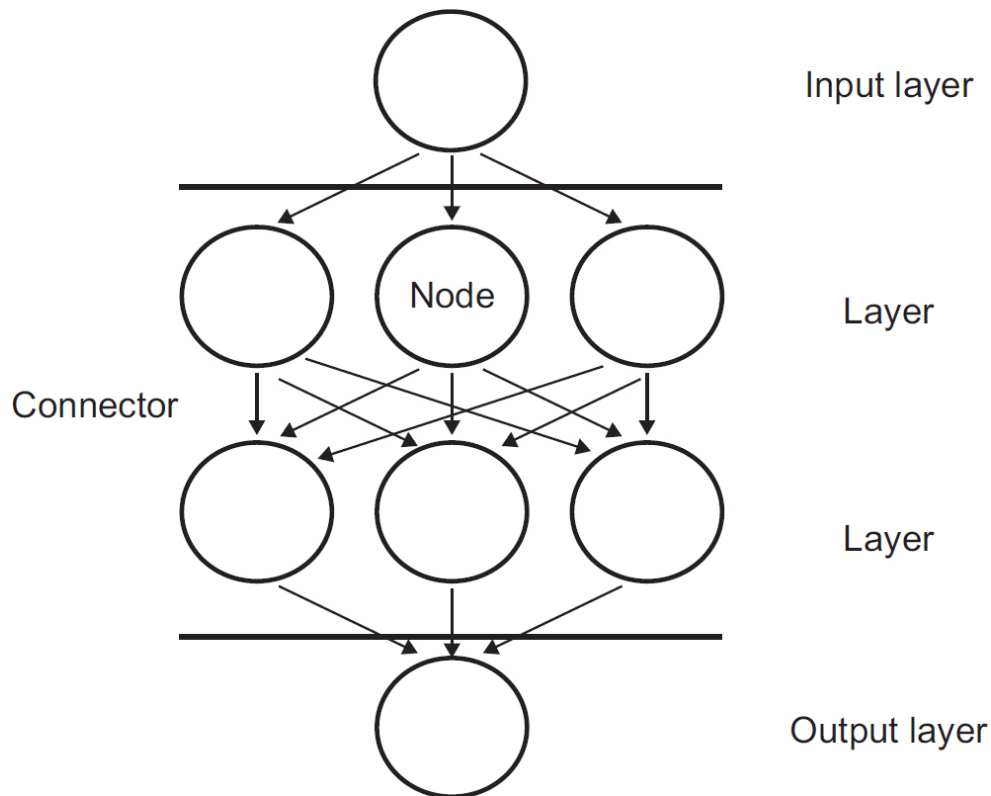
NEURAL NETWORKS

Neural networks is a branch of ML, and is fundamental to AI. They look to model how the brain works, with different connections between neurons firing in response to inputs, leading to outputs. If you see a ball coming towards you (input), you decide (neurons firing) whether to duck or catch it (the output).

The explanation of neural networks given here is a simple one, to provide a basic understanding. A neural network consists of layers of nodes (also known as neurons). The nodes in one layer are connected to the nodes in the adjoining layer. One layer is the input layer and one layer is the output layer. There may be one or more layers of nodes in between ([Figure 14.2](#)).

When a node receives an input, it will provide a value to the nodes in the next layer down. Each node in that layer will calculate a value based on the inputs from the previous layer and send this to the next layer. This repeats until the output layer is reached, which is where the result is returned.

Figure 14.2 A neural network



Each connection has a weight associated with it. The value from the sending node is adjusted by the weight of the connector, and it is the adjusted value that is received by the node in the next layer. The values received by a node are then added up and adjusted by a unique value for that node called the bias. This determines the value sent to the next layer down. In training the neural network, the weighting associated with each connector is changed, as is the bias. These are adjusted until the neural network returns an optimum result. This approach to learning is called back propagation. It is often used with

reinforcement learning which interacts with an environment to maximise rewards.

More complex versions of neural networks are used in generative AI solutions. These networks have many layers, and it is the huge number of layers that leads the use of them being described as deep learning.

ARTIFICIAL INTELLIGENCE

NASA describes AI in relation to complex tasks normally performed by humans. AI is sometimes split into narrow AI and general AI. This is to distinguish between AI that can do specific tasks a human can do and AI that can do all the tasks a human can do ([Figure 14.3](#)).

Narrow AI

Narrow AI is AI that is designed to perform a specific task. As such, it only has a narrow range of capabilities. In the last century, pre-2000, if you were asked to determine which roads had potholes it is likely you would need to look at the images of the road to identify them: a human task. We can now use ML to do this. This ML is performing a narrow AI task. It can identify potholes, but is unlikely to be able to spot missing roof tiles. It is

good at one specific task that would previously have involved a human.

General AI

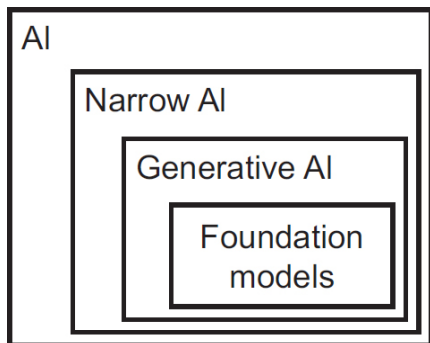
General AI is AI that can handle all the tasks that a human can undertake. At the time of writing, general AI has yet to be created. AI that goes beyond human capabilities is referred to as super AI.

Generative AI

Generative AI has a different meaning from general AI, although they sound very similar. This type of AI is called generative as it generates new data. Depending on the model being used, this can be in different forms including images, words and sounds. This AI is seen as being creative because it produces new material that, due to the data it is built on, is meaningful. Therefore, it might generate computer code that works or a picture of a subject and in a style that people relate to. Generative AI is not general AI, it still lacks all human capabilities. Take large language models: these are a type of generative AI that create text reflecting natural language. Although not covering all human capabilities, they can be applied to many different tasks requiring the generation of

natural language whether it be a chatbot to book a hair appointment or advice on data protection.

Figure 14.3 Hierarchy of AI definitions



FOUNDATIONAL MODELS

Foundational models are a form of generative AI. They are trained using extremely large quantities of diverse data. This creates a model that is pre-trained and can be adapted to different uses. As a result, new applications can be created much quicker and at significantly lower cost than would otherwise be the case. Foundational models work by predicting the next item in a sequence. In the case of language, this might be the next letter or word. The sequence then builds up into a comprehensible and meaningful language.

Risks of AI

We will take a step back from the end of the world as an AI risk and focus on the more technical risks relating to AI outputs. The main risks can be summarised as:

- **Bias:** Does the AI produce outputs that are in some way biased?
- **Hallucination:** Has the AI made something up, so the output is in some way incorrect?
- **Transparency:** Can you understand how the AI has reached the outcome it has?
- **Privacy:** Do you understand, and are you happy with, the way the data you have provided is being used?

In using AI within your work, it is important to understand your organisation's policy in relation to AI. This will be designed to help mitigate these risks.

Regulation of AI

There are different approaches to the regulation of AI. The UK has a 'pro-innovation' approach to AI regulation. This sets out five principles to be used by regulators, to regulate the use of AI within their specific sectors (Department for Science, Innovation and Technology, 2023). These are aligned to the risks outlined above and expressed as:

- safety, security and robustness;
- appropriate transparency and explainability;
- fairness;
- accountability and governance;
- contestability and redress.

The USA also has a pro-innovation approach to regulation. This can be compared to China, who regulates specific AI products (e.g. generative AI).

The EU was seen to take the lead on data protection and might be seen to be doing the same in relation to AI regulation. This is through the AI Act 2024, which provides a set of controls on AI based on the potential risk the AI poses. It is also proposing legislation to make it easier to seek redress where AI causes loss or damage (European Parliament, n.d.).

PRACTICAL TIPS

- Use the language of data science, ML and AI in a way that is meaningful to your audience.
- Consider the risks relating to each particular use of AI.
- Before adopting a complex or expensive solution, consider if a simpler, cheaper one would be equally effective.

- Remember that ethics and lawfulness relate to the use of data in AI.
- Check your company policy on AI use.

CASE STUDIES

Throughout this book we have included examples to help illustrate key points and build your understanding of them. These examples are by their nature simple and don't reflect the complexity of most data science projects. To create a richer picture of data science in action, here we have included several case studies of projects that have used data science. We are very grateful to the companies and individuals who have been willing to share their experiences. When reading these case studies, you will see many of the themes discussed in this book brought to life. Our contributors have also been kind enough to share their advice on data science and entrepreneurship, the latter reflecting the opportunities for innovation that data science creates within both markets and organisations.

INNOVATION FACTORY

Anwar is a reality TV star, with an interest in data science. It is best not to think *Love Island* in this context. Rather, think *Dragons Den* or *Shark Tank*, but earlier in the innovation cycle, when people take their vision and make it a reality – in other words, develop their product. He acts as an advisor on *Stars of Science*, both on the technical side and, equally important, the

business side. He asks the questions: who will buy the product and why?

Stars of Science helped to launch Anwar's first business (using sound detection to alert those with hearing impairment to emergencies). He recognised that being able to recognise patterns in sound had wider applications, and founded the Innovation Factory to build on this.

Birmingham, where the Innovation Factory is based, was at the heart of the motor industry, and it is still a city of cars. This can lead to congestion, with resulting air and noise pollution.

Anwar's theory was that vehicles' sounds could be used to both detect traffic and estimate pollution levels. This led to the creation of Traffic Ear, a cost-effective solution for monitoring both traffic and pollution.

Anwar knew he would need to collect lots of data on traffic to build the algorithms and train the models needed. He didn't quite envisage how hard this would be. The hardware was straightforward: acoustic sensors, a camera, an air pollution sensor and a central processing unit, along with some microprocessors, adaptors and Wi-Fi. Not that hard to put together until you need to fit all the components into a discreet

box and place it at the top of a lamp post, where it will be subject to the prevailing weather.

The challenge with the data lay in classifying the sounds so that the models could identify vehicles, and the emissions associated with them. Anwar, or one of his team, could have sat below the sensor and recorded each vehicle that passed to associate it to a sound. Nine months of sitting by a dual carriageway, through all weathers, didn't appeal to anyone. This led to the development of a novel solution to automate this: using a camera to record details of the vehicle, and using this data to label the recorded sounds. This allowed a classification model to be constructed, permitting different types of vehicles to be identified just by their sound. Other data, such as speed and air pollution, were also used to develop sound-based models.

These models don't exist in a vacuum, with their outputs being fed into a master model that then analyses them to provide holistic information on traffic and pollution levels. Anwar was keen to point out that the models used are bound by the data collected. In this case, the data set is limited to vehicles travelling up to 40 miles an hour (the speed limit on the roads where the data was collected). So, it would not be possible to extrapolate the data to estimate the pollution created by a bus travelling at 60 miles an hour.

The core technology developed for Traffic Ear is now being applied to other use cases. One is on the railways. How do you keep animals off the line and prevent trains striking them? The UK has a problem with deer, in Australia it's kangaroos and in India it's elephants. The answer is to use the Traffic Ear technology to detect animals and then use light and sound to scare them away. It's not that simple, as different animals have different behaviours and react in different ways. The species of animal needs to be detected along with what it is doing; is it just grazing or is it on the move? Anwar has developed a solution using generative AI. On the Traffic Ear technology detecting an animal, it plays the AI a few seconds of video and it can determine the species and what it is doing. This automatically triggers an appropriate response from the 20,000 that are available. And, yes, the animal's reaction is stored so that the data from that interaction can be used to improve the model.

Anwar highlighted the importance of prescriptive analytics, linking the data science outcomes to automated actions – in this case, the triggering of a light and sound pattern to deter the animal and reduce the risk of a train striking them.

When asked for advice, his key message was that data science, particularly AI, is a fast-developing field and you must invest in

your own learning. Your learning only sticks when you put it into practice.

SMART CONTAINER CO

Some people would say Steve had the dream job, simply cracking open beer and taking a few measurements. As for the beer, it had to be disposed of somehow. You can imagine the good cheer and wild parties that might follow. Oddly enough, Steve isn't that keen on beer, so the reality was rather different.

Smart Container Co, based in South Wales, helps companies to see what is happening to containers and products in the difficult-to-get-to parts of the supply chain. This includes brewers and the kegs in which they transport beverages. Their technology tracks and monitors kegs and the drinks inside them. Listening to their clients, they realised there were concerns that products could become over- or under-carbonated while in the keg, indicating problems with the drink's quality or a damaged keg (with the risk of explosion). This posed the question of how to measure the carbonisation of a liquid in a sealed, pressurised keg.

Steve stepped up to the role of resolving this locked-box puzzle. The hypothesis was that ultrasonic readings could be used to

determine fill level, and then this used to determine the level of carbonisation. A document review revealed very limited information on the subject. That which was found had a poor provenance (i.e. it wasn't from verified sources). This meant he needed to collect data himself.

To accurately collect data, Steve tried several different methods, using drinks in a range of containers. Eventually, he settled on filling and pressurising containers himself. In doing this, he had to source specialist equipment and learn how to use it.

Once a robust data set had been collected, Steve looked to understand the relationships between the data. He initially started out using a simple linear regression model. When this didn't reveal any substantive relationships, he investigated other models. This included multivariate and polynomial regression – to no avail, the hypothesis remained unproven.

Steve is very conscious of the old saying 'garbage in, garbage out'. He recognises the importance of data quality and the challenges of measuring it, including the tolerances of the machines used for this. He also advocates using someone as a sounding board to validate the approaches taken, especially if they have knowledge and experience you can tap into. Most of

all, he believes that data scientists need to be curious about what the data can reveal.

The few words of this case study don't reflect the time and effort that went into the collection and analysis of the data. Some would see the year of effort as a failure, as no relationship was found that could allow carbonisation levels to be determined. Steve and Smart Container Co don't see it that way. The results showed they were listening to their client, and they now know how not to measure carbonisation. This has allowed them to develop their thinking on how they could. Often, failure to prove the hypothesis is seen as a project failure, and buried. whereas it is in fact a valid outcome of a data science project and increases knowledge and understanding.

COGNITIVE BUSINESS

In the shadow of Ratcliffe-On-Soar, the last coal fired power station in the UK, you will find Ty and his company Cognitive Business. This area of the East Midlands is the historic epicentre of the UK power industry. Cognitive Business has emerged from this environment to provide data-driven solutions to the wind industry. Ty describes himself as a physicist who does lots of other things, including some data science.

Wind farms differ from power stations in one important respect: whereas each power station is unique, in a field of wind turbines, each will be similar. Ty recognised that this presented an opportunity to reduce costs and downtimes through predictive maintenance. Predictive maintenance involves working out when something is about to go wrong so that it can be fixed before it does. This was on the basis that the learning from one turbine could be intelligently applied to many.

The first challenge was finding data on wind farms. With some effort, Ty and his team tracked down open data on a small wind farm. This allowed the team to build a prototype that they could then share with a large energy generator. This generator saw the benefits, and gave access to the data from one of their farms, allowing a robust model to be developed through an iterative process.

Ty and his team make considerable use of machine learning to find patterns in the data and classify it. However, they start by visually inspecting the data and asking themselves questions around why changes they spot had occurred. Here, Ty's knowledge of physics comes in useful: industrial machines obey the laws of physics, and thus the laws of physics should connect

the data, these connections providing a focus for the automated investigations.

One of the challenges in predictive maintenance is the reliability of industrial machinery – it's very reliable. This means you don't get a rich data set of faults, which means data to train fault detection models on can be in short supply. To counter this, the team started by identifying the normal patterns within the data. Anomalies in the data started emerging, which could signify faults. These anomalies were used to guide engineering investigations, the outcome of which was then fed back into the models.

Although the turbines in a wind farm might notionally be the same, they are affected by external factors. This includes the wind direction, humidity and where they are physically located within the farm. To address this challenge, the team built millions of models to understand how the different factors affected the sensor data received. Clearly, each model could not be handcrafted, and so the process of model building and evaluation was automated.

As a result of the work by Ty and his team, previously unknown faults in wind farms can be identified and investigated. The fault does not have to have occurred in the past.

His view on data science is to focus on subjects you are passionate about. Find the data, use the free tools and build your skills.

GOOD WITH

Good With are a financial behavioural analytics company who are on a mission to change the way people are assessed for credit, to promote financial inclusion and support the mental well-being of groups who are excluded or underserved by financial services. Ellie and her colleagues are driven by their own experiences of financial exclusion, and bring their shared values and backgrounds in psychological sciences to their work. Ellie's interest in data science grew from applying statistical techniques to the research she undertook as a psychologist, which she later transferred to working in a data science team for a global financial company.

Good With have developed two core products: an alternative score for measuring financial capability and predicting borrowing risk, and a personalised learning journey designed to help people develop financial literacy and manage the emotional aspects of managing money. The personalisation of this framework is in part driven by personas (representations

of the target audience). The process applied to this work is outlined below.

Initial qualitative research identified four different financial personas for young people. Ellie wanted to validate these personas, using quantitative methods from data science. Data was collected from around 90 individuals using a questionnaire and by accessing their banking histories. The data was pre-processed to allow the correlation between key features to be explored. This included applying natural language processing to the text descriptions of financial transactions to classify spending, as well as looking at how often account balances fell below £0. It was unsurprising that, for example, high financial stress had a positive relationship with overdraft usage.

The next step was to see how the features grouped together, to identify whether the personas shown in the qualitative research were visible in the data. The pre-processed features used in the correlation analysis were loaded into an unsupervised ML clustering model, which identified four different groups. Remarkably, these groups were closely aligned to the original personas identified through the initial research. With the data to underpin the personas, it has been possible for Good With to develop the personalised learning pathways and make data available to lenders to allow them to make better

lending decisions, to the benefit of previously excluded customers.

Combining several different models in an ordered process results in a very transparent set of outputs. This means that any decision or recommendation made by the model has an audit trail and can be justified. Ellie explained this was important not only from a regulatory aspect (should someone challenge an automated decision), but also to build confidence in prospective customers.

Ellie recognises that a wide range of questions around financial and mental well-being could have been explored. In deciding what problems to explore the team asked the questions:

- What would the regulation allow them to do?
- How would understanding the problem lead to commercial benefits?
- How would the individual be affected?

Ellie uses the example of mental health: looking at people's financial data could be used to predict levels of depression. In a medical setting this might be beneficial. However, for Good With it didn't have value and might feel intrusive or be otherwise detrimental to the individual. Furthermore, special

category data (health) was involved, increasing the regulatory complications, meaning that this was an area they would not explore.

When working with data, Ellie highlights the importance of understanding the regulations. She has found that, although it might have slowed the work down, it meant the project didn't try to run before it could walk, leading to better problem definition and insight. She also emphasised the need to understand the implications of applying the data science in the real world, for example better lending decisions could increase lending, leading to lenders running out of money. However, her most important insight was that a data scientist isn't there to know the answer to a problem, but to bring their creativity to answering it.

SMARTabg

Dev's ambition is to become a winemaker. Although he adores travel and soaking up culture, he isn't looking to relocate to Chile or even France to do this. He'd like to produce fine English wines close to his home in Surrey. A step to fulfilling this dream is SMARTabg, a software company that includes financial trading software amongst its portfolio of products.

For smaller organisations involved in financial trading, there is a substantial cost in acquiring, storing and interpreting financial data. Dev recognised that they would benefit from someone doing this for them, providing those organisations with trading signals, depending on the market's movements.

Many of these trading signals are statistical models, for example comparing the moving average share price over different periods to determine if there is an upward or downward trend. Several signals combined create a strategic signal that can provide the basis for prescriptive analytics, when such strategic signals are used to automate trades. Here, probability comes into play: if five of five signals indicate a rising market, the probability of the market rising is higher than if three signals indicate it.

One of the biggest challenges facing Dev was ensuring the quality of the data he used to develop these signals. There is a plethora of sources of financial data; however, each source is of a different quality and has different costs associated with it. The higher the quality, the less data cleansing required, but the higher the cost. To determine the quality of different data sources, Dev took samples and ran a variety of tests including identifying outliers, which were investigated. This included looking at the standard deviation in price movements. This

sampling continued into the live platform to check and maintain the data quality.

Once the data was in place, around 200 models were built to enable the trading signals, and many of the models are common within the financial trading sector. How the data is stored is important for model development. For this project, Dev used both relational and NoSQL databases, which store unstructured data. One of the data sets (NoSQL) is needed for social context, because commentary by leading figures and bodies, such as the Bank of England, affects financial markets, and natural language processing can assist in interpreting this. The text of these comments is unstructured data, and thus doesn't fit into a relational data structure.

Over his career, Dev has spent time and effort building his domain knowledge through his work and qualifications. This helped him in this project, both in determining the quality of data and in identifying the models of most use to the traders they support. Although emphasising the importance of domain knowledge, he is keen to point out that in some instances it is good not to have any. This allows a focus on pattern detection unencumbered by existing understanding and preconceptions. In his view, data science is something that can be engaged with by people with many varied backgrounds.

CONCLUSION

This book is intended to introduce you to the world of data science. We have talked through the common techniques and models used within data science and more importantly we have described how to make data science a success within an organisation. There are plenty of pathways open to you to follow further, and our ambition is to shine a light on the options you can choose from.

We've explored applications of data science and its potential impact on society. We've delved into the technical aspects, from data collection and preparation to model building and evaluation. And we've also touched on the softer skills, such as communication and stakeholder management.

THE POWER OF DATA SCIENCE

Data science is more than just a collection of tools and techniques. It's a powerful approach that can be used to solve complex problems, uncover hidden insights and drive innovation. Whether you're a business analyst, a data scientist or just someone who's curious about the world around you, data science can help you to make better decisions and achieve your goals.

The human element

While data science is a technical field, it's important to remember that it's ultimately driven by people. The success of any data science project depends on the skills and creativity of the individuals involved. From data engineers to machine learning experts, each role is crucial in the process.

Data science has become a transformative force, revolutionising industries and reshaping societies. By harnessing the immense power of data, organisations can uncover valuable insights, make informed decisions and foster innovation.

The foundation of data science

At its core, data science is built on an understanding of the problem to be solved, the data available to us and an appreciation of how to use the answers we are shown. Taking on a role in data science can lead down many paths and requires a balance between using statistics or machine learning algorithms and data engineering/programming skills. These elements form the foundation that allows data scientists to extract meaningful insights from large, complex data sets. Their

communications skills allow them then to share this with their stakeholders.

Statistics and machine learning algorithms

Statistics provide the mathematical framework for analysing data. Techniques such as hypothesis testing, probability distributions and inferential statistics allow data scientists to make sense of data patterns and draw conclusions. Machine learning provides the tools to create models that 'learn' from data. These algorithms enable predictions, classifications, pattern recognition and decision-making.

Data engineering/programming skills

Much of this book has focused on the use of data science in an organisation, describing some of the modelling approaches. Preparing, storing, transforming and extracting are skills that are likely to involve databases and potentially a range of programming languages. They allow data scientists to manipulate data, build models and implement machine learning algorithms efficiently. Together, these skills empower data scientists to transform raw data into actionable insights that drive innovation in countless fields.

Where algorithms learn from data to make predictions or decisions, care should be taken to ensure that everyone understands both the data being used and the algorithm. The key groups of ML fall into:

Supervised learning: This uses labelled data to train models.

Examples:

- Linear regression for predicting house prices.
- Logistic regression for predicting whether a customer will churn.
- Decision trees and support vector machines for classification tasks.

Unsupervised learning: This works with unlabelled data to find patterns.

An example is clustering (e.g. grouping customers into segments).

Reinforcement learning: This involves training an agent to interact with an environment and maximise rewards.

Applications for this are robotics, game playing (e.g. AlphaGo) and autonomous vehicles.

A key objective of this book has been to introduce how to implement a data science project in real life. All the complex statistical models in the world do not make a great outcome on their own. They need to be used effectively in a business.

While data science has become a powerful tool to make smarter decisions, optimise processes and improve customer experiences, implementing a data science project can feel overwhelming, especially if your organisation is new to it. We have covered the stages of successfully implementing a data science project in a business in a simple and practical way.

As a reminder, the key steps should always cover:

Step 1: Identify a business problem (discover)

A successful data science project has a clear definition of the problem you're trying to solve. Make sure the problem is specific, measurable and aligned with your business goals.

Example:

Instead of saying, 'We want to improve sales', focus on a specific problem such as 'We want to predict which customers are likely to churn in the next three months.'

Tips:

- Talk to stakeholders (managers, employees or customers) to understand their pain points.
- Ask what decisions this project will improve.
- Prioritise problems that have a clear business impact and are feasible to solve with data.

Pitfall to avoid:

- Vague goals. If your problem is poorly defined, your data science project will lack direction and fail to deliver meaningful results.

Step 2: Assemble the right team (discover)

A successful data science project requires collaboration between different roles. A basic team structure should include:

- Business stakeholders: Define the problem and evaluate the results.
- Data scientists: Analyse the data and build models.
- Data engineers: Prepare and clean the data.
- IT/DevOps: Deploy solutions into production (e.g. on your website or app).

Tip:

- If your company is small, one person may wear multiple hats. That's OK. Just ensure every role is covered.

Pitfall to avoid:

- Lack of communication. If your team doesn't collaborate effectively, the project could fail to meet business needs.

Step 3: Gather and understand your data (source, prepare)

Once you've identified the problem, the next step is to collect the relevant data. Data is the foundation of any data science project, so focus on understanding its quality and availability.

Steps:

1. Locate your data: Identify where the relevant data exists (e.g. databases, spreadsheets, CRM systems).
2. Explore the data: Look for trends, missing values or inconsistencies.
3. Clean the data: Remove duplicate records, handle missing values and ensure formats are consistent.

Example:

If you're predicting customer churn, your data might include:

- customer demographics (age, location, etc.);
- purchase history;
- customer complaints.

Tips:

- Start with a small sample of data to explore before using the full data set.
- Use business knowledge to understand what the data represents.

Pitfall to avoid:

- Poor data quality. If your data is messy or incomplete, your results will be unreliable.

Step 4: Choose the right tools and techniques (explore)

Data science involves applying the right tools and techniques to analyse data and build models. The tools you choose will depend on your team's expertise and the complexity of the project.

Popular tools:

- Excel: Spreadsheet software that is versatile and accessible, a great starting point.
- Python or R: For data analysis, machine learning and visualisation.
- SQL: For querying databases.
- Tableau or Power BI: For creating dashboards and visualisations.
- Cloud platforms: AWS, Google Cloud or Azure for handling large-scale projects.

Tip:

- Use open-source tools (such as Python) to save money and leverage a large online community for support.

Pitfall to avoid:

- Overcomplicating the solution. Start simple, for example a basic statistical analysis may suffice instead of a complex ML algorithm.

Step 5: Build the model (create)

This is where the magic happens. Once your data is ready, you can start building a predictive model or analysing trends.

Steps:

1. Split the data into training data and testing data (e.g. 80 per cent training, 20 per cent testing).
2. Train your model on the training data.
3. Test the model on the testing data to evaluate its accuracy.

Example:

If you're predicting customer churn, you might use a classification algorithm such as logistic regression or decision trees to predict whether a customer will stay or leave.

Tip:

- Use simple models first (e.g. linear regression) and only move to complex ones if needed.

Pitfall to avoid:

- Overfitting. This happens when your model performs well on training data, but poorly on new data. Use testing data to check for this issue.

Step 6: Interpret results and communicate insights (analyse, communicate)

Once your model is ready, interpret the results and share them with stakeholders in a way they can easily understand.

Remember: the main goal is to solve the business problem.

Example:

Instead of saying ‘Our model has 85 per cent accuracy’, say ‘Our model predicts that 85 per cent of customers at risk of churning can be identified, allowing us to target them with retention offers.’

Tips:

- Use clear visuals such as charts and graphs.
- Focus on actionable insights (e.g. ‘Offer discounts to customers likely to churn’).

Pitfall to avoid:

- Too much technical jargon. Business stakeholders may not understand complex statistical terms.

Step 7: Deploy the solution (operationalise)

If your data science project is successful, the next step is to implement the solution in your business operations.

Example:

If you've built a churn prediction model, integrate it into your CRM system so customer service reps can identify high-risk customers and act accordingly.

Tip:

- Start with a pilot programme to test the solution on a small scale before full deployment.

Pitfall to avoid:

- Ignoring scalability. Ensure your solution can handle larger data sets or more users as your business grows.

Step 8: Monitor and improve (re-enter the data analysis lifecycle)

Data science is not a one-time effort. Once deployed, monitor the model's performance and update it as new data becomes available.

Tips:

- Schedule periodic reviews of the model.

- Collect feedback from users to ensure the solution is practical.

Pitfall to avoid:

- Set-it-and-forget-it mindset. Models can become outdated, or even just incorrect, over time as real-world data changes.

Final tips and insights:

- **Start small:** Begin with a simple project that solves a specific problem. As you gain experience, tackle more complex challenges.
- **Focus on the business need:** Always ask ‘How will this project save money, increase revenue or improve efficiency?’
- **Learn from failure:** Not all data science projects will succeed. Use failures as learning opportunities. Not all data science projects will give you ‘new’ answers, some may just tell you what you already knew – this is OK.

ETHICAL CONSIDERATIONS IN DATA SCIENCE

As data science continues to grow in influence, addressing its ethical implications is critical. Key considerations include:

- **Bias and fairness:** Ensuring algorithms do not perpetuate discrimination or inequality.

- Example: Avoid bias in hiring algorithms by ensuring diversity in training data.
- **Privacy:** Protecting sensitive data and adhering to regulations such as GDPR.
- Example: Anonymise data to prevent identification of individuals.
- **Transparency and accountability:** Making algorithms and their decision-making processes explainable. Holding individuals and organisations accountable for the impact of their models.

Ethical data science practices are essential to maintain trust and prevent harm in society.

THE FUTURE OF DATA SCIENCE

The field of data science is evolving rapidly, with several exciting trends on the horizon:

- **Advancements in AI and ML:** More sophisticated models, such as deep learning, will unlock new possibilities.
- **Big Data and cloud computing:** As data grows in volume, velocity and variety, scalable solutions such as cloud-based

platforms will become increasingly important.

- **Data science automation:** As tools become more and more accessible, data science will continue to become more commonplace. Democratising data science will make complex models and sophisticated outcomes available to non-experts.
- **Ethical AI:** Developing guidelines and regulations to ensure AI development aligns with societal values will be essential as more of our lives are predicted (controlled?) by data models.

By responsibly leveraging data science, we can unlock opportunities, drive meaningful innovation and create a better future for all.

A FINAL THOUGHT

In our introduction we saw how data science has changed sport. This is a small example of how data science is fundamentally transforming the way organisations and societies operate. From its foundations in statistics and ML to its applications in real-world decision-making, data science empowers us to uncover insights, solve problems and innovate. By embracing data science responsibly and addressing its ethical challenges, we can harness its power to build smarter

solutions, improve lives and shape a brighter future. As technology advances and new data sources emerge, the possibilities for innovation are endless. From personalised medicine to autonomous vehicles, data science is at the forefront of some of the most exciting developments of our time.

While data science can be a complex and challenging field, it's also incredibly rewarding. By mastering the fundamentals and staying up to date with the latest trends, you can unlock the full potential of data and make a real difference in the world.

GLOSSARY

3Vs: Relates to the volume, velocity and variety of Big Data.

5Vs: Adds veracity and variability to the 3Vs.

Accuracy: The completeness of a data set.

Administrative data: Data that has been generated through the activities of an organisation.

Aggregation: Combining data – often used to report on a group rather than individuals, where individuals might be recognised if reported on individually.

Agile: Often used to refer to iterative and incremental methods of project delivery – the term relates to software delivery methods based on principles set out in the Agile Manifesto.

AI Act 2024: European Union regulations for the use of artificial intelligence.

Algorithm: A sequence of steps or instructions to solve a problem, which might be found in a method.

Apriori algorithm: A method used in data mining to find frequent item sets and generate association rules.

ARIMA: A statistical method used for forecasting time series data by analysing past values – it uses a mix of moving averages and regression methods.

Array: A structured collection of items, often in rows and columns, used in programming and data analysis.

Artificial intelligence: Where a computer system performs a task normally done by a human.

Association rules: Patterns that show relationships between variables in data – often used in market basket analysis.

Bias: A systematic error that leads to incorrect results – it can happen in data collection or analysis.

Big Data: Used to describe the prevalence of data in today's society, having three key core concepts: volume, velocity, variety (the 3Vs).

Box and whisker plot: A graphical representation that shows the distribution of data based on five summary statistics.

Cartesian plane: A two-dimensional space defined by an x axis (horizontal) and a y axis (vertical) used to plot points.

Categorical: Data that can be placed into groups.

Classification: A method of grouping data into categories based on characteristics.

Clustering: A technique that groups similar data points together without predefined categories.

Correlation: A measure of how two variables are related – it shows whether they move together or in opposite directions.

CRUD: Create, read, update, delete – often referred to as the data lifecycle.

Data analysis lifecycle: A framework setting out the iterative steps needed to successfully deliver a data project.

Data classification: This can have different meanings:

- A way of categorising data to support good governance.
- To define the nature of the data, for example structured/unstructured.
- A method for analysing data.

Data governance: Sets out the strategy of managing data and operationalises it.

Data lake: A data store where the data has been stored in its raw format.

Data lifecycle: The lifecycle of a data item or data set, often referred to as CRUD.

Data mart: A subset of a data warehouse focused on a specific domain, for example sales.

Data Protection Act 2018: The UK law giving effect to GDPR.

Data science: Using analytical methods in an endeavour to gain insight from diverse data, or the art of finding patterns in data.

Data set: A collection of data.

Data storytelling pyramid: A framework for narrative consisting of background, trigger, key findings and action.

Data structure: Refers to the structure in which data is held, for example a table.

Data type: How data is held in a field, for example integer – it can also refer to whether the data is public, open, administrative or research.

Data warehouse: A data store where the data is used for reporting – the data is often prepared to enable this.

Decision tree: A flowchart-like structure that helps to make decisions based on different conditions.

Deep learning: A term used to describe the use of neural networks with many layers.

Derived data: Where new data is created from combining or transforming existing data.

Descriptive analytics: Analysing data to describe what happened – it uses past data to summarise results.

DIKW: A framework representing data, information, knowledge and wisdom.

Domain: The context in which the project is taking place – this informs the problem, the stakeholders and the interpretation of outputs.

ELT: The process of extracting, loading and transforming data.

Enrich: Where additional data is added to a data set.

Ethics: The principles by which we and society judge our own and others' behaviours – these change over time.

ETL: The process of extracting, transforming and loading data.

Export: Where data is taken from a data store.

Feature: Key characteristic or attribute of a data set.

Feature scaling: Creating a common scale for different features, that is, ensuring they have the same range.

Foundational model: A form of generative AI that has been pre-trained and can be adapted to different uses.

Four pillars of data storytelling: Design principles to aid the visualisation of data storytelling consisting of symbols, colour, captions and editorial layout.

GDPR: The General Data Protection Regulation contains the European Union's rules for use of data.

General AI: Artificial intelligence that can handle all tasks a human can undertake.

Generative AI: Artificial intelligence that generates new data or content.

Granularity: The level of detail in the data.

Hallucination: Where artificial intelligence makes something up, that is, returns an incorrect result.

Heat map: A graphical representation of data where values are depicted by colour – it helps to visualise data density.

Histogram: A graphical representation of the distribution of numerical data showing the frequency of data points in intervals.

Hybrid: A method of project delivery combining aspects of linear and iterative and incremental methods.

Hypothesis: A theory that provides a starting point for an investigation.

Hypothesis testing: A test to determine statistically the extent to which a hypothesis can be explained by the data.

ICO: The Information Commissioner's Office can investigate and take enforcement action for breaches of data protection laws.

Imputation: Where a value is given to a missing data item, often determined from the data around it.

Insight: An understanding of what the model has revealed upon evaluation.

Intellectual property: Refers to created works such as music, art and literature.

Internet of things: A network of interconnected devices embedded with sensors and software.

Interval: A type of quantitative data where the difference between values is meaningful, but there is no true zero – an example is temperature in Celsius.

Latency: The delay in transferring or processing data.

Lineage: The origin and history of the data set.

Linear regression: A statistical method that models the relationship between a dependent variable and one or more independent variables using a straight line.

Lists: A simple collection of items, usually in a specific order.

Load: Where data is placed in a data store.

Machine learning: Where software designed to perform a task improves the more it undertakes a task.

Metadata: Data about data.

Method: A technique used to analyse the data.

Model: A representation of a problem space made up of methods and data.

Narrow AI: Artificial intelligence that is designed to perform a specific task.

Neural network: A branch of machine learning that uses layers of nodes to take inputs and deliver an output.

Nominal: A type of qualitative data that categorises without a specific order, for example colour.

Obfuscation: The blurring of data to preserve confidentiality.

One-hot encoding: Converting categorical data into numerical data.

Open data: Public data of a known quality and a licence for its use.

Operational data: Data that has been generated through the activities of an organisation.

Ordinal: A type of qualitative data that has a clear order or ranking, for example ratings (such as 1 star to 5 stars).

Outlier: A data point that is significantly different from others – it can skew results.

Overfitting: A modelling error that occurs when a model is too complex and learns noise in the training data, making it perform poorly on new data.

Pattern: An arrangement or sequence of data normally found through examining a model.

PESTLE: A framework for identifying external factors that might affect a project, the external factors being political, economic, socio-cultural, technological, legal and environmental.

Power interest grid: A framework to help determine how to manage stakeholders based on their different levels of power and interest.

Predictive analytics: Using data and statistical algorithms to forecast what might happen in the future.

Prescriptive analytics: Analysing data to recommend actions – it tells you what to do based on predictions.

Principles of data: Used to describe high-level principles applied when discussing data – this has no formal definition but commonly encompasses the data analysis lifecycle and the concept of open, public, proprietary, administrative and research data.

Principles of user experience: Useful, desirable, usable, valuable, accessible, findable, credible.

Proprietary data: Data that is owned and controlled by an organisation.

Public data: Any data that is publicly accessible – this does not imply a right to use it.

***p*-value:** A number that helps to determine the significance of results in hypothesis testing – a low *p*-value indicates strong evidence against the null hypothesis.

Qualitative: Data that describes qualities or characteristics, for example colours, names and opinions.

Quantitative: Data that can be measured and expressed as numbers, for example height, weight and age.

Ratio: A type of quantitative data with meaningful differences and a true zero, for example weight and height.

Redaction: The blanking of data to preserve confidentiality.

Research data: Data that has been collected for a specific study or experiment.

Sampling: The process of selecting a small group from a larger population to study.

Significance level: A threshold in statistics to determine if results are likely due to chance – commonly set at 0.05 or 5 per cent.

Skew: A measure of how much a distribution leans to one side – positive skew means it leans right; negative skew means it leans left.

Stakeholder: Anyone with an interest (stake) in a project or its outcome.

Stakeholder wheel: A framework to help identify stakeholders.

Standard deviation: A measure of how spread out the data points are from the mean – a low standard deviation means data points are close to the mean.

Stem-and-leaf plot: A method of displaying quantitative data that retains the original data values while showing the shape of the distribution.

Storytelling: Using narrative to aid audience understanding.

Stratified: A sampling method where the population is divided into groups (strata) before sampling to ensure representation.

Summary statistic: A number that summarises or describes a set of data, like mean, median or mode.

Supervised learning: Where the data the model is trained on is labelled or tagged.

Synthetic data: Artificially generated data.

Tool: Software used to build and analyse models.

Train and test data sets: In machine learning, the training data set is used to train the model, while the test data set is used to evaluate its performance.

Transform: Where data form or structure is changed.

Transparency: The ability to understand how artificial intelligence has reached a result.

Trend: A pattern of changes over time.

Unsupervised learning: Where the model learns by looking for patterns in unlabelled data.

User experience honeycomb: A framework showing the seven principles of user experience.

Variability: The inconsistency in data – one of the 5Vs.

Variance: The average of the squared differences from the mean – it shows how much the data varies.

Variety: A concept of Big Data highlighting the diverse range of data generated.

Velocity: A concept of Big Data highlighting the speed at which data is created and processed.

Veracity: The quality of data – one of the 5Vs.

Volume: A concept of Big Data highlighting the vast amounts of data being created.

Waterfall: A linear method of project delivery where activities are cascaded, one activity being completed before another starts.

REFERENCES

American Optometric Association (2024) 'Color vision deficiency'. *American Optometric Association*.

<https://www.aoa.org/healthy-eyes/eye-and-vision-conditions/color-vision-deficiency>.

BBC (2018) 'Facebook value drops by \$37bn amid privacy backlash'. *BBC*. <https://www.bbc.co.uk/news/business-43462423>.

BBC (2020) 'A-levels and GCSE's: How did the exam algorithm work?' *BBC*. <https://www.bbc.co.uk/news/explainers-53807730>.

Cameron, J. (director) (1984) *The Terminator*. Orion Pictures.

Department for Science, Innovation and Technology (2023) 'A pro-innovation approach to AI regulation', Command Paper 815. HMSO. <https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach/white-paper>.

Dykes, B. (2019) *Effective Data Storytelling: How to Drive Change with Data, Narrative and Visuals*. Wiley.

European Parliament (n.d.) 'AI liability directive'. *Legislative Train Schedule*. European Parliament.

<https://www.europarl.europa.eu/legislative-train/theme-a-europe-fit-for-the-digital-age/file-ai-liability-directive>.

European Union (2025) 'Article 3: Definitions'. *EU Artificial Intelligence Act*. Future of Life Institute.

<https://artificialintelligenceact.eu/article/3/>.

Feigenbaum, A. and Alamalhodaie, A. (2020) *The Data Storytelling Workbook*. Routledge.

Halpert, M. and Moloney, M. (2022) 'Elon Musk taking legal action over Twitter account that tracks his private jet'. *BBC*.

<https://www.bbc.co.uk/news/world-us-canada-63978323>.

ICO (n.d.). 'Data Protection Act 2018 and UK General Data Protection: Regulation reprimand'. *Information Commissioner's Office*. <https://ico.org.uk/media/action-weve-taken/reprimands/4030350/20240702-cvhs-reprimand.pdf>.

Lewis, M. (2004) *Moneyball: The Art of Winning an Unfair Game*. W. W. Norton & Co.

McArthur, T. and Youngs, I. (2024) 'Hundreds lodge complaints over Oasis ticket prices'. *BBC*.

<https://www.bbc.co.uk/news/articles/c20r26p7d0ro>.

Melley, J. (2024) 'Census may have overestimated number of trans people in England and Wales'. *BBC*.

<https://www.bbc.co.uk/news/articles/c3030519849o>.

Miller, B. (director) (2011) *Moneyball*. Columbia Pictures.

Mitchell, T. (1997) *Machine Learning*. McGraw Hill.

Morville, P. (2004) 'User experience design'. *Semantic Studios*.

https://semanticstudios.com/user_experience_design/.

NASA (2024) 'What is artificial intelligence?'. *NASA*.

<https://www.nasa.gov/what-is-artificial-intelligence/>.

Newcombe, R. (2003) 'From client to project stakeholders: a stakeholder mapping approach'. *Construction Management and Economics*, 21(8), 841–848.

Palmer, J. (2013) 'Mobile location data “present anonymity risk”'. *BBC*. <https://www.bbc.co.uk/news/science-environment-21923360>.

Riker, C. (2024) *Conglomerate Ascendancy: A Deep Dive into the World of Business Goliaths*. Self-published.

Wikipedia (n.d.) 'DIKW pyramid'. *Wikipedia*.

https://en.wikipedia.org/wiki/DIKW_pyramid.

INDEX

Page numbers in italics refer to figures or tables.

2 common learning methods [129](#), [130](#)

2 common methods of stakeholder engagement [32](#)

3 key metrics for evaluating rules [118](#)

3 key project aspects [16](#)

3 lenses of data quality [47](#)

3 project certainty requirements [15](#)

3 stages of data movement [45](#), [45](#)

3 types of data analytics [67](#)

3 types of data store [44](#)

3Vs/5Vs of big data [40](#), [41](#)

4 ages of data (CRUD) [46](#), [46](#)

4 pillars of data storytelling [125](#), [126](#), [126](#)

4 principles of the BCS code of conduct [23–4](#)

4 types of quantitative/qualitative data [34](#), [34](#)

access controls [26](#)

aggregation [25](#), [35](#), [67](#)

Agile [17](#)

alternative hypothesis [71](#), [115](#)

Apriori algorithm [94–7](#)

area chart [105](#), [105](#), [106](#)

ARIMA (autoregressive integrated moving average) [81–4](#)

arithmetic average [58](#)

artificial intelligence (AI) [1](#), [69](#), [128](#), [129](#), [131–3](#)

foundational models [132](#)

generative [131](#), [132](#), [133](#)

hierarchy [132](#)

narrow [131](#), [132](#)

practical tips [133](#)

regulation [133](#)

risks [132–3](#)

association [66](#), [72](#), [74](#), [94–8](#), [118](#)

association rule mining [94](#), [97](#), [98](#)

averages [58–60](#)

backlogs [17](#)

bar charts [67](#), [99](#), [102–3](#), [103](#)

BCS, The Chartered Institute for IT [23](#)

bias [19](#), [35–7](#), [55](#), [56](#), [115](#), [131](#), [132](#), [146](#)

big data [40–1](#), [43](#), [147](#)

box and whisker plot [106](#), [107](#), [107](#)

Box–Jenkins model [81](#)

Caldicott Principles [23](#)

Cartesian coordinate system (Cartesian plane) [62–4](#), [62](#), [64](#), [65](#)

case studies [38](#), [134–9](#)

Cognitive Business [136–7](#)

Good With [137–9](#)

Innovation Factory [134–5](#)

SMARTabg [139](#)

Smart Container Co [135–6](#)

categories of stakeholder [6–7](#), [6](#)

central tendency [58](#), [59](#), [60](#), [70](#)

change control [16](#)

clustering [66](#), [72](#), [73](#), [89–94](#), [130](#), [138](#), [141](#)

communication [119–27](#)

audience [121–2](#)

DIKW model [119](#), [119](#)

domain context [122](#)

practical tips [127](#)

recommendations [127](#)

stakeholder communication plan [11–12](#), [12](#)

storytelling [123–7](#)

user experience [120–1](#)

competitors [7](#)

complementary events [61](#)

conditional probability [61](#)

confidence intervals [116](#)

confusion matrix [117](#)

content analysis [38](#)

continuous distribution (probability) [61](#)

convenience sampling [36](#)

correlation [35](#), [72](#), [74–6](#), [106](#), [107](#), [110](#), [124](#), [127](#), [138](#)

CRUD (create, record, update, delete) [46](#), [46](#)

customers [7](#)

customer segmentation [67](#), [89](#), [94](#)

data

access [25](#), [26](#)

administrative/operational [43](#)

aggregation [25](#), [67](#)

analysis [2](#), [7](#), [37](#), [38](#), [58](#), [60](#), [67](#), [69](#), [144](#)

analysis lifecycle [13–18](#), [27](#), [46](#), [48](#), [69](#), [113](#), [119](#), [120](#), [122](#), [146](#)

analyst [29](#)

analytics [2](#), [67](#)

anonymous [25–6](#)

architect [28](#), [30](#)

big [40–1](#), [43](#), [147](#)

classification [24](#), [25](#), [73](#)

collection [37](#), [38](#), [41–2](#), [44](#), [140](#)

compliance officer [29](#)

compression [89](#)

creation [42](#), [43](#)

duplication [54](#)

engineer [28](#), [140](#), [141](#), [142](#)

engineering skills [141–2](#)

enriched [56–7](#)

ethics [21](#)

exploratory data analysis (EDA) [37](#), [69](#), [89](#)

fields [15](#)

finding [43–4](#)

form [49](#)

governance [43](#), [45–6](#), [47](#), [48](#)

granularity [40](#), [49–51](#)

historical [67](#), [68](#), [80](#)

impact assessment [18](#)

labelled [141](#)

lake [44](#), [45](#)

lifecycle [46](#), [46](#)

mart [44](#), [45](#)

metadata [47–8](#)

minimisation [22](#)

moving [45](#)

NOIR (nominal, ordinal, interval, ratio) [34](#), [34](#)

numeric [53](#)

obfuscation/redaction [25](#)

open [42](#), [43](#), [47](#), [137](#)

operationalised [19](#)

operationalising [127](#)

personal [21](#), [23](#), [24](#), [25](#)

preparation [15](#), [44](#), [45](#), [49–57](#)

properties [33–9](#), [53](#)

proprietary [43](#)

public [42](#), [43](#)

qualitative [34](#), [38](#), [39](#)

quality [19](#), [41](#), [46–7](#), [47](#), [54](#), [56](#), [57](#), [136](#), [139](#), [143](#)

quality dashboard [56](#)

quantitative [34](#), [38](#), [108](#), [125](#)

quantity [39](#)

raw [28](#), [44](#), [45](#), [60](#), [81](#), [119](#), [141](#)

research [43](#)

scientist [3](#), [28](#), [29](#), [41](#), [45–6](#), [65–7](#), [80](#), [121](#), [124](#), [136](#), [139–42](#)

skewed [35–7](#), [36](#), [37](#), [54](#)

source/sourcing [15](#), [18](#), [19](#), [30](#), [32](#), [35–6](#), [40–8](#), [55](#), [57](#), [120](#), [139](#), [147](#)

special category [23](#)

spread [70](#)

storage [30](#), [41](#), [44](#)

stores [44](#), [45](#)

storytelling [123–7](#)

storytelling pyramid [123](#), [124](#)

structures [51–2](#), [54](#), [56](#), [139](#)

subjects [6](#), [7](#)

types [14](#), [33](#), [34](#), [42–3](#), [43](#), [52](#)

understanding [69–71](#)

use [18](#), [19](#), [21–6](#), [29](#), [40](#), [43](#), [44](#), [53](#), [57](#), [113](#), [116](#), [129–30](#)

virtual inspection [55](#), [67](#), [99–111](#)

visualisation [67](#), [100–2](#)

warehouse [44](#), [45](#)

Data Protection Act 2018 [21–2](#)

data science [140–7](#)

case studies [134–9](#)

definition [1](#), [2](#)

ethical considerations [146](#)

ethical issues [3](#), [4](#) [18–26](#)

foundation of [140–1](#)

future of [147](#)

human element [140](#)

key project steps [142–6](#)

machine learning, and [129](#), [130–1](#)

models [113](#)

DBSCAN (density-based spatial clustering of applications with noise) [89](#)

decision trees [68](#), [69](#), [85–9](#), [86](#)

delivery team [27–9](#)

demand forecasting [68](#)

dependent events [61](#), [116](#)

descriptive analytics [2](#), [29](#), [67](#), [69](#)

DIKW (data, information, knowledge, wisdom) pyramid [119](#), [119](#)

DIPA (data protection impact assessment) [18](#), [19](#)

discrete distribution (probability) [61](#)

domain context [29](#), [47](#), [122](#)

domain knowledge [1](#), [28](#), [97](#), [139](#)

dynamic pricing PESTLE [8](#), [9](#)

EDA (exploratory data analysis) [37](#), [69](#), [89](#)

effective enquiry [31–2](#)

ELT (extract, load, transform) [45](#), [45](#)

empirical probability [61](#)

employees [7](#)

ethics/lawfulness [18–26](#)

commercial constraints [24](#)

data [21](#)

data science [146](#)

definition [20](#)

governance [23–4](#)

intellectual property [24](#)

internal policy [23](#)

legislation/regulation [21–3](#)

practical tips [26](#)

questions [20–1](#)

risk mitigation [24–6](#)

stakeholder [20](#)

ETL (extract, transform, load) [45](#), [45](#)

event probability [61](#)

exploratory data analysis (EDA) [37](#), [69](#), [89](#)

exponential smoothing [81](#)

feasibility study [16](#), [27](#)

feature importance [118](#)

feature scaling [52](#), [53](#)

foundational models (AI) [132](#)

framing the problem [30–1](#), [32](#), [119](#)

gates (data analysis lifecycle) [18](#)

GDPR (General Data Protection Regulation) [21–3](#), [26](#), [146](#)

glossary [148–54](#)

governance [16](#), [23–4](#), [43](#), [45–6](#), [47](#), [48](#), [133](#)

granularity [40](#), [49–51](#)

harm [21](#)

heat map [107](#), [108](#)

hierarchical clustering [89](#)

hierarchy of AI definitions [132](#)

histograms [103](#), [104](#)

hybrid project delivery [17](#)

hypothesis [40](#), [47](#), [71–2](#)

alternative [71](#), [115](#)

development [31](#)

null [71](#), [72](#), [114](#), [115](#), [116](#)

testing [66](#), [72](#), [114–16](#), [122](#), [123](#), [12](#), [141](#)

types [71](#)

understanding [71–2](#)

ICO (Information Commissioner’s Office) [23](#)

independent events [61](#), [116](#)

intellectual property [24](#)

interactivity [100](#), [100](#), [101](#), [111](#)

interquartile range (IQR) [60](#)

interviews [32](#)

intuitive design [100](#)

iterative/incremental project delivery [17](#)

iterative process [38](#), [137](#)

k-means clustering [89](#), [90](#), [91](#)

KPI (key performance indicator) [33](#), [34](#), [67](#)

law of large numbers [61–2](#)

legislation/regulation [21–3](#)

LIME (Local Interpretable Model-agnostic Explanations) [118](#)

linear project delivery [16](#)

linear regression [68](#), [76–9](#), [79](#), [80](#), [81](#), [136](#), [141](#), [144](#)

line graphs [104](#), [104](#), [105](#), [109](#), [110](#), [111](#)

literature review [38](#)

logistic regression [85](#), [141](#), [144](#)

machine learning (ML) [68](#), [128–31](#), [137](#), [140](#), [141](#), [144](#)

management strategies [10](#), [11](#)

managers [7](#)

MAPE (mean absolute percentage error) [85](#)

mean (arithmetic average) [36](#), [37](#), [54](#), [56](#), [58–60](#), [70](#), [79](#), [81](#)

measures of spread [58](#), [60](#), [65](#)

median [37](#), [58](#), [59](#), [70](#), [106](#)

metadata [47–8](#)

missing values [54](#), [143](#)

mitigation strategies [24–6](#), [56–7](#)

mode [37](#), [58](#), [59–60](#), [70](#)

model(s) [31](#)

Apriori algorithm [94](#)

ARIMA [81–4](#)

Box–Jenkins [81](#)

choosing between [72–80](#)

classification [85](#)

data science [113](#)

data storytelling [123](#), [123](#), [124](#)

decision trees [68](#), [69](#), [85–9](#), [86](#)

DIKW [119](#), [119](#)

evaluation [113–18](#)

explainability tools [118](#)

false positive/negative [117](#)

foundational (AI) [132](#)

interpretation [118](#)

linear regression [76](#), [77](#), [81](#), [136](#)

selection [66–7](#)

statistical [59](#), [62](#), [68](#), [74](#), [76](#), [81](#), [86](#), [94](#), [139](#), [141](#)

support vector machine [85](#), [141](#)

training set/test set [116](#), [117](#)

understanding data [69–71](#)

value [123](#)

Musk, Elon [21](#)

negatively skewed data [36](#), [37](#)

neural networks [68](#), [130](#), [131](#), [131](#)

NOIR (nominal, ordinal, interval, ratio) data types [34](#), [34](#)

null hypothesis [71](#), [72](#), [114](#), [115](#), [116](#)

obfuscation/redaction [25](#)

observed frequency [61](#)

operationalised data [19](#), [29](#), [127](#)

outliers [35](#), [55](#), [56](#), [58–60](#), [60](#), [103](#), [106](#), [107](#), [110](#), [139](#)

overfitting [116](#), [117](#), [144](#)

owners [7](#)

partial dependence plots [118](#)

partners [7](#)

patterns over time [53–4](#)

Pearson's correlation coefficient [74](#)

performance metrics [116](#), [123](#)

PESTLE (political, economic, socio-cultural, technological, environmental) analysis [8–9](#), [9](#), [29](#)

plotting points [62](#), [63](#), [64](#), [65](#)

positively skewed data [36](#), [36](#)

power interest grid [9](#), [10](#), [10](#), [11](#), [11](#)

practical tips [3](#), [12](#), [17](#), [26](#), [32](#), [48](#), [57](#), [98](#), [127](#), [133](#)

predictive analytics [2](#), [67](#), [68](#), [69](#)

prescriptive analytics [2](#), [67](#), [68](#), [69](#), [135](#), [139](#)

pricing strategy [68](#)

principles of GDPR [22–3](#)

probability [58](#), [60–2](#), [65](#), [68](#), [72](#), [87](#), [113–14](#), [116](#), [128](#), [139](#), [141](#)

problem definition [30–1](#), [32](#), [37](#), [138](#)

product combinations [98](#)

project

failure [136](#)

manager [8](#), [16](#), [29](#)

resources [29–30](#)

roles [27](#), [28](#), [28](#), [29](#)

project delivery [13–17](#)

data analysis lifecycle [13–17](#)

hybrid [17](#)

iterative/incremental [17](#)

linear [16](#)

practical tips [17](#)

project discovery [27–32](#)

definition [27](#)

delivery team [27–9](#)

effective enquiry [31–2](#)

hypothesis development [31](#)

practical tips [32](#)

problem definition [30–1](#), [32](#), [37](#), [138](#)

resources [29–30](#)

properties of data [33–9](#)

bias [35–7](#)

qualitative research [37–8](#)

quantitative research [37](#)

‘right’ properties [33](#)

skewed data [35–7](#), [36](#), [37](#)

p-value [114](#), [115](#), [116](#)

qualitative data [34](#), [38](#), [39](#)

qualitative research [37–8](#), [138](#)

quality checks [55–6](#)

quality risks [54–5](#), [57](#)

quantitative data [34](#), [38](#), [108](#), [125](#)

quantitative research [37](#)

random forest [68](#), [85](#)

random sampling [36](#)

range [60](#)

regression [68](#), [72](#), [73](#), [76–81](#), [85](#), [118](#), [136](#), [141](#), [144](#)

regulators [7](#)

reinforcement learning [131](#), [141–2](#)

researcher [28](#), [71](#)

residual analysis [118](#)

resources [29–30](#)

responsive design [101](#)

risk

AI [128](#), [132–3](#)

assessment [68](#), [70](#)

mitigation [24–6](#)

quality [54–5](#), [57](#)

storytelling [125](#)

RMSE (root mean squared error) [85](#)

R-squared value [118](#)

sample/sampling [35–7](#), [39](#), [54–5](#), [114](#), [118](#), [143](#)

scale [52–3](#)

scatter plot [106](#), [106](#), [109](#), [110](#), [111](#)

scope [30](#)

scope, time, cost [16](#)

SHAP (SHapley Additive exPlanations) [118](#)

significance testing [116](#)

society [7](#)

sponsor [9](#), [11](#), [12](#), [14](#), [27](#)

sprints [17](#)

stakeholder [2](#), [3](#), [4–12](#), [29](#), [30](#), [37](#), [121](#), [142](#), [145](#)

categories of [6–7](#), [6](#)

communication plan [11–12](#), [12](#)

definition [4](#)

engagement [31](#), [32](#)

ethics [20](#)

example [5–6](#), [5](#)

expectations [13](#), [17](#)

identifying [4–7](#), [20](#)

internal [8](#)

investigating [8–9](#)

key [4](#)

managing [9–12](#), [13](#), [140](#)

needs [127](#)

potential [8](#)

power/interest [9–11](#)

practical tips [12](#)

stakeholder wheel [6–7](#), [6](#), [20](#)

strategies for management [10](#), [11](#)

standard deviation [37](#), [60](#), [70](#), [71](#), [139](#)

statistical analysis [37](#), [58](#), [144](#)

stem-and-leaf plot [108](#), [109](#)

steps in analysis lifecycle [13](#), [14](#)

storytelling [123–7](#)

strategies for stakeholder management [10](#), [11](#)

stratified sampling [36](#)

subject matter expert [28](#)

supervised learning [73](#), [85](#), [89](#), [129](#), [130](#), [141](#)

suppliers [7](#)

supply chain optimisation [68](#)

support vector machine [85](#), [141](#)

support, confidence, lift [118](#)

testing hypothesis [66](#), [72](#), [114–16](#), [122](#), [123](#), [12](#), [141](#)

test set [116](#)

thematic analysis [38](#)

time-boxing [30](#)

time series analysis [73](#), [80–5](#)

time series forecasting [68](#), [81](#)

training set [89](#), [116](#)

underfitting [116](#), [117](#)

unsupervised learning [89](#), [130](#), [141](#)

user experience (UX) [99](#), [100–2](#), [120](#), [121](#), [127](#)

user experience honeycomb [120](#)

user research [37](#), [38](#)

variance [54](#), [60](#), [70](#), [79](#)

visualisations [99–111](#)

accessibility [101](#)

audience [99](#)

bar charts [102–3](#), [103](#)

chart types [99](#)

combining chars [109–11](#)

data [67](#), [100–2](#)

interactivity [100](#), [100](#), [101](#), [111](#)

intuitive design [100](#)

labelling [102](#), [111](#)

responsive design [101](#)

user experience [100–2](#)

visualising distance [64](#)

volume, velocity, variety (3Vs of big data) [40](#), [41](#)

Waterfall [16](#)

workshops [32](#)

DATA SCIENCE FOUNDATIONS

Navigating digital insight

Stephen Mariadas and Ian Huke

While data science is powering most modern organisations its technical intricacies often remain confined to a select few.

Data Science Foundations provides foundational knowledge for newcomers to the field, building their confidence and understanding. This will enable effective project contributions through informed decision-making and action. Written in an accessible manner, this book provides explanations of complex concepts and methods in layman's terms supported by examples. It also includes a holistic view of data science covering the technical, ethical and delivery challenges illustrated by case studies.

Aligned with industry-recognised qualifications, this is the definitive guide for aspiring data scientists, providing a solid foundation in data analysis techniques.

- Written for the beginner, providing clear explanations in layman's terms
- Explanations supported by practical examples
- Includes real-world case studies
- Content aligned to data analysis qualifications
- Covers the technical, ethical and delivery challenges

ABOUT THE AUTHORS

Steve Mariadas is CEO of an Institute of Technology, awarded a Technical Teaching Fellowship by the Royal Commission on delivering data skills to industry. Steve is passionate about building data talent in everyone. Ian Huke, an independent consultant, qualified as a lawyer before moving into software. He has worked for start-ups and large organisations, engaging in all aspects of change.

A must-read companion for anyone eager to dive into data science.

Jake Wilson PhD, Data Scientist, GHGSat Inc

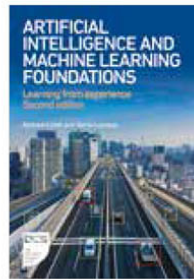
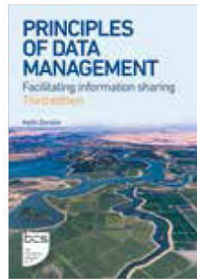
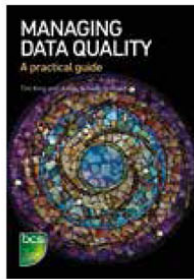
This book will definitely be the one I hand to our team members who are about to embark on a new project. Highly recommended.

John Opsblum, Co-Founder and Chief Consulting Officer, Illuminet Solutions

An excellent, thoughtful, and concise grounding in data science, packed with relatable examples, practical tips and a comprehensive framework for achieving successful data science projects.

Paul Williamson, Senior Business Analyst (Private Equity)

You might also be interested in:



**Business,
Information Technology**

Cover photo: iStock © Natalia Lagutkina



BCS, The Chartered Institute for IT
3 Newbridge Square, Swindon, SN1 1BY, UK
+44 (0) 1793 417 417
publishing@bcs.uk • www.bcs.org/contact

EU GPSR Authorised Representative
LOGOS EUROPE, 9 rue Nicolas Poussin,
17000, La Rochelle, France
Contact@logoseurope.eu

